Quantum Field Theory

Lecture Notes

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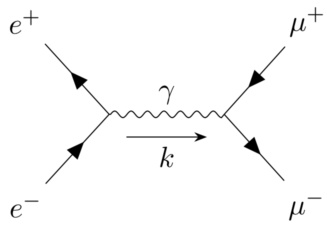

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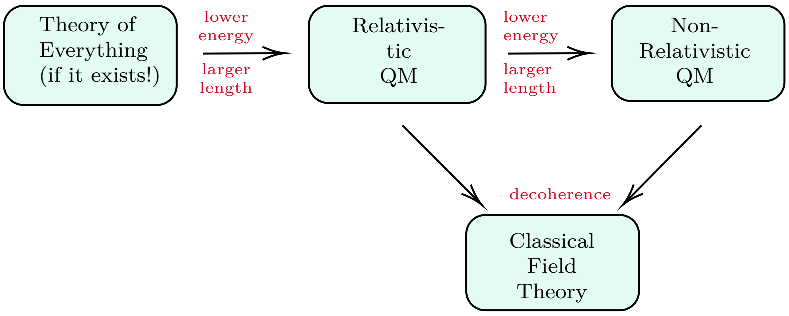
Lecture 01: Intro and Quantum SHO

People do 'weird' stuffs for earning,
You can do the same for learning!

Many physical theories have been proposed and it is imperative to see how each theory is linked with the previous in some way.



From the proposed theory of everything, we can transform to a *relativisitc* quantum field theory which will be the topic of discussion for most part. When subjected to *decoherence*, a relativistic QFT collapse to a classical QFT as noise hides the 'quantumness' from consideration. Hence we will start our discussion from classical fields and then move to quantum fields.

Since the foundational aspect of QFT (and many other topics in Physics) is the Harmonic Oscillator, let us discuss the quantum Harmonic oscillator for introduction. For that, note that the classical Hamiltonian for the harmonic oscillator is given by:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

We do not like the things like m and ω which prevent us from seeing things clearly 😅. Hence, we invoke the holy action of making things dimensionless. For that, we define:

$$X = \sqrt{\frac{m\omega}{\hbar}}x \qquad \longrightarrow \qquad x^2 = \frac{\hbar}{m\omega}X^2$$
$$P = \frac{1}{\sqrt{m\omega\hbar}}p \qquad \longrightarrow \qquad p^2 = m\omega\hbar P^2$$

Substituting these in the Hamiltonian, we have:

$$H = \frac{\hbar\omega}{2}(X^2 + P^2)$$

Note that, we are still within the classical domain. Now, let us elevate x and p to operators and we define:

$$[x,p] = i\hbar\,\mathbb{1} \implies [X,P] = \sqrt{\frac{\cancel{m\omega}}{\hbar}}\frac{i\hbar\,\mathbb{1}}{\sqrt{\cancel{m\omega\hbar}}} = i$$

Introducing the commutator bracket brings us to the quantum world. Now, we invoke our very own ladder operators:

$$\hat{a} = \frac{1}{\sqrt{2}}(\hat{X} + i\hat{P}) \qquad : \text{annihilation operator}$$
$$\hat{a}^\dagger = \frac{1}{\sqrt{2}}(\hat{X} - i\hat{P}) \qquad : \text{creation operator}$$

Then we will have [1]:

$$\left[a, a^\dagger\right] = \frac{1}{2}\left[X + iP, X - iP\right] = \frac{-i}{2}([X,P] - [P,X]) = -i \times i = 1$$

[1] henceforth, forsaking the hat symbol and identity operator $\mathbb{1}$, since they cause nothing but trouble, when the context is clear

Also, note that:

$$a^\dagger a = \frac{1}{2}(X^2 + P^2 + i\underbrace{(XP - PX)}_{i}) = \frac{1}{2}(X^2 + P^2) - \frac{1}{2} \implies H = \frac{\hbar\omega}{2}(a^\dagger a + \frac{1}{2})$$

Let us consider the *complete set of commuting observables* (CSCO) for this problem. Evidently, the set $\{H\}$ itself satisfies the condition since the eigenvalues are all non-degenerate (hence, we can label each state with only one index). To understand why, let us consider the action of the the annihilation operator on a (normalised) state $|\psi\rangle$. For that we note the following:

$$\left[a^\dagger a, a\right] = -a \implies [H, a] = -\hbar\omega\ a$$
$$\left[a^\dagger a, a\right] = a^\dagger \implies \left[H, a^\dagger\right] = \hbar\omega\ a^\dagger$$

Now, we have:

$$Ha\,|\psi\rangle - aH\,|\psi\rangle = [H, a]\,|\psi\rangle = -\hbar\omega a\,|\psi\rangle \implies H(a\,|\psi\rangle) = (E - \hbar\omega)(a\,|\psi\rangle)$$

Thus, if $|\psi\rangle$ has an energy eigenvalue E, then $a\,|\psi\rangle$ will have an energy eigenvalue $E - \hbar\omega$. Thus, starting from any eenrgy state, we can change to another state with energy reduced by one unit of $\hbar\omega$, using the annihilation operator. Similarly, we will have:

$$H(a^\dagger\,|\psi\rangle) = (E + \hbar\omega)(a^\dagger\,|\psi\rangle)$$

Let us denote the states $a^\dagger\,|\psi\rangle$ and $a\,|\psi\rangle$ by $|\psi_+\rangle$ and $|\psi_-\rangle$ respectively. Then, we will have

$$\langle\psi_-|\psi_-\rangle = \langle\psi|\,a^\dagger a\,|\psi\rangle = \langle\psi|\left(\frac{H}{\hbar\omega} - \frac{1}{2}\right)|\psi\rangle$$

Now, since $|\psi_-\rangle$ is a valid vector of the Hilbert space, its norm must be non-negative and finite. Hence, we obtain the condition:

$$0 \leq \frac{E}{\hbar\omega} - \frac{1}{2} < \infty \implies \frac{\hbar\omega}{2} \leq E$$

Hence, we get a lower bound on the energy eigenvalue, that is, there exists a state $|\psi_{\text{min}}\rangle$ such that $H\,|\psi_{\text{min}}\rangle = E_{min}\,|\psi_{\text{min}}\rangle$ where $E_{min} = \frac{\hbar\omega}{2}$. Then, starting from this state, if we apply the creation operator, we will get successively increasing energies (and hence the system is non-degenerate). We thus obtain:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad n = 0, 1, \ldots$$

Lecture 02: Dimensional Deep Dive

For the quantum harmonic oscillator, we had seen that how making things dimensionless eases calculations a bit. We consider three fundamental constants, each occupying their own special place in their field of action.

$$c \equiv [LT^{-1}] \qquad : \text{relativity}$$
$$\hbar \equiv [L^2MT^{-1}] \qquad : \text{quantum}$$
$$G \equiv [L^3M^{-1}T^{-2}] \qquad : \text{gravity}$$

Lengths are tractable for us, since we can see the ‘length’, same goes for mass, atleast we can ‘feel’ it. However, time is an *enigma*. *We can't hold two ends of time at the same time* unlike holding two ends of a rod to measure its length. The absolute truth is: *Time passes!*

Note that, in physics, we are mostly concerned with equations like $E = mc^2$ and $E = \hbar\omega$. To that extend, we define the natural units:

$$c = 1$$
$$\hbar = 1$$

We want these quantitites to be numerically equal to 1 and dimensionless. Note that, we had also done this kind of things before. When writing Newton's law, we had said that

$$F \propto m, F \propto a \implies F \propto ma \implies F = kma \qquad \text{for some } k$$

Now, we chose unit and dimension of force in a way such that $k = 1$ (dimensionless and unit value) which gave us the celebrated law.

Note that:

- Making c dimensionless:
$$[LT^{-1}] \equiv 0 \implies [L] = [T]$$
- Making $\hbar$ dimensionless:
$$[L^2MT^{-1}] = [L^2ML^{-1}] = [LM] \equiv 0 \implies [L] = [M^{-1}]$$

Hence, we see that length and time are equivalent while mass and length have inverse relation. In this natural units, we have that energy is equivalent to mass and any other unit can be represented in terms of mass. Thus, by our convention, we choose mass or energy as the only important dimension.

Note that
$$\hbar c \equiv Jm = 1 \qquad \text{(in natural units)}$$

From this, we can say heuristically that increasing length scale is decreases the energy (mass) scale. We mainly use the length scale in context of *de Broglie wavelength*.

This is perhaps not the only way to define natural units. In cosmology, G plays a more important role and hence it is better to set $G = 1$, leaving aside $\hbar$. Using this *natural units*, we find:

$$[L] = [M]$$
$$[L] = [T]$$

Here, we see that length and mass scale are directly related. Well, in this regard, we treat the length scale to be that of the Schwarzschild radius of a blackhole [1] which intuitively grows with mass. Since the above two natural units are widely distinct, there is as such no problem, however, in the unfortunate case where we have to consider both de Broglie wavelength and the Schwarzschild radius (that is, in the infamous domain of quantum gravity 😨), one needs to be very careful.

Theorem 1 ($\pi-$Theorem):
Let $q_1, \ldots, q_n$ be n variables which are physically relevant to a problem and which are related by an expression, that is,

$$F(q_1, \ldots, q_n) = 0 \iff q_i = \widetilde{F}(q_1, \ldots \hat{q}_i, \ldots q_n)$$

If k is the number of fundamental dimensions required to describe the n variables, then we can group these in $(n-k)$ groups of dimensionless variables $\Pi_1, \ldots, \Pi_{n-k}$ such that for some f, we have:

$$f(\Pi_1, \ldots, \Pi_{n-k}) = 0 \iff \Pi_i = \widetilde{f}(\Pi_1, \ldots \hat{\Pi}_i, \ldots \Pi_{n-k})$$

.

[1] This crap is the radius of an object such that if the body is squeezed to a radius lesser than the Schwarzschild radius, the gravitational attraction between the constituents of the body causes its irreversible collapse, turning it to a black hole ⚫

The theorem seems a bit vague (and pointless too). Let us take a physical example. Consider a spherical ball in a viscous fluid. The variable in the problem are:

$$\begin{aligned} \text{Drag force:} \quad & q_1 \to \quad F \quad [MLT^{-2}] \\ \text{Sphere diameter:} \quad & q_2 \to \quad d \quad [L] \\ \text{Fluid density:} \quad & q_3 \to \quad \rho \quad [ML^{-3}] \\ \text{Fluid velocity:} \quad & q_4 \to \quad v \quad [LT^{-1}] \\ \text{Fluid viscosity:} \quad & q_5 \to \quad \eta \quad [ML^{-1}T^{-1}] \end{aligned}$$

So, there are 5 such parameters and only three units viz. M, L, T are needed to describe them. Hence we will have two Π groups. It is a good thing to choose the repeating variables (variables which will be in both groups) that relate to mass, geometry and the kinematics of the problem. Also, note that since the Π groups are dimensionless, we can take the non-repeating variable's power to be 1. Hence, in this problem we choose them to be ρ, d, v. Thus, we will have:

$$\begin{aligned} \Pi_1 &= \rho^{a_1} d^{a_2} v^{a_3} F {\equiv} [ML^{-3}]^{a_1}[L]^{a_2}[LT^{-1}]^{a_3}[MLT^{-2}] = [M^{a_1+1}L^{-3a_1+a_2+a_3+1}T^{-a_3-2}] \\ \Pi_2 &= \rho^{b_1} d^{b_2} v^{b_3} \eta \equiv [ML^{-3}]^{b_1}[L]^{b_2}[LT^{-1}]^{b_3}[ML^{-1}T^{-1}] = [M^{b_1+1}L^{-3b_1+b_2+b_3-1}T^{-b_3-1}] \end{aligned}$$

Hence we obtain two sets of equations:

$$\begin{aligned} a_1 + 1 = 0 &\implies a_1 = -1 \\ -a_3 - 2 = 0 &\implies a_3 = -2 \\ -3a_1 + a_2 + a_3 + 1 = 0 &\implies 3 + a_2 - 2 + 1 = 0 \implies a_2 = -2 \\ b_1 &= -1 \\ b_3 &= -1 \\ -3b_1 + b_2 + b_3 - 1 = 0 &\implies 3 + b_2 - 2 = 0 \implies b_2 = -1 \end{aligned}$$

Then we obtain the Π groups as:

$$\Pi_1 = \frac{F}{\rho d^2 v^2} \qquad \Pi_2 = \frac{\eta}{\rho d v} = \frac{1}{\frac{\rho d v}{\eta}}$$

We identity $\frac{\rho d v}{\eta}$ to be the *Reynold's numer* $\mathcal{R}$. Then we can say, for some ϕ,

$$\frac{F}{\rho d^2 v^2} = \phi(\mathcal{R})$$

To some extraterrestrial being, the physical laws will (hopefully) be valid, however, they might not understand the human-made units (like metres and seconds) in which we measure these quantitites.

We need something natural, based on Nature and hence we used the natural units. For this purpose, we will use things like $c, \hbar, G, k_b \ldots$ and we set all of them to 1. Doing this will lead to change in all scales. For that, we define the Planck units, made of fundamental constants of Nature:

- **Planck mass**: $E_p = \sqrt{\frac{\hbar c}{G}}$
- **Planck length**: $l_p = \sqrt{\frac{\hbar G}{c^3}}$
- **Planck time**: $t_p = \sqrt{\frac{\hbar c}{G^5}}$

Let us now analyse the physical regimes in which the fundamental constants become important. For that, we will use a tuple $(G, \frac{1}{c}, \hbar)$ (note that all the elements of the tuple are written in terms of very small quantitites of the SI scale).

$$
\begin{aligned}
(0,0,0) &\to \text{Classical mechanics} \\
(G,0,0) &\to \text{Newtonian gravity} \\
\left(0,\tfrac{1}{c},0\right) &\to \text{Special relativity} \\
(0,0,\hbar) &\to \text{Basic quantum mechanics} \\
\left(G,\tfrac{1}{c},0\right) &\to \text{General relativity} \\
\left(0,\tfrac{1}{c},\hbar\right) &\to \text{QFT and relativistic QM} \\
(G,0,\hbar) &\to \text{Non-relativistic gravity} \\
\left(G,\tfrac{1}{c},\hbar\right) &\to \text{Quantum gravity}
\end{aligned}
$$

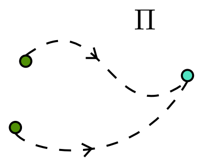
Lecture 03: From Classical to Quantum

Inspecting the transition from a classical to quantum description is necessary for understanding QFT, which is just a *sophisticated mechanics.*

We know that the position $\mathbf{q} \equiv \{q_1, \ldots, q_n\}$ and momentum $\mathbf{p} \equiv \{p_1, \ldots, p_n\}$ uniquely define the state of a *classical* particle.

$$(\mathbf{q}, \mathbf{p}) \in \Pi \ (2n \text{ dimensional phase space}) \qquad q_i, p_i \in \mathbb{R}$$

The Hamiltonian $H(\mathbf{q}, \mathbf{p})$ is just a way of performing time evolution on the system. Using the Hamiltonian, we can 'fibrate' the phase space, that is, starting from one point, we can go to some other point. Also, note that, changing the parameters in the Hamiltonian (eg. ω in SHO Hamiltonian), we change the fibration patterns(eg. *Lissajou's* figures).



In the quantum world, the phase space changes to the Hilbert space $\mathcal{H}$ while a point in the phase space (q, p) changes to a vector $|\psi\rangle$ in the Hilbert space.

Also, these real variables become Hermitian (self-adjoint) operators [1]and the classical Poisson bracket now transforms to the commutator.

$$
\begin{aligned}
\text{CM:} \qquad &\{q, p\} = 1 \\
\text{QM:} \qquad &[\hat{q}, \hat{p}] = i
\end{aligned}
$$

3.1. Time Evolution

From classical Hamilton's equations of motion, we have:

$$
\begin{aligned}
\dot{q}_i &= \{q_i, H\} = \frac{\partial H}{\partial p_i} \\
\dot{p}_i &= \{p_i, H\} = -\frac{\partial H}{\partial q_i}
\end{aligned}
$$

[1] A self-adjoint operator $\mathcal{O}$ is such that $\mathcal{O} = \mathcal{O}^\dagger$ in all respect, that is, $\mathcal{O}$ and $\mathcal{O}^\dagger$ have the same domain and action. In general, $\mathbb{D}(\mathcal{O}) \subseteq \mathbb{D}(\mathcal{O}^\dagger)$ where $\mathbb{D}(\cdot)$ represents the domain of some operator. If the domains are not equal but action is same on a restricted domain (mainly occurs in infinite-dimensional spaces), then those operators are not self-adjoint but called *symmetric/Hermitian* (in some places Hermiticity also requires a symmetric operator to be bounded).

If $\mathbf{z} = \begin{pmatrix} \mathbf{p} \\ \mathbf{q} \end{pmatrix}$ is a $2n$ dimensional vector, then $\dot{\mathbf{z}} = \{\mathbf{z}, H\}$. Now, moving to quantum mechanics, we know the celebrated Schrödinger equation, which specifies the time evolution of a state vector $|\psi(t)\rangle \in \mathcal{H}$:

$$i\frac{\partial |\psi(t)\rangle}{\partial t} = H |\psi(t)\rangle$$

Thus, in both classical and quantum case, the time derivative of a quantity is equal to some action of the Hamiltonian on that quantity (classical $\rightarrow$ Poisson bracket, quantum $\rightarrow$ Multiplication with Hamiltonian)[1].

The Hamiltonian in the position basis becomes:

$$H = \frac{-\hbar^2}{2m}\nabla^2 + V(\mathbf{x})$$

Thus, the Schrödinger equation now acts on the wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x}|\psi\rangle$:

$$i\frac{\partial \psi(\mathbf{x}, t)}{\partial t} = -\frac{1}{2m}\nabla^2\psi(\mathbf{x}, t) + V(\mathbf{x})\psi(x, t)$$

And the energy and momentum operators become $E \rightarrow i\frac{\partial}{\partial t}$ and $p \rightarrow -i\nabla$.

3.1.1. Free Particle

The dispersion relation for a non-relativistic free particle is:

$$E = \frac{\mathbf{p}^2}{2m}$$

The typical solution can be written as plane-waves, $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} = e^{i(\mathbf{p}\cdot\mathbf{x}-Et)} \equiv \exp(-ip^\mu x_\mu)$ where we have used the repeated summation notation and taken the metric $\eta_{\mu\nu} = (+, -, -, -)$.

For a relativistic free particle, the dispersion relation becomes:

$$E^2 = m^2 + p^2$$

Instead of the wavefunction $\psi(\mathbf{x}, t)$, we will use $\phi(\mathbf{x}, t)$, since the resulting equation actually acts on a scalar-field and ϕ is common notation for a scalar-field. Substituting the energy and momentum operators here, we get:

$$\begin{aligned} i^2\frac{\partial^2\phi(\mathbf{x}, t)}{\partial t^2} &= (-i)^2\nabla^2\phi(\mathbf{x}, t) + m^2\phi(\mathbf{x}, t) \\ \implies \frac{\partial^2\phi(\mathbf{x}, t)}{\partial t^2} - \nabla^2\phi(\mathbf{x}, t) &= -m^2\phi(\mathbf{x}, t) \\ \implies \left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2\right)\phi(\mathbf{x}, t) = 0 \end{aligned}$$

The above equation is called the *Klein-Gordon equation.* In covariant notation, we can compactly write it as:

$$\left(\partial_\mu\partial^\mu + m^2\right)\phi(x^\nu) = 0$$

The KG equation is Lorentz invariant, that is, under a Lorentz transformation, $x^\mu \rightarrow x'^\nu$, $\phi'(x'^\nu) = \phi(x^\mu)$

[1] A better analogy would be to use density matrices which gives the von Neumann equation which uses commutator bracket

3.1.2. Continuity Equation

Taking the complex cojungate of the Schrödinger's equation and then after some algebraic manipulation, we obtain:

$$\frac{\partial(\psi^*\psi)}{\partial t} = \frac{i}{2m}\boldsymbol{\nabla}\cdot[\psi^*(\boldsymbol{\nabla}\psi) - \psi(\boldsymbol{\nabla}\psi^*)]$$

Here, we identify:

$$\rho := \psi^*\psi \qquad\qquad \mathbf{J} = \frac{-i}{2m}[\psi^*(\boldsymbol{\nabla}\psi) - \psi(\boldsymbol{\nabla}\psi^*)]$$

which yields the well-known form of the *continuity equation*:

$$\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot\mathbf{J} = 0 \qquad \longrightarrow \qquad \partial_\mu J^\mu = 0$$

where $J^\mu = (\rho, \mathbf{J})$ is the four-current. Note that in this case, ρ is a positive-definite quantity and can indeed have the interpretation of probability. Also, note that if ψ somehow becomes real-valued, then $\mathbf{J} = 0$ which implies that ρ is constant in time (though it can change in space).

Doing the same thing to Klein-Gordon equation yields:

$$\frac{\partial}{\partial t}\left(\phi^*\frac{\partial\phi}{\partial t} - \phi\frac{\partial\phi^*}{\partial t}\right) - \boldsymbol{\nabla}\cdot(\phi^*\boldsymbol{\nabla}\phi - \phi\boldsymbol{\nabla}\phi^*) = 0$$

From this equation, we can identify:

$$\rho := \left(\phi^*\frac{\partial\phi}{\partial t} - \phi\frac{\partial\phi^*}{\partial t}\right) \qquad\qquad \mathbf{J} := -(\phi^*\boldsymbol{\nabla}\phi - \phi\boldsymbol{\nabla}\phi^*)$$

Thus here, the 4-current becomes $J^\mu = (\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*)$ Note the apparent problems with this identification:

- It is not apriori obvious that ρ is positive-definite and hence has problem with probability interpretation.
- The equation is second-order in time and hence ρ seems to have a term evolving forward and one term evolving backward in time.
- The dispersion relation does not have a single solution; the solutions are $\pm E$

The thing is, KG equation treats time and space on equal footing (unlike Schrödinger equation where time was in first order and space was in second order). Hence we have to consider all possibilities of moving back and forth in both space and time.

Lecture 04: Classical Fields

Let us start with a familiar setting, the spring-mass system in one-dimensions. So, each mass is connected with isotropic springs of natural length a and spring constant k, and at some instance, i^{th} spring is displaced from its equilibrium position by an amount η_i as shown below:

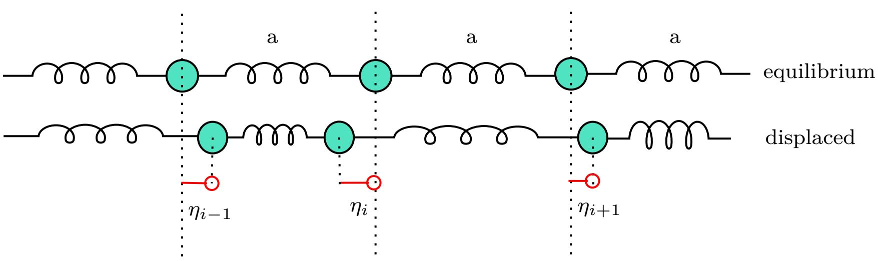


Figure 1: A spring-mass system; each mass is displaced by some amount from its equilibrium position

Then the kinetic and potential energies can be written as:

$$T = \frac{1}{2}\sum_i \dot{\eta}_i^2 \qquad V = \frac{1}{2}\sum_i k(\eta_{i+} - \eta_i)^2$$

From this, the Langrangian of the system is obtained as:

$$L = T - V = \frac{1}{2}\sum_i a\left[({}^m/{}_a)\dot{\eta}_i^2 - ka\left(\frac{\eta_{i+1} - \eta_i}{a}\right)^2\right] = a\sum_i L_i$$

The resulting ELEOM are:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\eta}_i}\right) = \frac{\partial L}{\partial \eta} \implies ({}^m/{}_a)\ddot{\eta}_i - ka\left(\frac{\eta_{i+1} - \eta_i}{a^2}\right) + ka\left(\frac{\eta_i - \eta_{i-1}}{a^2}\right) = 0$$

In the *continuum* limit, that is, $a \to 0$ (so, the system kind of becomes like a rod), we have:

$$\frac{m}{a} \longrightarrow \mu \qquad : \text{mass per unit length of the rod}$$

$$ka \longrightarrow Y \qquad : \text{Young's modulus of the rod}$$

$$i \longrightarrow x \qquad : \text{position label}$$

$$\eta_i \longrightarrow \eta(x) \qquad : \text{field}$$

$$\frac{\eta_{i+1} - \eta_i}{a} \longrightarrow \frac{d\eta}{dx} \qquad : \text{derivative}$$

$$\sum_i \longrightarrow \int dx \qquad : \text{integral}$$

Hence in the continuum limit, the ELEOM becomes:

$$\lim_{a\to 0}\left(\mu\frac{\mathrm{d}^2\eta}{\mathrm{d}t^2} - \frac{Y}{a}\frac{\mathrm{d}\eta}{\mathrm{d}x}\bigg|_x + \frac{Y}{a}\frac{\mathrm{d}\eta}{\mathrm{d}x}\bigg|_{x-a}\right) = 0 \implies \mu\frac{\mathrm{d}^2\eta}{\mathrm{d}t^2} - Y\frac{\mathrm{d}^2\eta}{\mathrm{d}x^2} = 0$$

Defining the velocity as $v := \sqrt{\frac{Y}{\mu}}$, we obtain the wave-equation:

$$\frac{1}{v^2}\frac{\mathrm{d}^2\eta}{\mathrm{d}t^2} - \frac{\mathrm{d}^2\eta}{\mathrm{d}x^2} = 0$$

Note that, in this case, position becomes a mere label like in the discrete case, i just labelled the index of the i^{th} mass. Also, technically, η can also depend on time, so it is better to write it as $\eta(x,t)$.

Position and time, both are kind of labels and thus are independent. Hence, total and partial derivatives coincide in this case. Note that in continuum, the Langrangian itself becomes an integral of the form:

$$L = \frac{1}{2}\int \left[\mu\dot{\eta}^2 - Y\left(\frac{\mathrm{d}\eta}{\mathrm{d}x}\right)^2\right] dx$$

The bracketted term is the Lagrangian density, denoted by $\mathcal{L}$ and hence, the Lagrangian is the space integral of the Langrangian density:

$$L = \int \mathcal{L}\ dx$$

Henceforth, we allow ourselves to be *sloppy* enough and refer to Lagrangian density as simply the Langrangian when context is clear.

4.1. Our actions tell a lot!

Let us now consider a general setting. So, the Lagrangian density can be a function of the field, the spatial and temporal derivatives and space and time themselves. In compact notation, we write:

$$\mathcal{L} \equiv \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

Remember, that we define the classical action S as:

$$S = \int_{t_1}^{t_2} dt\ L = \int_{t_1}^{t_2} dt \int d^3\mathbf{x}\ \mathcal{L}(\phi, \partial_\mu\phi, x^\mu) = \int_{\mathcal{R}\subset\mathbb{R}^4} d^4\mathbf{x}\ \mathcal{L}(\phi, \partial_\mu\phi, x^\mu)$$

The integration can indeed be over a region $\mathcal{R}$ which is a subset of $\mathbb{R}^4$ in general. Mostly, we take $\mathcal{R} = [t_1, t_2] \times \mathbb{R}^3$.

We now want to use Hamilton's principle, which states that, under fixed boundary conditions, the system takes a path in configuration space for which the action is stationary, that is,

$$\delta S = 0$$

Note that the action is a functional which takes a field ϕ as input and returns a real number[1]. Let us consider a variation in the field itself: $\phi \longrightarrow\ \phi' = \phi + \delta\phi$ such that $\delta\phi$ vanishes at the boundary of the integration domain. Then we have the following:

$$\delta S = S[\phi'] - S[\phi] = \int d^4\mathbf{x}\ \big(\mathcal{L}(\phi', \partial_\mu\phi', x^\mu) - \mathcal{L}(\phi, \partial_\mu\phi, x^\mu)\big)$$

Now, we try to find what the above quantity is:

$$\begin{aligned}\mathcal{L}(\phi', \partial_\mu\phi', x^\mu) &= \mathcal{L}(\phi + \delta\phi, \partial_\mu(\phi+\delta\phi), x^\mu)\\ &= \mathcal{L}(\phi, \partial_\mu\phi, x^\mu) + \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi)\end{aligned}$$

Also note that $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$. Hence, from the above expression, we obtain that:

$$\delta S = \int d^4\mathbf{x}\ \left(\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi)\right)$$

Using *integration by parts* on the second term, we obtain:

$$\int d^4\mathbf{x}\ \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) = -\int d^4\mathbf{x}\ \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta\phi + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right)\Bigg|_{\text{boundary}}$$

The variation in the field vanishes at the boundary by our assumption. Then we obtain, for the variation of the action:

$$\delta S = \int d^4\mathbf{x}\ \left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\right)\delta\phi = 0$$

Since the variation in the field was arbitrary, we obtain the Euler-Lagrange's equations for the field as:

$$\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) = 0$$

In general, we want the action to be *Lorentz* invariant, in order to maintain the universality of physical laws. Since the volume element of Minkowski space $d^4\mathbf{x}$ is invariant under Lorentz transformations, this imply that $\mathcal{L}$ should also be constructed of Lorentz-invariant quantities.

[1]For functionals, a common notation for the input is to use square brackets, that is, if A is a linear functional taking function f as input, $A[f] \in \mathbb{R}$

The field ϕ (and all its powers) is a Lorentz *scalar* and is hence invariant. The derivative $\partial_\mu\phi$ is not invariant, however, $\partial^\mu\partial_\mu\phi$ is. Bilinears like $(\partial^\mu\phi)(\partial_\mu\phi)$ are also invariant. Hence, a simple Lagrangian should be constructed of these quantities. Let us consider the Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\eta^{\mu\nu}(\partial_\nu\phi)(\partial_\mu\phi) - \frac{1}{2}m^2\phi^2$$

Then, from the Lagrange's equations,

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial\phi} &= -m^2\phi \\
\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} &= \frac{1}{2}\eta^{\mu\nu}(\partial_\nu\phi + \partial_\mu\phi\delta^\mu{}_\nu) \\
&= \frac{1}{2}\eta^{\mu\nu}(2\times\partial_\nu\phi) \\
&= \eta^{\mu\nu}\partial_\nu\phi \\
&= \partial^\mu\phi
\end{aligned}$$

Then we finally obtain:

$$-m^2\phi - \partial_\mu\partial^\mu\phi = 0 \implies (\partial_\mu\partial^\mu + m^2)\phi = 0$$

This is the KG equation that we had seen earlier. Thus, the above Lagrangian models the KG equation.

When we have multiple fields $\{\phi_i\}$, then for each field, we will obtain:

$$\frac{\partial\mathcal{L}}{\partial\phi_i} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right) = 0$$

NOTE: In terms of the total Lagrangian, the ELEOM for the field is:

$$\frac{\delta L}{\delta\phi} - \partial_t\left(\frac{\delta L}{\delta\dot{\phi}}\right) = 0$$

Lecture 05 & 06: Looking into Hamiltonians

Previously, we saw a ton of crap about the Lagrangians and how the equations of motions are obtained from the action being stationary. Now, we will look into the *Hamiltonian* formalism which will allow us to quantise the field.

In basic classical mechanics, we had seen that given a Lagrangian $L(q_i, \dot{q}_i, t)$, we obtained the Hamiltonian by the Legendre transformation as:

$$H(q_i, p_i, t) = p_i\dot{q}_i - L(q_i, \dot{q}_i, t)$$

where $p_i := \frac{\partial L}{\partial\dot{q}_i}$ was defined to be the generalised momenta, conjugate to the generalised coordinate q_i. Similarly, for classical fields, we can define:

$$\pi^i(x^\mu) := \frac{\partial\mathcal{L}}{\partial(\partial_0\phi_i)} = \frac{\partial\mathcal{L}}{\partial\dot{\phi}_i}$$

This is the *canonical momentum* corresponding to the field. From the ELEOM, we obtain $\dot{\pi}^i = \frac{\delta L}{\delta\phi}$ Using this, the *Hamiltonian density* can be defined as:

$$\mathcal{H}(\phi^a, \pi_a, \partial_i\phi_a, x^\mu) := \pi^a\dot{\phi}_a - \mathcal{L}(\phi_a, \partial_\mu\phi_a, x^\mu)$$

In the final expression on the LHS, we have to replace $\dot{\phi}_a$ in terms of π^a to get the Hamiltonian density (Replacing this will get rid of all the time derivative and hence, in the Hamiltonian, we used ∂_i, the spatial derivatives only). Using this then, we get the total Hamiltonian as:

$$H[\pi^a, \phi_a] = \int d^3\mathbf{x}\ \mathcal{H}(\phi_a, \pi_a, \partial_i\phi_a, x^\mu)$$

6.1. Functional Derivatives

Since action, Lagrangian, etc. are functionals, an apt way to describe their variation is through the *functional derivative.* These come under the domain of functional analysis (which is Math ☠️), however, a short introduction might be necessary.

Let us take a functional $\mathcal{F}[f]$ and consider its variation $\delta\mathcal{F}$ when $f \to f + \delta f$, where $\delta f = \epsilon\eta$. Here ϵ is a small parameter and η is a generic function, often called the *test function.* Now,

$$\mathcal{F}[f+\epsilon\eta] = \mathcal{F}[f] + \left.\frac{\mathrm{d}\mathcal{F}[f+\epsilon\eta]}{\mathrm{d}\epsilon}\right|_{\epsilon=0} \epsilon + \mathcal{O}(\epsilon^2)$$

We define the functional derivative as the first-order coefficient of the expansion, that is,

$$\frac{\delta\mathcal{F}}{\delta f} = \lim_{\epsilon\to 0}\frac{\mathcal{F}[f+\epsilon\eta]-\mathcal{F}[f]}{\epsilon}$$

Functional derivatives seem to follow similar rules as in ordinary differential calculus (viz. linearity, Liebnitz rule, chain rule) and hence, can be treated to be operationally same as the ordinary derivative. However, exceptions are always there.

The differential (variation) of a functional is given as:

$$\delta\mathcal{F}[\phi] := \int \frac{\delta\mathcal{F}[\phi]}{\delta\phi}\delta\phi$$

Now let us try to relate functional derivative with normal derivative. For that, let us suppose that the space is discretised into cells, each of volume ΔV^i and each cell is assigned the *average value* of ϕ within the volume, that is,

$$\phi_i(t) = \frac{1}{\Delta V^i}\int\limits_{\Delta V^i} d^3\mathbf{x}\ \phi(\mathbf{x},t)$$

Then, L now depends on the discrete components, ϕ_i and $\dot{\phi}_i$ and the variation can be written as in terms of partial derivatives (since ϕ_i are now discrete):

$$\delta L = \sum_i \left(\frac{\partial L}{\partial\phi_i}\delta\phi_i + \frac{\partial L}{\partial\dot{\phi}_i}\delta\dot{\phi}_i\right) = \sum_i \frac{1}{\Delta V^i}\left(\frac{\partial L}{\partial\phi_i}\delta\phi_i + \frac{\partial L}{\partial\dot{\phi}_i}\delta\dot{\phi}_i\right)\Delta V^i$$

Now, consider the Lagrangian variation using the definition of functional derivative:

$$\delta L = \int d^3\mathbf{x}\left(\frac{\delta L}{\delta\phi}\delta\phi + \frac{\delta L}{\delta\dot{\phi}}\delta\dot{\phi}\right)$$

From the two above equations, we can make the following identifications:

$$\frac{\delta L}{\delta\phi} = \lim_{\Delta V^i\to 0}\frac{1}{\Delta V^i}\frac{\partial L}{\partial\phi_i}$$
$$\frac{\delta L}{\delta\dot{\phi}} = \lim_{\Delta V^i\to 0}\frac{1}{\Delta V^i}\frac{\partial L}{\partial\dot{\phi}_i}$$

In the LHS, while taking functional derivative, $\mathbf{x}$ belongs to the i^{th} cell. Thus, we see that the functional derivative is proportional to the partial derivative. Now, given a Lagrangian, we can write the variation under $\phi \to \phi + \delta\phi$ using the Lagrangian density:

$$\delta L = \int d^3\mathbf{x}\ \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_i\phi)}\delta(\partial_i\phi) + \frac{\partial\mathcal{L}}{\partial\dot{\phi}}\delta\dot{\phi}$$

Integration by parts and ignoring boundaries on the second term, we obtain:

$$\delta L = \int d^3\mathbf{x}\ \left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_i\left(\frac{\partial\mathcal{L}}{\partial(\partial_i\phi)}\right)\right)\delta\phi + \frac{\partial\mathcal{L}}{\partial\dot{\phi}}\delta\dot{\phi}$$

Comparing this with the definition of functional derivative, we obtain:

$$\frac{\delta L}{\delta\phi} = \frac{\partial\mathcal{L}}{\partial\phi} - \partial_i\left(\frac{\partial\mathcal{L}}{\partial(\partial_i\phi)}\right) \qquad \frac{\delta L}{\delta\dot{\phi}} = \frac{\partial\mathcal{L}}{\partial\dot{\phi}}$$

6.2. Hamilton's Equations of Motion

To obtain HEOM, we start with the variation of the total Hamiltonian,

$$\delta H = \int d^3\mathbf{x}\; \delta\big(\pi^\alpha \dot{\phi}_\alpha - \mathcal{L}\big) = \int d^3\mathbf{x}\; \big(\pi^\alpha \delta\dot{\phi}_\alpha + \dot{\phi}_\alpha \delta\pi^\alpha\big) - \delta L$$

Now, from ELEOM:

$$\delta L = \int \frac{\delta L}{\delta \phi_\alpha}\delta\phi_\alpha + \frac{\delta L}{\delta\dot{\phi}_\alpha}\delta\dot{\phi}_\alpha = \int \partial_t\Big(\frac{\delta L}{\delta\dot{\phi}_\alpha}\Big)\delta\phi_\alpha + \pi^\alpha\delta\dot{\phi}_\alpha = \int \dot{\pi}^\alpha\delta\phi_\alpha + \pi^\alpha\delta\dot{\phi}_\alpha$$

Using this, the variation in Hamiltonian becomes:

$$\delta H = \int d^3\mathbf{x}\; (\dot{\phi}_\alpha\delta\pi^\alpha - \dot{\pi}^\alpha\delta\phi_\alpha)$$

Thus, from above, we can identity:

$$\frac{\delta H}{\delta\phi_\alpha} = -\dot{\pi}^\alpha \qquad\qquad \frac{\delta H}{\delta\pi^\alpha} = \dot{\phi}_\alpha$$

Also, previously we saw the relation between functional derivative of the Lagrangian and partial derivative of the Lagrangian density. That could well be extended to the Hamiltonian and we obtain:

$$\dot{\pi}^\alpha = -\frac{\delta H}{\delta\phi_\alpha} = -\frac{\partial\mathcal{H}}{\partial\phi_\alpha} + \partial_i\Big(\frac{\partial\mathcal{H}}{\partial(\partial_i\phi_\alpha)}\Big) \qquad\qquad \dot{\phi}_\alpha = \frac{\delta H}{\delta\pi^\alpha} = \frac{\partial\mathcal{H}}{\partial\pi^\alpha} - \partial_i\Big(\frac{\partial\mathcal{H}}{\partial(\partial_i\pi^\alpha)}\Big)$$

In general, the Hamiltonian density does not depend on $\partial_i\pi^\alpha$, so we have $\dot{\phi}_\alpha = \frac{\partial\mathcal{H}}{\partial\pi^\alpha}$

6.3. Poisson Bracket

In the Hamiltonian formulation of classical mechanics, we defined the Poisson bracket of two quantities A, B by:

$$\{A, B\} = \frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i}$$

Similarly, for fields also, given two functionals $\mathcal{F}[\phi,\pi]$ and $\mathcal{G}[\phi,\pi]$, we define the Poisson bracket by:

$$\{\mathcal{F},\mathcal{G}\} := \int d^3\mathbf{x}\; \frac{\partial\mathcal{F}}{\partial\phi_\alpha}\frac{\partial\mathcal{G}}{\partial\pi^\alpha} - \frac{\partial\mathcal{F}}{\partial\pi^\alpha}\frac{\partial\mathcal{G}}{\partial\phi_\alpha}$$

Given a function $\mathcal{F}[\phi^\alpha,\pi^\alpha,t]$, note:

$$\begin{aligned}
\dot{\mathcal{F}} &= \partial_t\mathcal{F} + \int d^3\mathbf{x}\;\Big(\frac{\delta\mathcal{F}}{\delta\phi_\alpha}\frac{\partial\phi_\alpha}{\partial t} + \frac{\delta\mathcal{F}}{\delta\pi_\alpha}\frac{\partial\pi_\alpha}{\partial t}\Big)\\
&= \partial_t\mathcal{F} + \int d^3\mathbf{x}\;\Big(\frac{\delta\mathcal{F}}{\delta\phi_\alpha}\frac{\delta H}{\delta\pi^\alpha} - \frac{\delta\mathcal{F}}{\delta\pi_\alpha}\frac{\delta H}{\delta\phi_\alpha}\Big) \qquad \text{(from HEOM)}\\
&= \partial_t\mathcal{F} + \{\mathcal{F}, H\}
\end{aligned}$$

This specifies the *time-evolution* of the functionals using the Poisson bracket. Now, let us consider the Poisson bracket of the fields ϕ_α and π^α. However, since the Poisson bracket was defined for functionals only, there is some problem. This can be resolved by noting that any function can be written as a functional depending on itself as:

$$\phi_\alpha(\mathbf{x},t) = \int d^3\,\mathbf{x}'\phi_\alpha(\mathbf{x}',t)\delta^3(\mathbf{x}-\mathbf{x}')$$

From the definition of the functional derivative, it is easy to note that:

$$\frac{\delta\phi_\alpha(\mathbf{x},t)}{\delta\phi_\beta(\mathbf{x}',t)} = \delta_{\alpha\beta}\delta^3(\mathbf{x}-\mathbf{x}')$$

Also, ϕ and π are *independent functionals* and hence,

$$\frac{\delta \pi(\mathbf{x},t)}{\delta \phi(\mathbf{x}',t)} = 0 \qquad \frac{\delta \phi(\mathbf{x},t)}{\delta \pi(\mathbf{x}',t)} = 0 \ \forall \ \mathbf{x},\mathbf{x}',t$$

Now, we can compute the Poisson bracket of the fields themselves and we obtain:

$$\begin{aligned}
\{\phi_\alpha(\mathbf{x},t), \pi^\beta(\mathbf{x}',t)\} &= \int d^3\left(\mathbf{x}'' \frac{\delta \phi_\alpha(\mathbf{x},t)}{\delta \phi_\mu(\mathbf{x}'',t)} \frac{\delta \pi^\beta(\mathbf{x}',t)}{\delta \pi^\mu(\mathbf{x}'',t)} - 0\right) \\
&= \int d^3\left(\mathbf{x}'' \delta_{\alpha\mu}\delta_{\beta\mu}\delta^3(\mathbf{x}-\mathbf{x}'')\delta^3(\mathbf{x}'-\mathbf{x}'')\right) \\
&= \delta_{\alpha\beta}\delta^3(\mathbf{x}-\mathbf{x}')
\end{aligned}$$

We also have the other two Poisson brackets which are trivial:

$$\{\phi_\alpha, \phi_\beta\} = 0 \qquad \{\pi^\alpha, \pi^\beta\} = 0$$

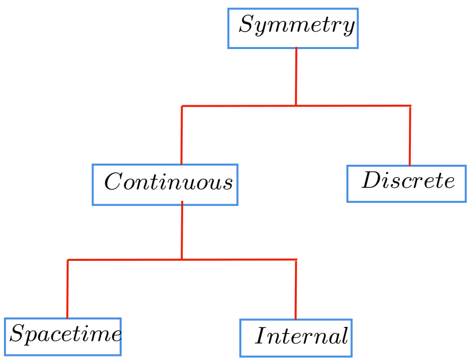
Lecture 07 & 08: Representations of Rotation and Lorentz group

A *group* is a tuple $(G, *)$ which follows certain rules like associativity, existence of identity, inverses, etc. A *Lie Group* [1] is some continuous group with properties so special that it becomes very important (and cool) to study them.

I think a short, vague introduction cannot do justice to these elegant topics which are subjects on their own right and require proper analysis. Hence, I might skip them here and instead try to include them in the link [2] below as and when I read about them!

Lecture 09: Noether's Theorem

To understand Noether's Theorem, let us first understand what symmetry actually is. Symmetry of a system can be thought of as some transformation under which the properties of the system remain invariant.



Discrete symmetries describes non-continuous changes in the system, like *parity* (kind of spatial reflection) and *time-reversal*. *Internal* symmetries include cases where the fields themselves vary while *space-time* symmetry describes the system when the space-time points vary. *Local* symmetries are transformations allowing us to change the system differently at different spacetime points while *global* symmetries act in the same way at every point.

[1] A Lie Group is a group that is also a manifold.
[2] A very basic and nasty introduction can be found here!

Theorem 2 (Noether's Theorem):
For every continuous global symmetry there exists a conserved current J^μ, that is,

$$\partial_\mu J^\mu = 0$$

Proof. Note that, under any symmetry transformation, we required our equations of motion to be invariant and hence require our action $\mathcal{S}$ to be 'invariant' since action is a fundamental quantity to our system from which we derived our EOM. Note that while deriving the equations of motions, we ignored the boundary term of the action. Thus, the action can vary by some arbitrary boundary term. Using this, we define a *Noether symmetry* to be a transformation which keeps the action invariant upto an integral of a total divergence, that is,

$$\delta\mathcal{S} = \int_{\partial\mathcal{R}} d^4\mathbf{x}\ \partial_\mu K^\mu \implies \delta\mathcal{L} \equiv \epsilon\ \partial_\mu K^\mu$$

The implication comes from the fact that divergence should be in first order with respect to some infinitesimal variation ϵ. Let us consider the case of *internal symmetries* first.

Consider a variation in the field $\phi = \phi_0 + \delta\phi$. Then,

$$\delta\mathcal{L} = \left.\frac{\partial\mathcal{L}}{\partial\phi}\right|_{\phi_0}\delta\phi + \left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\phi_0}\delta(\partial_\mu\phi) = \left.\frac{\partial\mathcal{L}}{\partial\phi}\right|_{\phi_0}\delta\phi + \left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\phi_0}\partial_\mu(\delta\phi)$$

Now, suppose there exists a solution $\widetilde{\phi}$ to the equation of motion of the system. Since the above equation is valid for any ϕ_0, it is also valid for $\widetilde{\phi}$ and hence, we get:

$$\begin{aligned}
\delta\mathcal{L} &= \left.\frac{\partial\mathcal{L}}{\partial\phi}\right|_{\widetilde{\phi}}\delta\phi + \left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\widetilde{\phi}}\partial_\mu(\delta\phi) \\
&= \partial_\mu\left.\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\right|_{\widetilde{\phi}}\delta\phi + \left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\widetilde{\phi}}\partial_\mu(\delta\phi) \quad \text{(using ELEOM)} \\
&= \partial_\mu\left(\left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\widetilde{\phi}}\delta\phi\right) \quad \text{(using product rule)}
\end{aligned}$$

Let the variation of the Lagrangian be a global symmetry, denoted by, $\delta_\alpha\phi$. Under this symmetry, we want our action $\mathcal{S}$ to change atmost by a boundary term and hence we have,

$$\partial_\mu\left(\left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\widetilde{\phi}}\delta\phi\right) = \partial_\mu K^\mu \quad\implies\quad \partial_\mu\left(\left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\widetilde{\phi}}\delta\phi - K^\mu\right) = 0$$

Let us define the quantity in the bracket by J^μ which is called the *Noether current*. Thus,

$$J^\mu := \left.\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right|_{\widetilde{\phi}}\delta\phi - K^\mu$$

from where we get $\partial_\mu J^\mu = 0 \implies$ Noether current is conserved.

The above proof worked for internal symmetries where the field itself was varied. Now, let us look into the *spacetime* symmetry, where the spacetime points are varied. For simplicity, we will deal with scalar fields only. Let us consider an infinitesimal coordinate transformation,

$$x'^\mu = x^\mu + \delta x^\mu$$

The variation in the field can be given by:

$$\delta\phi = \phi'(\mathbf{x}') - \phi(\mathbf{x}) = \phi'(x'^\mu) - \phi'(x^\mu) + \phi'(x^\mu) - \phi(x^\mu) = \delta_0\phi(x^\mu) + \phi'(x'^\mu) - \phi'(x^\mu)$$

where we have defined $\delta_0\phi(x^\mu) := \phi'(x^\mu) - \phi(x^\mu)$. We did this to distinguish between variation of the dependent and independent quantitites. Using Taylor expansion, we have:

$$\phi'(x'^\mu) = \phi'(x^\mu + \delta x^\mu) = \phi'(x^\mu) + \frac{\partial\phi'}{\partial x^\mu}\delta x^\mu \implies \phi'(x'^\mu) - \phi'(x^\mu) = \partial_\mu\phi'\delta x^\mu$$

Now, $\partial_\mu\phi'\delta x^\mu \approx \partial_\mu\phi\delta x^\mu$, since if we do the Taylor expansion of $\delta_0\phi(x'^\mu)$, we would obtain a term $\partial_\nu\delta_0$ which is a variation of second order and hence, can be neglected. Thus, we obtain:

$$\phi'(x'^\mu) - \phi'(x^\mu) = \partial_\mu\phi\delta x^\mu \implies \delta\phi = \delta_0\phi(x^\mu) + \partial_\mu\phi\delta x^\mu$$

The action is not in general invariant under both changes in coordinates and fields. Indeed, the volume element (integral measure) $\mathrm{d}^4\mathbf{x}$ transforms as:

$$\mathrm{d}^4\mathbf{x}' = J\ \mathrm{d}^4\mathbf{x}$$

where J is the determinant of the Jacobian matrix. Now, we find some expression for the Jacobian:

$$x'^\mu = x^\mu + \delta x^\mu \implies \partial_\nu x'^\mu = \delta^\mu{}_\nu + \partial_\nu(\delta x^\mu)$$

Now, we use an identity $\det(\mathbb{1} + \epsilon A) = 1 + \epsilon\,\mathrm{Tr}\,A + \mathcal{O}(\epsilon^2)$ on the above equation to get an approximation for the Jacobian as:

$$J = \det\left(\frac{\partial x'^\mu}{\partial x^\nu}\right) = \det(\delta^\mu{}_\nu + \partial_\nu(\delta x^\mu)) \approx (1 + \partial_\nu(\delta x^\mu)) \implies \delta J = \partial_\mu(\delta x^\mu)$$

Now, we consider the variation of the action itself and we see:

$$\begin{aligned}
\delta\mathcal{S} &= \int \delta(\mathrm{d}^4\mathbf{x})\ \mathcal{L} + \int \mathrm{d}^4\mathbf{x}\ \delta\mathcal{L} \\
&= \int \mathrm{d}^4\mathbf{x}\ \partial_\mu\delta x^\mu\mathcal{L} + \int \mathrm{d}^4\mathbf{x}\ \delta\mathcal{L} \\
&= \int \mathrm{d}^4\mathbf{x}\ \left(\partial_\mu\delta x^\mu\mathcal{L} + \frac{\partial\mathcal{L}}{\partial\phi}\delta_0\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0(\partial_\mu\phi) + \frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu\right)
\end{aligned}$$

We had written the variation of the Lagrangian with respect to both dependent and independent variables. Now the first and last term can be written together using product rule and hence we have:

$$\begin{aligned}
\delta\mathcal{S} &= \int \mathrm{d}^4\mathbf{x}\ \left(\partial_\mu(\delta x^\mu\mathcal{L}) + \frac{\partial\mathcal{L}}{\partial\phi}\delta_0\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\delta_0\phi\right) \\
&= \int \mathrm{d}^4\mathbf{x}\ \left(\partial_\mu(\delta x^\mu\mathcal{L}) + \frac{\partial\mathcal{L}}{\partial\phi}\delta_0\phi + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0\phi\right) - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta_0\phi\right) \\
&= \int \mathrm{d}^4\mathbf{x}\ \partial_\mu\left(\delta x^\mu\mathcal{L} + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0\phi\right)\right) + \delta_0\phi\left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\right)
\end{aligned}$$

For our field satisfying the equation of motion, the last term drops and we have our variation of the action and this should be equal to the boundary term, hence we have:

$$\partial_\mu\left(\delta x^\mu\mathcal{L} + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0\phi\right)\right) = \partial_\mu K^\mu$$

In the case for multiple fields, the *Noether current* turns out to be:

$$J^\mu := \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\delta_0\phi_i + \delta x^\mu\mathcal{L} - K^\mu \qquad \partial_\mu J^\mu = 0 \quad \square$$

The above condition is called *current conservation* which we had also seen earlier for internal symmetries. However, the second term, which is purely an artifact of the spacetime symmetry, did not occur there in the conserved current for internal symmetries. Associated with each current, we can define the charge enclosed in a volume V as:

$$Q_V := \int_V \mathrm{d}^3\mathbf{x}\ J^0(\mathbf{x}, t) = \int_V \mathrm{d}^3\mathbf{x}\ \rho$$

Then, we can write the conservation equation as:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} + \boldsymbol{\nabla}\cdot\mathbf{J} = 0 \implies \frac{\mathrm{d}Q}{\mathrm{d}t} = -\int_V \mathrm{d}^3\mathbf{x}\ \boldsymbol{\nabla}\cdot\mathbf{J} = -\int_{\partial V} \mathbf{J}\cdot\mathrm{d}\mathbf{S}$$

The last equality comes from Gauss' law and ∂V is the boundary of the region. If we extend the volume V to cover the entire space and assume that at spatial infinities, current is zero, then we have $\frac{\mathrm{d}Q}{\mathrm{d}t} = 0$ and hence, current is conserved. In most cases, we would have K^μ to be zero (and thus, let us take it to be zero). Let us see some examples for Noether's theorem:

9.1. Examples: Spacetime Symmetries

Let us start with some examples of Noether's theorem where spactime symmetry is explicitly there. We will consider translations and rotations.

9.1.1. Translational Invariance

Classically, we know that translational invariance leads to *energy conservation*. Let us see what happens in case of fields. We thus consider an infinitesimal global translation:

$$x^\mu \to x'^\mu = x^\mu + \epsilon a^\mu$$

Then $\delta x^\mu = \epsilon a^\mu$. As calculated before, the variation in the Lagrangian is given by:

$$\delta\mathcal{L} = \partial_\mu\left(\delta x^\mu \mathcal{L} + \delta_0\phi_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right)$$

Note that $\delta x^\mu = \epsilon a^\mu$ and $\delta_0\phi_i = \phi_i'(x^\mu) - \phi_i(x^\mu)$. Note that the new field at x^μ is just the previous field at $x^\mu - \epsilon a^\mu$ and hence we have from a first-order expansion that $\delta_0\phi_i = -\epsilon a^\nu\partial_\nu\phi$. Substituting this above, we obtain:

$$\begin{aligned}\partial_\mu\left(\delta x^\mu \mathcal{L} + \delta_0\phi_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right) &= \partial_\mu\left(\epsilon a^\mu\mathcal{L} - \epsilon a^\nu\partial_\nu\phi_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right) \\ &= \partial_\mu\left(\epsilon a^\nu\delta^\mu{}_\nu\mathcal{L} - \epsilon a^\nu\partial_\nu\phi_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right) \\ &= -\epsilon a^\nu\partial_\mu\left(\partial_\nu\phi_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} - \delta^\mu{}_\nu\mathcal{L}\right)\end{aligned}$$

From the above, we can thus define the conserved current as:

$$T^\mu{}_\nu := \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\partial_\nu\phi_i - \delta^\mu{}_\nu\mathcal{L}$$

Now, note the following:

$$T^{\mu\nu} = g^{\alpha\nu}T^\mu{}_\alpha = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}g^{\alpha\nu}\partial_\alpha\phi_i - g^{\alpha\nu}\delta^\mu{}_\alpha\mathcal{L} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\partial^\nu\phi_i - g^{\mu\nu}\mathcal{L}$$

The above quantity can be shown to be a tensor and is called the *energy-momentum stress tensor*. Why?

Note that for each value of ν we have a current four vector. Let us calculate the charge for the time-translation invariance part, that is, $\nu = 0$. From this, we have:

$$P^0 = \int \mathrm{d}^3\mathbf{x}\; T^{00} = \int \mathrm{d}^3\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_i)} \partial^0 \phi_i - g^{00}\mathcal{L} \right) = \int \mathrm{d}^3\mathbf{x}\; \left(\pi^i \dot{\phi}_i - \mathcal{L}\right) = \int \mathrm{d}^3\mathbf{x}\; \mathcal{H} = H$$

We see that corresponding to the converved current from time translation invariance, we have a conserved charge $P^0 = H$. Thus, the total Hamiltonian is conserved under time translation and hence our classical intuition worked here as well!

Now, let us see what happens for space translations, for which we consider $\nu \equiv j = 1, 2, 3$. Then we have:

$$P^j = \int \mathrm{d}^3\mathbf{x}\; T^{0j} = \int \mathrm{d}^3\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_i)} \partial^j \phi_i - \cancelto{0}{g^{0j}}\,\mathcal{L} \right) = \int \mathrm{d}^3\mathbf{x}\; \left(\pi^i \partial^j \phi_i\right)$$

The last thing is the j^{th} component of the total momentum which is conserved in this case. Thus, we see that for spatial translation invariance, the total momentum is conserved, which once again matches with the classical analog.

9.1.2. Lorentz Invariance

An infinitesimal Lorentz transformation (which covers both *boosts* and *rotations*) can be written as:

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$$

where $\omega^\mu{}_\nu$ is an infinitesimal deviation from the identity. Now, we know for any Lorentz transformation,

$$\Lambda^\mu{}_\tau \eta^{\tau\sigma} \Lambda^\nu{}_\sigma = \eta^{\mu\nu} \iff \Lambda^{\mathrm{T}} \eta \Lambda = \eta$$

If we put the infinitesimal transformation in the above equation and retain only terms upto first order in ω, we would have $\omega^{\mu\nu} = \omega^{\nu\mu}$, that is, the infinitesimal deviation is an anti-symmetric matrix. The spacetime coordinates transforms as:

$$x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu = \delta^\mu{}_\nu x^\nu + \omega^\mu{}_\nu x^\nu = x^\mu + \omega^\mu{}_\nu x^\nu \implies \delta x^\mu = \omega^\mu{}_\nu x^\nu$$

And the variation in ϕ is given as:

$$\delta_0 \phi_i = \phi'(x^\mu) - \phi(x^\mu) = \phi(x^\mu - \omega^\mu{}_\nu x^\nu) - \phi(x^\mu) = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi_i$$

Now, we are ready to put in the variation of the Lagrangian:

$$\begin{aligned}
\partial_\mu\left(\delta x^\mu \mathcal{L} + \delta_0 \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\right) &= \partial_\mu\left(\omega^\mu{}_\nu x^\nu \mathcal{L} - \omega^\sigma{}_\nu x^\nu \partial_\sigma \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\right) \\
&= \partial_\mu\left(\eta^{\mu\rho}\omega_{\rho\nu} x^\nu \mathcal{L} - \eta^{\sigma\lambda}\omega_{\lambda\nu} x^\nu \partial_\sigma \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\right) \\
&= \partial_\mu\left(\eta^{\mu\rho}\omega_{\rho\nu} x^\nu \mathcal{L} - \omega_{\lambda\nu} x^\nu \partial^\lambda \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\right) \\
&= \partial_\mu\left(\eta^{\mu\rho}\omega_{\rho\nu} x^\nu \mathcal{L} - \omega_{\rho\nu} x^\nu \partial^\rho \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\right) \\
&= \partial_\mu\left(-\omega_{\rho\nu} x^\nu \times \left(-\eta^{\mu\rho}\mathcal{L} + \partial^\rho \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\right)\right) \\
&= \partial_\mu(-\omega_{\rho\nu} x^\nu T^{\rho\mu}) \\
&= \partial_\mu(\omega_{\nu\rho} x^\nu T^{\rho\mu})
\end{aligned}$$

Since the current must be conserved for all the six elements of the anti-symmetric matrix (since anti-symmetric so there are total $\frac{4\times(4-1)}{2} = 6$ independent entries), the part of $x^\nu T^{\rho\mu}$ which is anti-symmetric in ρ and μ must be conserved. After antisymmetrising and considering current for each component, we have:

$$M^{\nu\mu\rho} := x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu} \qquad \partial_\mu M^{\nu\mu\rho} = 0$$

Let us now see the conserved charges for this case. Corresponding to spatial rotations, we have:

$$\mathcal{J}^{ij} = \int \mathrm{d}^3\mathbf{x}\ M^{i0j} = \int \mathrm{d}^3\mathbf{x}\ (x^i T^{0j} - x^j T^{0i})$$

This can be related to the angular-momentum since T^{0i} was related to P^i, the i^{th} component of the momentum. Hence, rotational invariance leads to the *conservation of angular momentum*. Let us see what happens when we give *boosts*.

$$\mathcal{J}^{0j} = \int \mathrm{d}^3\mathbf{x}\ M^{00j} = \int \mathrm{d}^3\mathbf{x}\ (x^0 T^{0j} - x^j T^{00})$$

As $t = x^0$ and we know that the current $\mathcal{J}^{0j}$ must be conserved, we have:

$$\frac{\mathrm{d}\mathcal{J}^{0j}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}^3\mathbf{x}\ (x^0 T^{0j} - x^j T^{00}) = \int \mathrm{d}^3\mathbf{x}\ T^{0j} + \int \mathrm{d}^3\mathbf{x}\ t\ \partial_t T^{0j} - \frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}^3\mathbf{x}\ x^j T^{00} = 0$$

In terms of P^j, the above equation can be written as:

$$P^j + t\frac{\mathrm{d}P^j}{\mathrm{d}t} - \frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}^3\mathbf{x}\ x^j T^{00} = 0$$

Previously we had seen that P^j was the conserved charge corresponding to spatial translation and hence P^j is a constant, which gives us:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}^3\mathbf{x}\ x^j T^{00} = \mathcal{C}\ (const.)$$

The above relation can be seen as the centre of energy (or mass) of the field travels with constant velocity as T^{00} can be treated as the Hamiltonian ($\approx$ energy) density.

9.2. Examples: Internal Symmetries

Let us now look at internal symmetries, where the spacetime coordinates themselves do not change but the fields change due to some global transformations.

9.2.1. Global $\mathrm{U}(1)$ Symmetry of Complex Scalar Field

Consider a complex scalar field ϕ with the following Lagrangian:

$$\mathcal{L} = (\partial_\mu\phi)(\partial^\mu\phi^*) - m^2\phi\phi^*$$

Let us consider a $U(1)$ transformation [1]. Note that elements of the group $U(1)$ are represented by $e^{i\alpha}$, that is, they lie on the unit circle. Then, for an infinitesimal change $\psi \to e^{i\alpha}\psi$, $e^{i\alpha} = 1 + i\alpha + \mathcal{O}(\alpha^2)$[2] and we have:

$$\phi' = (1 + i\alpha)\phi \implies \delta_\alpha\phi = i\alpha$$
$$\phi^{*\prime} = (1 - i\alpha)\phi^* \implies \delta_\alpha\phi^* = -i\alpha$$

Under this transformation, we see that the Lagrangian is invariant, that is,

$$\mathcal{L}' = (\partial_\mu\phi')(\partial^\mu\phi'^*) - m^2\phi'\phi'^* = e^{i\alpha}e^{-i\alpha}(\partial_\mu\phi)(\partial^\mu\phi'^*) - e^{i\alpha}e^{-i\alpha}m^2\phi'\phi'^* = \mathcal{L}$$

From this, we can take $K^\mu = 0$. Using the formula for the conserved current, we have:

$$J^\mu = i\alpha\phi(\partial^\mu\phi^*) - i\alpha\phi^*(\partial^\mu\phi) \sim i(\partial^\mu\phi^*\phi - \partial^\mu\phi\phi^*)$$

The converved charge is then given by the volume integral:

$$Q = \int \mathrm{d}^3\mathbf{x}\ J^0 = -i\int \mathrm{d}^3\mathbf{x}\ (\phi\dot{\phi}^* - \dot{\phi}\phi^*)$$

[1] This is an *internal* symmetry as spacetime points are not varied.

[2] This is a *global* symmetry as α does not depend on the spacetime coordinates

Lecture 10 & 11: Field Quantisation

So far we had worked with classical fields. In the introduction, we saw that, quantum theories due to decoherence, transforms to classical theories, with an inevitable loss of information. Thus, finding the quantum theory leading to the classical model (working backwards) is not *unique*. More than one quantum field theory might have the same classical limit. Thus, while doing the quantisation process, we always have to make an *educated guess*!

Now, let us make the KG field *quantum* by quantising the canonical coordinates, similar to as we did for the harmonic oscillator.

We have our classical field $\phi(\mathbf{x}, t)$ and its canonical conjugate momentum $\pi(\mathbf{x}, t)$ which become operators $\hat{\phi}(x)$ and $\hat{\pi}(x)$[1]. The classical Poisson bracket changes to the commutator, that is, $\{\cdot,\cdot\} \to i[\cdot,\cdot]$.

A quantum treatment of a classical Hamiltonian is sometimes messy since we have to maintain the Hermiticity condition and there can be multiple ways to do that. However, almost always, we just put the hats on appropriate places and wish we are correct!

Classically, we had $\{\phi(\mathbf{x}), \pi(\mathbf{y})\} = \delta^3{}_{\mathbf{x}-\mathbf{y}}$ and by the same logic:

$$[\hat{\phi}(\mathbf{x}), \hat{\pi}(\mathbf{y})] = i\delta^3{}_{\mathbf{x}-\mathbf{y}}$$

Let us take their Fourier transforms,

$$\widetilde{\phi}(\mathbf{k}, t) = \int \mathrm{d}^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}}\phi(\mathbf{x}, t)$$
$$\widetilde{\pi}(\mathbf{k}, t) = \int \mathrm{d}^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}}\pi(\mathbf{x}, t)$$

Let us now consider the Klein-Gordon equation,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 \iff (\partial_\mu\partial^\mu + m^2)\phi = 0$$

Expanding the field in Fourier space, we would obtain the following:

$$\left(\frac{\partial^2}{\partial t^2} + (|\mathbf{p}|^2 + m^2)\right)\tilde{\phi}(\mathbf{p}, t) = 0$$

Note that this is the equation for a classical harmonic oscillator with frequency $\omega_\mathbf{p} = \sqrt{|\mathbf{p}|^2 + m^2}$ and the general solution $\widetilde{\phi}(\mathbf{x}, t)$ will be a linear combination of $e^{i\omega_\mathbf{p}}$ and $e^{-i\omega_\mathbf{p}}$. Thus we have,

$$\widetilde{\phi}(\mathbf{p}, t) = A(\mathbf{p})e^{i\omega_\mathbf{p}t} + B(\mathbf{p})e^{-i\omega_\mathbf{p}t} \qquad \omega_\mathbf{p} = \sqrt{|\mathbf{p}|^2 + m^2}$$

Now, in the quantum domain, $A(\mathbf{p})$ and $B(\mathbf{p})$ will become the creation and annihilation operators $a_\mathbf{p}$ and $a_\mathbf{p}^\dagger$ and in that spirit, let us define the field operators in terms of the annihilation and creation operators for each Fourier mode. Note that, for simplicity we define the field operators at $t = 0$.

$$\hat{\phi}(\mathbf{x}) := \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{p}}}\left(\hat{a}_\mathbf{p}e^{i\mathbf{p}\cdot\mathbf{x}} + \hat{a}_\mathbf{p}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}}\right)$$
$$\hat{\pi}(\mathbf{x}) := \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}(-i)\sqrt{\frac{\omega_\mathbf{p}}{2}}\left(\hat{a}_\mathbf{p}e^{i\mathbf{p}\cdot\mathbf{x}} - \hat{a}_\mathbf{p}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}}\right)$$

where $\omega_\mathbf{p} = \sqrt{|\mathbf{p}|^2 + m^2}$ is the frequency of the mode with momentum $\mathbf{p}$. Replacing $\mathbf{p}$ with $-\mathbf{p}$ in the second term, the above can also be written in a more compact way as:

$$\hat{\phi}(\mathbf{x}) := \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{p}}}\left(\hat{a}_\mathbf{p} + \hat{a}_{-\mathbf{p}}^\dagger\right)e^{i\mathbf{p}\cdot\mathbf{x}}$$
$$\hat{\pi}(\mathbf{x}) := \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}(-i)\sqrt{\frac{\omega_\mathbf{p}}{2}}\left(\hat{a}_\mathbf{p} - \hat{a}_{-\mathbf{p}}^\dagger\right)e^{i\mathbf{p}\cdot\mathbf{x}}$$

[1] Note that we will write four vectors with non-bold letter, as x will denote the four components of the four-vector. To denote the three vectors, we will write in bold-face, that is, $\mathbf{x}$

Let us check the commutation relation:

$$\left[\hat{\phi}(\mathbf{x}), \hat{\pi}(\mathbf{y})\right] = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{(-i)}{2}\sqrt{\frac{\omega_\mathbf{p}}{\omega_{\mathbf{p}'}}}\left(\left[\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}}, \hat{a}_{\mathbf{p}'} - \hat{a}^\dagger_{-\mathbf{p}'}\right]\right)e^{i(\mathbf{p}\cdot\mathbf{x}+\mathbf{p}'\cdot\mathbf{y})} = i\delta^3{}_{\mathbf{x}-\mathbf{y}}$$

Note that the above relation can be satisfied if we take:

$$\left[\hat{a}_\mathbf{p}, \hat{a}^\dagger_{\mathbf{p}'}\right] = (2\pi)^3\delta^3{}_{\mathbf{p}-\mathbf{p}'}\mathbb{1}$$

Now, let us do a horrenduous calculation, which will then lead to a horrenduous result and after some reasoning based on physical grounds, we will obtain a cute result. From the Hamiltonian density, we have the total Hamiltonian as:

$$\hat{H} = \int \mathrm{d}^3\mathbf{x}\ \mathcal{H} = \int \mathrm{d}^3\mathbf{x}\left(\frac{1}{2}\hat{\pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2\right)$$

Let us calculate this term by term. The first term gives us:

$$\begin{aligned}
\int \mathrm{d}^3\mathbf{x}\ \pi^2(\mathbf{x}) &= \int \mathrm{d}^3\mathbf{x}\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}{-2}(\hat{a}_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{\mathbf{p}'} - \hat{a}^\dagger_{-\mathbf{p}'})e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\mathrm{d}^3\mathbf{p}'\frac{\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}{-2}(\hat{a}_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{\mathbf{p}'} - \hat{a}^\dagger_{-\mathbf{p}'})\delta^3{}_{\mathbf{p}+\mathbf{p}'} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left[-\frac{\omega_\mathbf{p}}{2}\right](\hat{a}_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{-\mathbf{p}} - \hat{a}^\dagger_{\mathbf{p}}) \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left[-\frac{\omega_\mathbf{p}}{2}\right](\hat{a}_\mathbf{p}\hat{a}_{-\mathbf{p}} - \hat{a}_\mathbf{p}\hat{a}^\dagger_{\mathbf{p}} - \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}})
\end{aligned}$$

From the last term, we have:

$$\begin{aligned}
\int \mathrm{d}^3\mathbf{x}\ \phi^2(\mathbf{x}) &= \int \mathrm{d}^3\mathbf{x}\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{1}{2\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}(\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{\mathbf{p}'} + \hat{a}^\dagger_{-\mathbf{p}'})e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\mathrm{d}^3\mathbf{p}'\frac{1}{2\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}(\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{\mathbf{p}'} + \hat{a}^\dagger_{-\mathbf{p}'})\delta^3{}_{\mathbf{p}+\mathbf{p}'} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left[\frac{1}{2\omega_\mathbf{p}}\right](\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{\mathbf{p}}) \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left[\frac{1}{2\omega_\mathbf{p}}\right](\hat{a}_\mathbf{p}\hat{a}_{-\mathbf{p}} + \hat{a}_\mathbf{p}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}})
\end{aligned}$$

Now comes the second term. For this, note that we have $\nabla(e^{i\mathbf{k}\cdot\mathbf{x}}) = i\mathbf{k}(e^{i\mathbf{k}\cdot\mathbf{x}})$. Using this, we obtain:

$$\begin{aligned}
\int \mathrm{d}^3\mathbf{x}\ (\nabla\phi)\cdot(\nabla\phi) &= \int \mathrm{d}^3\mathbf{x}\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{1}{2\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}\left[i\mathbf{p}(\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})e^{i\mathbf{p}\cdot\mathbf{x}}\right]\cdot\left[i\mathbf{p}'(\hat{a}_{\mathbf{p}'} + \hat{a}^\dagger_{-\mathbf{p}'})e^{i\mathbf{p}'\cdot\mathbf{x}}\right] \\
&= -\int \mathrm{d}^3\mathbf{x}\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{1}{2\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}\left[(\mathbf{p}\cdot\mathbf{p}')(\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{\mathbf{p}'} + \hat{a}^\dagger_{-\mathbf{p}'})\right]e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\
&= -\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\mathrm{d}^3\mathbf{p}'\frac{1}{2\sqrt{\omega_\mathbf{p}\omega_{\mathbf{p}'}}}\left[(\mathbf{p}\cdot\mathbf{p}')(\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{\mathbf{p}'} + \hat{a}^\dagger_{-\mathbf{p}'})\right]\delta^3{}_{\mathbf{p}+\mathbf{p}'} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_\mathbf{p}}\left[(\mathbf{p}^2)(\hat{a}_\mathbf{p} + \hat{a}^\dagger_{-\mathbf{p}})(\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{\mathbf{p}})\right] \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathbf{p}^2}{2\omega_\mathbf{p}}\left(\hat{a}_\mathbf{p}\hat{a}_{-\mathbf{p}} + \hat{a}_\mathbf{p}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}\right)
\end{aligned}$$

Adding all the terms with correct pre-factors, we obtain the following:

$$\begin{aligned}
\hat{H} &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left\{\left[-\frac{\omega_{\mathbf{p}}}{4}\right]\left(\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} - \hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} - \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}\right)\right\} + \left\{\left[\frac{m^2}{4\omega_{\mathbf{p}}}\right]\left(\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}\right)\right\} \\
&\qquad + \left\{\frac{\omega_{\mathbf{p}}^2 - m^2}{4\omega_{\mathbf{p}}}\left(\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}\right)\right\} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\omega_{\mathbf{p}}}{4}\left(\cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} + \cancel{\hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}} - \cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} - \cancel{\hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}}\right) \\
&\qquad + \frac{m^2}{4\omega_{\mathbf{p}}}\left(\cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \cancel{\hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}} + \cancel{\hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \cancel{\hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}} - \cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} - \cancel{\hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}} - \cancel{\hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}}} - \cancel{\hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}}\right) \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\omega_{\mathbf{p}}}{2}\left(\hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}}\right) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\omega_{\mathbf{p}}}{2}\left(\hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} + \hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{p}}\right)
\end{aligned}$$

Now, using the commutator as obtained before, we get:

$$\hat{H} = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\omega_{\mathbf{p}}}{2}\left((2\pi)^3{\delta^3}_0 + 2\hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{p}}\right) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\omega_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{p}} + \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\omega_{\mathbf{p}}}{2}{\delta^3}_0$$

This is the end of the horrenduous calculation and now we discuss the horrenduous term (the second term) which was obtained as a result of applying the commutator. Since this contains a delta-function (distribution?) and the integral is over all momentum, this definitely cannot be a finite quantity, hence it diverges. This term basically represents the zero-point energies of the harmonic oscillators (since there are infinite number of harmonic oscillators, we have a divergent zero-point energy). The fortunate thing is that this is a constant.

When physically measuring energies, only differences can be calculated and hence this term gets cancelled and thus, no divergence actually happens.[1]

Thus, in all practicality, we can rescale our energy and we obtain the Hamiltonian as:

$$\hat{H} = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\omega_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{p}}$$

Now, let us see the commutator of the ladder operators with the Hamiltonian. We have:

$$\left[\hat{H}, \hat{a}^\dagger_{\mathbf{p}}\right] = \int \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}\omega_{\mathbf{p'}}\left[\hat{a}^\dagger_{\mathbf{p'}}\hat{a}_{\mathbf{p'}}, \hat{a}^\dagger_{\mathbf{p}}\right] = \int \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}\omega_{\mathbf{p'}}\hat{a}^\dagger_{\mathbf{p'}}\left[\hat{a}_{\mathbf{p'}}, \hat{a}^\dagger_{\mathbf{p}}\right] = \int \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}\omega_{\mathbf{p'}}\hat{a}^\dagger_{\mathbf{p'}}(2\pi)^3{\delta^3}_{\mathbf{p}-\mathbf{p'}} = \omega_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} \tag{1}$$

$$\left[\hat{H}, \hat{a}_{\mathbf{p}}\right] = \int \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}\omega_{\mathbf{p'}}\left[\hat{a}^\dagger_{\mathbf{p'}}\hat{a}_{\mathbf{p'}}, \hat{a}_{\mathbf{p}}\right] = \int \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}\omega_{\mathbf{p'}}\left[\hat{a}^\dagger_{\mathbf{p'}}, \hat{a}_{\mathbf{p}}\right]\hat{a}_{\mathbf{p'}} = -\int \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}\omega_{\mathbf{p'}}\hat{a}_{\mathbf{p'}}(2\pi)^3{\delta^3}_{\mathbf{p}-\mathbf{p'}} = -\omega_{\mathbf{p}}\hat{a}_{\mathbf{p}} \tag{2}$$

11.1. Conserved Charges

Note that we already found the Hamiltonian which is one of the conserved charges corresponding to T^{00}. Let us now focus on the momentum, that is, $\hat{P}^j$ corresponding to T^{0j}

$$\hat{P}^j = \int \mathrm{d}^3\mathbf{x}\ \hat{T}^{0j} = -\int \mathrm{d}^3\mathbf{x}\ \hat{\pi}\partial_j\hat{\phi}$$

Note that, while changing from the classical expression to quantum by 'putting hats', we had several choices for the ordering of $\hat{\pi}$ and $\hat{\phi}$ but here, we specify the one which works!

We now have two options, either to do the calculation again or just specify the result of this calculation

[1]This infinite term is sometimes attributed to the *cosmological constant.* It also has some role in the *Casimir effect.*

since this had already been done by stalwarts before. We choose the former and thus, we have:

$$\begin{aligned}\int \mathrm{d}^3\mathbf{x}\ \hat{\pi}\partial_j\hat{\phi} &= \int \mathrm{d}^3\mathbf{x}\ \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3}(-i)\sqrt{\frac{\omega_\mathbf{p}}{2}}\frac{1}{\sqrt{2\omega_{\mathbf{p'}}}}\Big(\hat{a}_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}}\Big)ip'_j\Big(\hat{a}_{\mathbf{p'}} + \hat{a}^\dagger_{-\mathbf{p'}}\Big)e^{i(\mathbf{p}+\mathbf{p'})\cdot\mathbf{x}} \\ &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\sqrt{\frac{\omega_\mathbf{p}}{2}}\frac{1}{\sqrt{2\omega_\mathbf{p}}}\Big(\hat{a}_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}}\Big)p'_j\Big(\hat{a}_{-\mathbf{p}} + \hat{a}^\dagger_\mathbf{p}\Big) \\ &= \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}p_j(\hat{a}_\mathbf{p}\hat{a}_{-\mathbf{p}} + \hat{a}_\mathbf{p}\hat{a}^\dagger_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}} - \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_\mathbf{p})\end{aligned}$$

Now, note the following:

$$\mathcal{I} = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}p_j(\hat{a}_\mathbf{p}\hat{a}_{-\mathbf{p}} - \hat{a}^\dagger_{-\mathbf{p}}\hat{a}^\dagger_\mathbf{p}) \quad \overset{\mathbf{p}\to-\mathbf{p}}{=} \quad \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}(-p_j)(\hat{a}_{-\mathbf{p}}\hat{a}_\mathbf{p} - \hat{a}^\dagger_\mathbf{p}\hat{a}^\dagger_{-\mathbf{p}}) = -\mathcal{I}$$

The last equality comes since the creation and annihilation operators commute for different values of the momentum. Thus, this integral is zero and the expression for the total momentum becomes:

$$\hat{P}^j = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}p_j\Big(\hat{a}_\mathbf{p}\hat{a}^\dagger_\mathbf{p} - \hat{a}^\dagger_{-\mathbf{p}}\hat{a}_{-\mathbf{p}}\Big) = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}p_j\Big(\hat{a}_\mathbf{p}\hat{a}^\dagger_\mathbf{p} + \hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p}\Big) = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}p_j\hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p}$$

In the last term, we introduced the commutator and removed the constant infinity term.

11.2. Normal Ordering

The way we are doing, like removing the infinities, is a common thing in QFT. This purely results from the fact the things commute in the classical case and hence there is always an order ambiguity when we move to quantum.

For example, if in classical scenario, we have some term like xp, when elevating (idk if its really elevating) these things to operators, we can have multiple possibilities, viz. $\hat{x}\hat{p}$, $\hat{p}\hat{x}$ or even $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ (placing things symmetrically). Each one would present some additional constant on the final result, since the commutator of $\hat{p}$ and $\hat{x}$ is just a constant.

To be consistent, we generally use *normal ordering*, that is, keeping all the creation operators to the left. For that we define the *normal ordering operator*, $\widehat{\mathcal{N}}$. For example,

$$\widehat{\mathcal{N}}[\hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p}] = \hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p} \qquad \widehat{\mathcal{N}}[\hat{a}_\mathbf{p}\hat{a}^\dagger_\mathbf{p}] = \hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p} \qquad \widehat{\mathcal{N}}[\hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p}\hat{a}_\mathbf{p}\hat{a}^\dagger_\mathbf{p}] = \hat{a}^\dagger_\mathbf{p}\hat{a}^\dagger_\mathbf{p}\hat{a}_\mathbf{p}\hat{a}_\mathbf{p} \quad \ldots$$

The $\widehat{\mathcal{N}}$ is not an operator in the sense that is doesnt act on a state to provide some new information. Instead, normal ordering is an interpretation that we use to eliminate the meaningless infinities that occur in field theories. So the rule goes that 'if you want to tell someone about a string of operators in quantum field theory, you have to normal order them first'. Thus, in operators like $\hat{P}$ and $\hat{H}$, in the final expression, we don't use the commutator but the normal operator operator, which will then completely subdue the issue of the appearance of Dirac-delta.

11.3. Constructing the Hilbert Space

The entire basis for this is the harmonic oscillator. In quantum SHO, we started with the vacuum and then acted creation operator successively on the vacuum to create excited states. Here, we will also do the same. Let us define the vacuum state $|\Omega\rangle$ such that,

$$\hat{a}_\mathbf{p}\,|\Omega\rangle = 0\ \forall\ \mathbf{p}$$

With such a definition, on physical grounds also, we obtain,

$$\hat{H}\,|\Omega\rangle = 0 \implies \text{Ground state has zero energy}$$

The vacuum can be written explicitly as:

$$|\Omega\rangle = \bigotimes_{\mathbf{p}} |\Omega\rangle_{\mathbf{p}} \qquad |\Omega\rangle_{\mathbf{p}} \to \text{vac. of SHO with mom. } \mathbf{p}$$

The Hilbert space is then built as the space generated by all finite linear combinations of vectors of the form:

$$|\mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n\rangle \equiv \hat{a}^\dagger_{\mathbf{p}_1} \hat{a}^\dagger_{\mathbf{p}_2} \ldots \hat{a}^\dagger_{\mathbf{p}_n} |\Omega\rangle$$

Till now, this is not exactly the Hilbert space, since we did not account for the norm of the vectors, where a huge subtlely lies.

Note that $|\mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n\rangle$ is an eigenstate of both $\hat{H}$ and $\hat{P}$ since,

$$\hat{H} |\mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n\rangle = (\omega_{\mathbf{p}_1} + \ldots + \omega_{\mathbf{p}_n}) |\mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n\rangle$$
$$\hat{P} |\mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n\rangle = (\mathbf{p}_1 + \ldots + \mathbf{p}_n) |\mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n\rangle$$

Since the creation operators commute, the multi-particle states are symmetric under exchange of particles, that is, the particles follow *Bose-Einstein* statistics.

For a general many-particle system, this 'pre'-Hilbert space is termed as *Fock space* $\mathfrak{F}$ and is formally defined with respect to direct-sums,

$$\mathfrak{F} = \bigoplus_{n=0}^{\infty} \mathscr{H}^{\otimes n}$$

where $\mathscr{H}^{\otimes n}$ is the n-times tensor product of the Hilbert space. In simple terms, $\mathscr{H}^{\otimes 0}$ is the space spanned by the vacuum state, $\mathscr{H}^{\otimes 1}$ is the space spanned by $|\mathbf{p}\rangle$. For $n \geq 2$, we have to consider the fermionic and bosonic properties too, that is, in our case, since these are bosons, $\mathscr{H}^{\otimes 2} = \text{Sym.}(\mathscr{H} \otimes \mathscr{H})$ is the space spanned by $|\mathbf{p}_1, \mathbf{p}_2\rangle$ such that Bose statisitcs is followed and so on...

Lecture 12: Making things invariant

The states that we had defined earlier, $|\mathbf{p}_1, \mathbf{p}_2, \ldots\rangle$ are *improper*, in the sense that they cannot be normalised. The obvious choice, $\langle \mathbf{p}|\mathbf{p}'\rangle = (2\pi)^3 \delta^3{}_{\mathbf{p}-\mathbf{p}'}$ is not physical due to the presence of the Dirac-delta. We can form physical states by 'smearing out' these deltas, by defining:

$$|\psi\rangle = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \psi(\mathbf{p}) |\mathbf{p}\rangle = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \psi(\mathbf{p}) \hat{a}^\dagger_{\mathbf{p}} ||\Omega\rangle\rangle$$

where $\psi(\mathbf{p})$ is an $\mathbb{L}^2$ function. However, this smearing process is not apriori Lorentz invariant. Thus, we can force the process to follow Lorentz invariance since the inception only, so that we do not run into trouble later.

12.1. Invariant Inner Product

For simplicity, consider a Lorentz boost along the x direction. Then,

$$p'^1 = \gamma(p^1 + \beta E) \qquad E' = \gamma(E + \beta p^1)$$

where $\gamma = 1/\sqrt{1-(v/c)^2}$ and $\beta = v/c$ are defined as usual. Now, note that $p^\mu p_\mu = m^2 \implies E^2 = (p^0)^2 + (p^1)^2 + (p^2)^2 + m^2 \implies 0 < E = \sqrt{(p^0)^2 + (p^1)^2 + (p^2)^2 + m^2}$ under the Minkowski metric. As the transformation is only along x direction, the momentum in x and y do not change. Thus, we have:

$$\frac{\partial E}{\partial p^j} = \frac{1}{2E} \times (2p^j) \implies \frac{\partial E}{\partial p^1} = \frac{p^1}{E}$$

Now, note the transformation of the Dirac-delta:

$$\int \mathrm{d}^3\mathbf{y}\ \delta^3{}_{\mathbf{x}-\mathbf{y}} = \int \mathrm{d}^3\mathbf{y}'\ \delta^3{}_{\mathbf{x}'-\mathbf{y}'} = \int \left|\frac{\partial \mathbf{y}'}{\partial \mathbf{y}}\right| \mathrm{d}^3\mathbf{y}\ \delta^3{}_{\mathbf{x}'-\mathbf{y}'}$$

Comparing, we see that the Dirac-delta transforms as ${\delta^3}_{\mathbf{x}-\mathbf{y}} = \left|\frac{\partial \mathbf{y}'}{\partial \mathbf{y}}\right| {\delta^3}_{\mathbf{x}'-\mathbf{y}'}$ where $\left|\frac{\partial \mathbf{y}'}{\partial \mathbf{y}}\right|$ is the determinant of the Jacobian matrix J of the transformation. Hence, for the Lorentz transformation, we have:

$$\begin{aligned} {\delta^3}_{\mathbf{p}-\mathbf{k}} &= \delta(p^1 - k^1)\delta(p^2 - k^2)\delta(p^3 - k^3) \\ &= \mathfrak{J}\ \delta(p'^1 - k'^1)\delta(p'^2 - k'^2)\delta(p'^3 - k'^3) \end{aligned}$$

Now, let us calculate the Jacobian matrix, whose elements will be given by $J^{ij} = \frac{\partial p'^i}{\partial p^j}$ (note that since we are concerned only with 3-momentum delta function, only spatial indices will appear). Note that $\frac{\partial p'^2}{\partial p^j} = \delta_{2j}$ and $\frac{\partial p'^3}{\partial p^j} = \delta_{3j}$. Only for $j = 1$, we have:

$$\frac{\partial p'^1}{\partial p^j} = \gamma\delta_{1j} + \gamma\beta\frac{\partial E}{\partial p^j}$$

and from the dispersion relation, we have $\frac{\partial E}{\partial p^j} = \frac{p^j}{E}$. The matrix is then:

$$J \equiv \begin{pmatrix} \gamma\left(1 + \beta\frac{p^1}{E}\right) & \gamma\beta\frac{p^2}{E} & \gamma\beta\frac{p^3}{E} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies \mathfrak{J} = \det J = \gamma\left(1 + \beta\frac{p^1}{E}\right) = \frac{\gamma}{E}\left(E + \beta p^1\right) = \frac{E'}{E}$$

Thus, we obtain the transformation of the delta function as:

$${\delta^3}_{\mathbf{p}-\mathbf{k}} = \frac{E'}{E}{\delta^3}_{\mathbf{p}'-\mathbf{k}'} \implies E{\delta^3}_{\mathbf{p}-\mathbf{k}} \text{ is invariant under L.T}$$

Then, we can now define the Lorentz-invariant inner product such that:

$$\langle \mathbf{p}|\mathbf{k}\rangle = (2E_{\mathbf{p}})(2\pi)^3{\delta^3}_{\mathbf{p}-\mathbf{k}} \equiv (2\omega_{\mathbf{p}})(2\pi)^3{\delta^3}_{\mathbf{p}-\mathbf{k}}$$

In line with this, we can therefore define a single-particle state as:

$$|\mathbf{p}\rangle = \sqrt{2\omega_{\mathbf{p}}}\hat{a}^\dagger_{\mathbf{p}}\,|\Omega\rangle$$

Technically, we should have made some distinct notation to differentiate between the states with and without proper normalisation, however, henceforth we will always use normalised states, so confusion might not be there.

12.2. Invariant Measure

Define the *projector* onto the single-particle state,

$$\mathbb{1} := \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\,|\mathbf{p}\rangle\langle\mathbf{p}|$$

Here we have used the states with proper normalisation, obeying Lorentz invariance. The LHS is Lorentz invariant, since the operator essentially checks whether we have single particle or not, which is a frame independent fact. Thus, the only thing to change is the *integral measure*!

Consider the following integral,

$$\int \mathrm{d}^4p\ \delta((p^0)^2 - \mathbf{p}^2 - m^2)\Theta(p^0)$$

Note that, d^4p is Lorentz invariant. Also, from the dispersion relation, we have:

$$p^\mu p_\mu = (p^0)^2 - \mathbf{p}^2 = m^2$$

The dispersion relation is also Lorentz invariant, since it contains scalars $p^\mu p_\mu$ and m^2. Solving for p^0, we have $p^0 = \pm\sqrt{\mathbf{p}^2 + m^2}$. Now, the choice of which branch to choose (poisitive or negative) is also Lorentz invariant, since here we are considering only *orthochronous transformations*, which does not change the time component of the momentum. Thus, overall the integral is Lorentz-invariant.

$$\begin{aligned}\int \mathrm{d}^4p\ \delta((p^0)^2-\mathbf{p}^2-m^2)\Theta(p^0) &= \int \mathrm{d}^4p\ \delta((p^0)^2-E_{\mathbf{p}}^2)\Theta(p^0)\\ &= \int d^3\mathbf{p}\int_{-\infty}^{\infty}\mathrm{d}p^0\ \delta((p^0)^2-E_{\mathbf{p}}^2)\Theta(p^0)\\ &= \int d^3\mathbf{p}\int_{-\infty}^{\infty}\mathrm{d}p^0\ \frac{1}{2E_{\mathbf{p}}}\big[\delta(p^0+E_{\mathbf{p}})+\delta(p^0-E_{\mathbf{p}})\big]\Theta(p^0)\\ &= \int\frac{\mathrm{d}^3\mathbf{p}}{2E_{\mathbf{p}}}\end{aligned}$$

Hence, whatever measure we were taking earlier, that is, $\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}$ was not Lorentz-invariant, however, as seen above,

$$\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2E_{\mathbf{p}}}\ \text{is manifestly Lorentz invariant}$$

Nice! So, henceforth we will always try to write the integrals in manifestly Lorentz invariant form, for example, the field expression becomes:

$$\hat{\phi}(x)=\int\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_{\mathbf{p}}}\left(\sqrt{2\omega_{\mathbf{p}}}\hat{a}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{x}}+\sqrt{2\omega_{\mathbf{p}}}\hat{a}^\dagger_{\mathbf{p}}e^{-i\mathbf{p}\cdot\mathbf{x}}\right)$$

We could technically absorb the $\sqrt{2\omega_{\mathbf{p}}}$ in the definition of the creation and annihilation operators which would make the new field expressions similar to the old ones, but it is not really needed!

Finally, we can rewrite the identity operator on one-particle states as,

$$\mathbb{1}:=\int\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_{\mathbf{p}}}\left|\mathbf{p}\right\rangle\left\langle\mathbf{p}\right|$$

12.3. Position states?

Consider the field operator acting on the vacuum. Since annihilation operators kill the vacuum, we only have:

$$\hat{\phi}(x)\left|\Omega\right\rangle=\int\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_{\mathbf{p}}}e^{-i\mathbf{p}\cdot\mathbf{x}}\left|p\right\rangle$$

Apart from the factor $\frac{1}{2\omega_{\mathbf{p}}}$, this looks more or less like some position state since it is linear superposition of all well-defined momentum states. Also, note that:

$$\begin{aligned}\left\langle\Omega\right|\hat{\phi}(\mathbf{x})\left|\mathbf{p}'\right\rangle &= \left\langle\Omega\right|\int\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_{\mathbf{p}}}\left(\sqrt{2\omega_{\mathbf{p}}}\hat{a}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{x}}+\sqrt{2\omega_{\mathbf{p}}}\hat{a}^\dagger_{\mathbf{p}}e^{-i\mathbf{p}\cdot\mathbf{x}}\right)\sqrt{2\omega_{\mathbf{p}}}\hat{a}^\dagger_{\mathbf{p}'}\left|\Omega\right\rangle\\ &= \int\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left(e^{i\mathbf{p}\cdot\mathbf{x}}\right)\left\langle\Omega\right|\hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}'}\left|\Omega\right\rangle\\ &= \int\frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left(e^{i\mathbf{p}\cdot\mathbf{x}}\right)\left\langle\Omega\right|(2\pi)^3\delta^3_{\mathbf{p}-\mathbf{p}'}+\hat{a}^\dagger_{\mathbf{p}'}\hat{a}_{\mathbf{p}}\left|\Omega\right\rangle\\ &= e^{i\mathbf{p}'\cdot\mathbf{x}}\end{aligned}$$

We can interpret this as the position-space representation of the single-particle state $\left|\mathbf{p}\right\rangle$, exactly as in non-relativistic quantum mechanics. Thus, we have:

$$\left\langle\Omega\right|\hat{\phi}(\mathbf{x})\left|\mathbf{p}\right\rangle=\left\langle x|p\right\rangle=e^{i\mathbf{p}\cdot\mathbf{x}}$$

Lecture 13: Heisenberg Picture

Till now, we saw that the operator $\hat{\phi}(\mathbf{x})$ has no time dependence (we defined it at $t = 0$). However, space and time should be on equal footing. Hence, it is better to define these in the Heisenberg picture, where the operators evolve in time while the states don't. Recall that a Heisenberg operator is defined from the Schödinger operator via,

$$\mathcal{O}_H = e^{+iHt}\mathcal{O}_s e^{-iHt} \implies \frac{\mathrm{d}\mathcal{O}_H}{\mathrm{d}t} = i\left[H, \mathcal{O}_H\right] + \frac{\partial \mathcal{O}_H}{\partial t}$$

We consider here only free fields, that is, there is no interactions. However, in general, the Hamiltonian can be decomposed as

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int.}}$$

where $\hat{H}_0$ is the *free field*. In that case, it would be better to work in the interaction picture. However, for now we assume $\hat{H}_{\text{int.}} = 0$. Note that from Eq. 1 and Eq. 2, we can obtain the time evolution of the ladder operators as:

$$\frac{\mathrm{d}\hat{a}^\dagger_{\mathbf{p}}}{\mathrm{d}t} = i\omega_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}} \implies \hat{a}^\dagger_{\mathbf{p}}(t) = \hat{a}^\dagger_{\mathbf{p}}(0)e^{i\omega_{\mathbf{p}}t} \implies e^{iH_0t}\hat{a}^\dagger_{\mathbf{p}}e^{-iH_0t} = \hat{a}^\dagger_{\mathbf{p}}e^{i\omega_{\mathbf{p}}t}$$
$$\frac{\mathrm{d}\hat{a}_{\mathbf{p}}}{\mathrm{d}t} = -i\omega_{\mathbf{p}}\hat{a}_{\mathbf{p}} \implies \hat{a}_{\mathbf{p}}(t) = \hat{a}_{\mathbf{p}}(0)e^{-i\omega_{\mathbf{p}}t} \implies e^{iH_0t}\hat{a}_{\mathbf{p}}e^{-iH_0t} = \hat{a}_{\mathbf{p}}e^{-i\omega_{\mathbf{p}}t}$$

Now, we calculate the field operator in the Heisenberg picture which gives:

$$\begin{aligned}
\hat{\phi}(\mathbf{x}, t) &= e^{iH_0t}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\mathbf{p}}}}\left(\hat{a}^\dagger_{\mathbf{p}}e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{a}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{x}}\right)e^{-iH_0t} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\mathbf{p}}}}\left(e^{iH_0t}\hat{a}^\dagger_{\mathbf{p}}e^{-iH_0t}e^{-i\mathbf{p}\cdot\mathbf{x}} + e^{iH_0t}\hat{a}_{\mathbf{p}}e^{-iH_0t}e^{i\mathbf{p}\cdot\mathbf{x}}\right) \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\mathbf{p}}}}\left(\hat{a}^\dagger_{\mathbf{p}}e^{i\omega_{\mathbf{p}}t}e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{a}_{\mathbf{p}}e^{-i\omega_{\mathbf{p}}t}e^{i\mathbf{p}\cdot\mathbf{x}}\right) \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\mathbf{p}}}}\left(\hat{a}^\dagger_{\mathbf{p}}e^{ip\cdot x} + \hat{a}_{\mathbf{p}}e^{-ip\cdot x}\right)
\end{aligned}$$

where $p \cdot x = p^\mu x_\mu = \omega_{\mathbf{p}}t - \mathbf{p}\cdot\mathbf{x}$ is the inner product of four momentum and four position in Minkowski space. Similarly, the conjugate momentum operator will be obtained as:

$$\hat{\pi}(\mathbf{x}, t) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}(-i)\sqrt{\frac{\omega_{\mathbf{p}}}{2}}\left(\hat{a}_{\mathbf{p}}e^{-ip\cdot x} - \hat{a}^\dagger_{\mathbf{p}}e^{ip\cdot x}\right)$$

Lecture 14 & 15: Propagators and Causality

In the previous section, we saw the time-evolution of the field operator in the Heisenberg pictre, essentially calculating an expression for $\hat{\phi}(\mathbf{x}, t)$. We will denote it simply as $\hat{\phi}(x)$ where x is the position four-vector while 3-vectors will be bold-faced.

Given a particle with momentum $|\mathbf{p}\rangle$, the transition to some other momentum state $|\mathbf{k}\rangle$, where $\mathbf{k} \neq \mathbf{p}$, is given by:

$$\langle \mathbf{k}|\mathbf{p}\rangle = (2\pi)^3\delta^3_{\mathbf{k}-\mathbf{p}} = 0$$

Since the momentum states are eigenstates of the Hamiltonian, transition is not possible. However, same cannot be said for position states. To understand if transition to another state is possible, we need something measurable. The obvious choice $\langle x|y\rangle = \langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle$ is not always physical, since the field operators are not Hermitian in general (eg. complex scalar field). Nevertheless, let us see what happens when we calculate this quantity.

Formally, the inner product is called the *Wightman function* and in terms of the operators, it is defined as:

$$D(x-y) := \langle x|y\rangle = \langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle$$

where x and y are four-positions and $\hat{\phi}$ is the field operator in the Heisenberg picture. The Wightman function is simply the 2-point correlation function. Let us calculate an explicit expression for this:

$$\begin{aligned}
D(x-y) &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{p}}}\frac{1}{\sqrt{2\omega_{\mathbf{p}'}}}\langle\Omega|\left\{\hat{a}_\mathbf{p}e^{-ip\cdot x}+\hat{a}^\dagger_\mathbf{p}e^{ip\cdot x}\right\}\left\{\hat{a}_{\mathbf{p}'}e^{-ip'\cdot y}+\hat{a}^\dagger_{\mathbf{p}'}e^{ip'\cdot y}\right\}|\Omega\rangle \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{p}}}\frac{1}{\sqrt{2\omega_{\mathbf{p}'}}}e^{-ip\cdot x}e^{ip'\cdot y}\langle\Omega|\,\hat{a}_\mathbf{p}\hat{a}^\dagger_{\mathbf{p}'}\,|\Omega\rangle \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{p}'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{p}}}\frac{1}{\sqrt{2\omega_{\mathbf{p}'}}}e^{-ip\cdot x}e^{ip'\cdot y}(2\pi)^3{\delta^3}_{\mathbf{p}-\mathbf{p}'} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_\mathbf{p}}e^{-ip\cdot(x-y)}
\end{aligned}$$

From the above expression, it is evident that $D(x-y)$ is Lorentz invariant since the measure and integrand both are invariant. Let us consider the case for different intervals.

Case 1: (x-y) is a *time-like* interval, that is, $(x-y)^2 > 0$ under Minkowski metric $(+,-,-,-)$

We can find a frame where the spatial interval is zero, that is, $\mathbf{x}'-\mathbf{y}' = 0$. Now, since $x^0 > y^0$ in our original frame and we are considering *orthochronous* Lorentz transformation, $(x')^0 > (y')^0$ too, since orthochronous transformations cannot flip the sign of the time-component. Defining $t = (x')^0 - (y')^0$, we have:

$$\begin{aligned}
D(x-y) = D(x'-y') &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_\mathbf{p}}e^{-ip^0t} \\
&= \int_0^\infty \frac{4\pi p^2dp}{(2\pi)^3}\frac{1}{2\sqrt{p^2+m^2}}e^{-i\sqrt{p^2+m^2}t} \\
&= \frac{1}{(2\pi)^2}\int_0^\infty dp\frac{p^2}{\sqrt{p^2+m^2}}e^{-i\sqrt{p^2+m^2}t}
\end{aligned}$$

Let us take $\epsilon = \sqrt{p^2+m^2} \implies \epsilon^2 = p^2+m^2 \implies \epsilon\mathrm{d}\epsilon = p\mathrm{d}p$. Substituting this in the integral, we obtain:

$$\begin{aligned}
D(x-y) &= \frac{1}{(2\pi)^2}\int_0^\infty dp\frac{p^2}{\sqrt{p^2+m^2}}e^{-i\sqrt{p^2+m^2}t} \\
&= \frac{1}{(2\pi)^2}\int_m^\infty \frac{1}{\cancel{\epsilon}}e^{-i\epsilon t}(\epsilon^2-m^2)\frac{\cancel{\epsilon}\mathrm{d}\epsilon}{\sqrt{\epsilon^2-m^2}} \\
&= \frac{1}{(2\pi)^2}\int_m^\infty e^{-i\epsilon t}\sqrt{\epsilon^2-m^2}\mathrm{d}\epsilon
\end{aligned}$$

Now, let us take $\epsilon = m\cosh\theta \implies \mathrm{d}\epsilon = m\sinh\theta\mathrm{d}\theta$. Substituting this in the integral above, we obtain:

$$\begin{aligned}
D(x-y) &= \frac{1}{(2\pi)^2}\int_0^\infty e^{-im\cosh\theta t}\sinh^2\theta\mathrm{d}\theta \\
&= \frac{1}{2(2\pi)^2}\int_0^\infty e^{-im\cosh\theta t}(\cosh 2\theta - 1)\mathrm{d}\theta \\
&= \frac{1}{2(2\pi)^2}\left(\int_0^\infty e^{-im\cosh\theta t}\cosh 2\theta\mathrm{d}\theta - \int_0^\infty e^{-im\cosh\theta t}\mathrm{d}\theta\right) \\
&= \frac{1}{2(2\pi)^2}(K_2(imt) - K_0(imt))
\end{aligned}$$

where $K_\alpha(x) = \int_0^\infty e^{-x\cosh\theta}\cosh\alpha\theta\ \mathrm{d}\theta$ is the integral representation of the *modified Bessel function of the second kind*. The integral can also be solved in the asymptotic limit, when $t >> \frac{1}{m}$. Note that for large t and large ϵ, the phase in the exponent is large and there is rapid oscillations, leading to the overall integral being zero. Hence, the contribution to the integral will only come from around $\epsilon \approx m$.

Thus, taking $\epsilon = m + \xi, \xi << m$, we have:

$$D(x-y) = \frac{e^{-imt}}{(2\pi)^2}\int_0^\infty e^{-i\xi t}\sqrt{(2m+\cancel{\xi})\xi}\mathrm{d}\xi = \frac{e^{-imt}\sqrt{2m}}{(2\pi)^2}\int_0^\infty \sqrt{\xi}e^{-i\xi t}\mathrm{d}\xi = e^{-imt}\frac{\sqrt{2m}}{(2\pi)^2 t^{3/2}}\int_0^\infty \sqrt{z}e^{-iz}\mathrm{d}z$$

The integral is found out to be $\frac{\Gamma(3/2)}{i^{3/2}} = \frac{\sqrt{\pi}}{2}e^{-3\pi i/4}$, by the analytic continuation of the Gamma function. Anyways, what we found is that, upto some factor,

$$D(x-y) \sim e^{-imt}$$

Case 2: (x-y) is a *space-like* interval, that is, $(x-y)^2 < 0$ under the Minkowski metric $(+,-,-,-)$

We can find a frame where the temporal interval is zero, that is, $(x')^0 - (y')^0 = 0$. Let us define $\mathbf{r} = \mathbf{x} - \mathbf{y}$. Then we have:

$$D(x-y) = D(x'-y') = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}$$

Consider $\mathbf{r}$ along the polar axis (if not, the rotate the polar axis), then we have:

$$\begin{aligned}
D(x-y) &= \frac{1}{2(2\pi)^2}\int_0^\infty \mathrm{d}p\int_{-1}^1 \mathrm{d}(\cos\theta)\ \frac{p^2}{\sqrt{p^2+m^2}}e^{ipr\cos\theta} \\
&= \frac{1}{2(2\pi)^2}\frac{1}{ir}\int_0^\infty \mathrm{d}p\frac{p}{\sqrt{p^2+m^2}}(e^{ipr} - e^{-ipr}) \\
&= \frac{1}{2ir(2\pi)^2}\int_{-\infty}^\infty \mathrm{d}p\frac{pe^{ipr}}{\sqrt{p^2+m^2}} \\
&= \frac{\mathcal{I}}{2ir(2\pi)^2}
\end{aligned}$$

Since square root is a multi-valued function, the integrand has branch points at $p = \pm im$ and we choose the branch cuts to be emanating from the poles and going to infinity. Let $r > 0$, so we chose the contour to close in the upper-half plane, as per Jordan's lemma. The contour, which avoids the branch cut, is shown below.

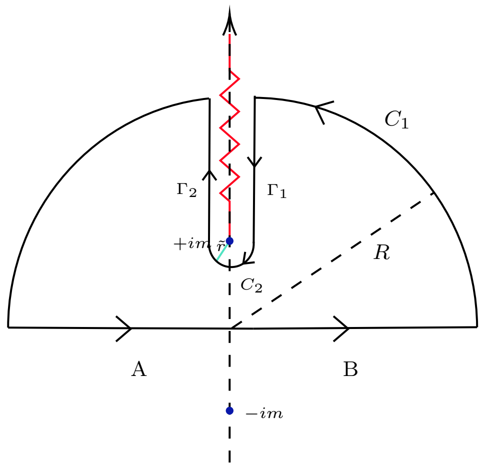
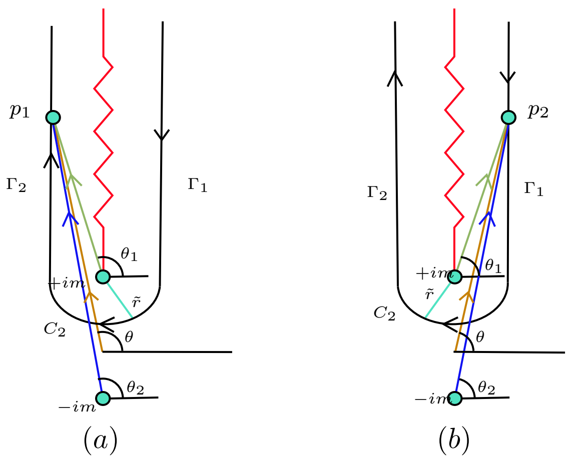


Figure 2: Contour for the integral (r>0). The blue dots represent the branch points while red kinked line denotes the branch cut.

The integrals along the arcs vanish for $R \to \infty$ and $\tilde{r} \to 0$ [1]. The only contribution to the integral comes because of the discontinuity across the branch cut. Since the contour encloses no poles, then by the Residue Theorem, we have:

$$\mathcal{I} + \int_{\Gamma_1} + \int_{\Gamma_2} = 0$$

Let us now see what these two integrals along both side of the branch cut evaluates to. For that, consider the diagram below:



Note that, on the left-side of the cut, θ_1 and θ_2 both are $\pi/2$. On the right side, since $p + im$ does not encircle the branch cut, θ_2 remain $\pi/2$ however, θ_1 becomes $\pi/2 - 2\pi = -3\pi/2$, since it encircles the branch cut. We write it as following:

$$\sqrt{p^2+m^2} = \sqrt{|p+im||p-im|}\exp\left(i\frac{\theta_1+\theta_2}{2}\right) = \begin{cases} \sqrt{|p+im||p-im|}i & \text{right} \\ -\sqrt{|p+im||p-im|}i & \text{left} \end{cases}$$

Then, the square root becomes negative of each other on either side of the branch cut (on right, side,).

[1]There is a subtlety tbh; for Jordan's lemma to be applicable, the integrand other than the exponential should vanish as $R \to \infty$. However, it doesn't happen here. Yet, by some transformation with Dirac delta, we can make this thing work afaik (Refer Padmanabhan, Problem1)!

Replacing $p = i\rho$, we have:

$$0 = \mathcal{I} + \int_{\infty}^{m} \frac{i\rho e^{-\rho r}}{\sqrt{\rho^2 - m^2}} + \int_{m}^{\infty} \frac{-i\rho e^{-\rho r}}{\sqrt{\rho^2 - m^2}}$$

$$\implies \mathcal{I} = 2i \int_{m}^{\infty} \frac{\rho e^{-\rho r}}{\sqrt{\rho^2 - m^2}}$$

Substituting $\rho = m\cosh\lambda$, we have the expression:

$$\mathcal{I} = 2im \int_{0}^{\infty} \cosh\lambda e^{-mr\cosh\lambda} = 2imK_1(mr)$$

Therefore, the final value of the correlation function becomes:

$$D(x-y) = \frac{m}{4\pi^2 r} K_1(mr) \xrightarrow{r\to\infty} \frac{m}{4\pi^2 r}\sqrt{\frac{\pi}{2mr}} e^{-mr} \sim e^{-mr}$$

We see that for spacelike separations, the amplitude falls off as $e^{-mr} \neq 0$. This shows that there is a non-zero probability amplitude for a particle to propagate over a spacelike interval, which violates causality (as it implies faster than light travel). As we had said earlier, this quantity is not very physical. A more appropriate quantity to analyse will be the commutator of the fields, that is,

$$\begin{aligned}
\left[\hat{\phi}(x), \hat{\phi}(y)\right] &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{\mathrm{d}^3\mathbf{p'}}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p'}}}} \left\{ \left[\hat{a}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{p'}}\right] e^{-ip\cdot x} e^{ip'\cdot y} + \left[\hat{a}^\dagger_{\mathbf{p'}}, \hat{a}_{\mathbf{p'}}\right] e^{ip\cdot x} e^{-ip'\cdot y} \right\} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} \left(e^{-ip\cdot(x-y)} - e^{ip\cdot(x-y)} \right) \\
&= D(x-y) - D(y-x)
\end{aligned}$$

Thus, we define the propagator as:

$$\triangle(x-y) = D(x-y) - D(y-x)$$

In this case, for a space-like interval, we can find a transformation which will take $(x-y) \mapsto (y-x)$, that is, If a particle can travel in a spacelike direction from $x \to y$, it can just as easily travel from $y \to x$ (since there is not Lorentz invariant way to order events), as a result of which, $\triangle(x-y) = 0$, no longer causing any violation of causality. For time-like separation, we cannot find such a transformation and as a result, the commutator is non-zero.

15.1. Propagators

Note that the commutator is basically a number (with an implicit identity operator sitting quietly) and thus,

$$\langle\Omega| \left[\hat{\phi}(x), \hat{\phi}(y)\right] |\Omega\rangle = D(x-y) - D(y-x) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} \left(e^{-ip\cdot(x-y)} - e^{ip\cdot(x-y)} \right)$$

Let us now calculate a 4-D integral of the following form, taking suppose $x_0 > y_0$:

$$\begin{aligned}
\mathcal{I} &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{\mathrm{d}p^0}{2\pi i} \frac{-e^{-ip\cdot(x-y)}}{p^2 - m^2} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \int \frac{\mathrm{d}p^0}{2\pi i} \frac{-e^{-ip^0(x^0-y^0) + i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{(p^0)^2 - |\mathbf{p}|^2 - m^2} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \int \frac{\mathrm{d}p^0}{2\pi i} \frac{-e^{-ip^0(x^0-y^0) + i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{(p^0)^2 - \omega_{\mathbf{p}}^2} \\
&= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \int \frac{\mathrm{d}p^0}{2\pi i} \frac{-e^{-ip^0(x^0-y^0)}}{[(p^0) - \omega_{\mathbf{p}}][(p^0) + \omega_{\mathbf{p}}]}
\end{aligned}$$

Note that the integral has two poles at $p^0 = \pm\omega_{\mathbf{p}}$, which lies on the real line. Since we are having $x^0 > y^0$, to apply Jordan's lemma, we need to close the contour in the lower half-plane. For this case, we choose to include both the poles into the contour, hence, applying the Residue Theorem, we have:

$$\int \mathrm{d}p^0 \frac{e^{-ip^0(x^0-y^0)}}{[(p^0)-\omega_{\mathbf{p}}][(p^0)+\omega_{\mathbf{p}}]} = -2\pi i\left(\frac{e^{-i\omega_{\mathbf{p}}(x^0-y^0)}}{2\omega_{\mathbf{p}}} + \frac{e^{i\omega_{\mathbf{p}}(x^0-y^0)}}{-2\omega_{\mathbf{p}}}\right)$$

$-2\pi i$ comes since the contour is taken clockwise. Substituting this value in the above integral, we obtain:

$$\begin{aligned}\mathcal{I} &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\left(\frac{e^{-i\omega_{\mathbf{p}}(x^0-y^0)}}{2\omega_{\mathbf{p}}} + \frac{e^{i\omega_{\mathbf{p}}(x^0-y^0)}}{-2\omega_{\mathbf{p}}}\right) \\ &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\left(\frac{e^{-i\omega_{\mathbf{p}}(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{2\omega_{\mathbf{p}}} + \frac{e^{i\omega_{\mathbf{p}}(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{-2\omega_{\mathbf{p}}}\right) \\ &= \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2\omega_{\mathbf{p}}}\left(e^{-ip\cdot(x-y)} - e^{ip\cdot(x-y)}\right)\end{aligned}$$

In the second term, we had taken $\mathbf{p} \to -\mathbf{p}$. Since $\omega_{\mathbf{p}} = \omega_{-\mathbf{p}}$ and the volume element remains invariant, we obtained the second term in the final expression.

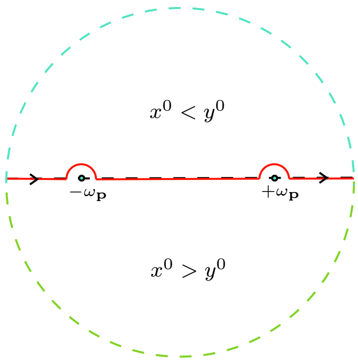


Figure 3: The contour choice for the *retarded propagator*. The green contour gives non-zero value while the blue contour gives zero since no pole is inside the contour. The read curve is common to both the contours

Thus, we see from the final expression that,

$$\langle\Omega|\left[\hat{\phi}(x),\hat{\phi}(y)\right]|\Omega\rangle = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\int \frac{\mathrm{d}p^0}{-2\pi i}\frac{e^{-ip\cdot(x-y)}}{(p^0)^2 - |\mathbf{p}|^2 - m^2}$$

Note that, instead of deforming the contour, we could have just shifted the poles downwards (by some amount $i\epsilon$ where $\epsilon > 0$) from the real axis, so that they fall into the contour which belongs completely to the lower half-plane. Thus, we can write the expression equivalently as:

$$\langle\Omega|\left[\hat{\phi}(x),\hat{\phi}(y)\right]|\Omega\rangle = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\int \frac{\mathrm{d}p^0}{-2\pi i}\frac{e^{-ip\cdot(x-y)}}{(p^0+i\epsilon)^2 - |\mathbf{p}|^2 - m^2} = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\int \frac{\mathrm{d}p^0}{-2\pi i}\frac{e^{-ip\cdot(x-y)}}{(p^0+i\epsilon)^2 - \omega_{\mathbf{p}}^2}$$

Note that, if we had taken $x^0 < y^0$, we would have closed the contour in the upper half-plane for Jordan's lemma to be valid and we would have got the integral as zero, since the contour did not enclose any of the poles. This is called the *Retarded Propagator* and is written compactly as,

$$D_R(x-y) = \Theta(x^0-y^0)\langle\Omega|\left[\hat{\phi}(x),\hat{\phi}(y)\right]|\Omega\rangle$$

where $\Theta(\cdot)$ represents the Heaviside-theta function. Similar to the above, we can define the *Advanced Propagator* which vanishes for $x^0 > y^0$. In this case, the poles are shifted upwards by $i\epsilon$. We can define this as:

$$D_A(x-y) = \Theta(y^0-x^0)\langle\Omega|\left[\hat{\phi}(x),\hat{\phi}(y)\right]|\Omega\rangle$$

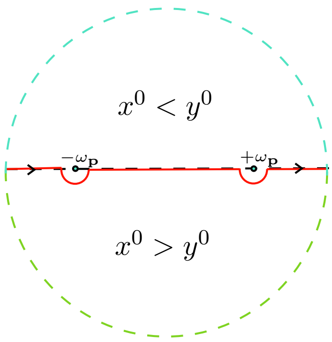


Figure 4: Contour choice for the advanced propagator.

There exists another pole prescription which gives rise to the celebrated *Feynman Propagator* which we investigate now.

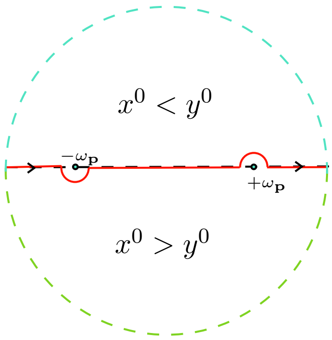


Figure 5: Pole choice for the Feynman propagator. We choose one pole to be inside and one pole to be outside the contour.

The Feynman propagator is defined as:

$$D_F(x-y) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \int \frac{\mathrm{d}p^0}{-2\pi i} \frac{e^{-ip\cdot(x-y)}}{(p^0)^2 - \omega_{\mathbf{p}}^2 + i\epsilon}$$

In this case, the contour is chosen such that one pole is shifted up and one pole is shifted down. Let the poles be,

$$p^0 = -\omega_{\mathbf{p}} + i\epsilon' \qquad p^0 = \omega_{\mathbf{p}} - i\epsilon'$$

Then we have the following,

$$(p^0 + \omega_{\mathbf{p}} - i\epsilon')(p^0 - \omega_{\mathbf{p}} + i\epsilon') = (p^0)^2 - (\omega_{\mathbf{p}} - i\epsilon')^2 = (p^0)^2 - \omega_{\mathbf{p}}^2 + \epsilon'^2 + 2i\omega_{\mathbf{p}}\epsilon'$$

We can neglect the ϵ'^2 term compared to the other term since it's a infinitesimal quantity. Also, for the same reason, we can define $\epsilon \equiv 2\omega_{\mathbf{p}}\epsilon'$ in the integral. The Feynman propagator can then be written as,

$$\begin{aligned} D_F(x-y) &= \Theta(x^0 - y^0) \langle\Omega| \hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle + \Theta(y^0 - x^0) \langle\Omega| \hat{\phi}(y)\hat{\phi}(x) |\Omega\rangle \\ &= \langle\Omega| \mathcal{T}\hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle \end{aligned}$$

where $\mathcal{T}$ is the *time-ordering* operator, which brings the operator to the left with the 'greater' time, similar to the normal ordering operator encounteres previously.

Feynman propagator is very important and has many applications as will be discussed later.

Lecture 16: Introducing the Dirac Equation

We saw that, when quantised, the Klein-Gordon equation provided a negative probability density, which was problematic since it had no obvious interpretation. Hence we said that, the KG equation cannot describe a single-particle system, instead, it describes a spin-0 particle *field*. All the problems arose since, from the relativistic dispersion relation,

$$E^2 = p^2 + m^2 = p_x^2 + p_y^2 + p_z^2 + m^2 \tag{3}$$

we obtain a differential equation which is second order in both space and time.

Let us now look for something which we force to be first order in space and time from the beginning. The most sane choice is to just take the square-root of the dispersion relation, however, square-root of an operator is ill-defined and has many other subtleties. We want everything to be as simple as possible. Hence, we try to *linearise* the square-root. We thus write,

$$E = \sqrt{p^2 + m^2} = \alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3 + \beta m$$

We first promote the momentum and energy to operators, that is, $p \to -i\nabla$ and $E \to i\partial_t$. Then the Schrödinger equation becomes,

$$i\partial_t \psi = \left(-i\alpha_i \frac{\partial}{\partial x^i} + \beta m\right)\psi \qquad \Longleftrightarrow \qquad H\psi = E\psi$$

Since we want this to match with the actual dispersion relation Eq. 3, let us see what happens when we apply the Hamiltonian twice on the time-independent equation (which would give us an E^2).

Note that, since the *linearisation* of the square-root is extremely non-trivial, α_i and β might not be just numbers, they could be literally anything. Thus, we do not make any apriori assumptions![1]

$$H^2 \equiv = \left(-\alpha_i^2 \frac{\partial^2}{\partial x^{i2}} + \beta m^2 - (\alpha_i\alpha_j + \alpha_j\alpha_i)\frac{\partial^2}{\partial x^i \partial x^j} - im(\beta a_i + \alpha_i\beta)\frac{\partial}{\partial x^i}\right)$$

If the relativistic dispersion relation has to be satisfied, we find some constraints on α_i and β:

$$\begin{aligned} \alpha_i^2 &= \mathbb{1} \qquad & \alpha_i\alpha_j + \alpha_j\alpha_i &= 2\delta_{ij}\mathbb{1} \\ \beta^2 &= \mathbb{1} \qquad & \beta\alpha_i + \alpha_i\beta &= 0 \end{aligned}$$

where $\mathbb{1}$ represents the identity operator in appropriate space. Having α_i and β to be just simple scalar numbers clearly do not satisfy the equations, hence let us assume for now, that these are matrices. We then additionally impose the contraint that $\alpha_i^\dagger = \alpha_i$ and $\beta^\dagger = \beta$ since energy and momentum operator (and hence the Hamiltonian) are Hermitian. With this assumption, let us note some nice properties of these matrices.

- Let v be an eigenvector of $\alpha_i \implies \alpha_i v = \lambda v \implies \underbrace{\alpha_i^2}_{\mathbb{1}} v = \lambda^2 v \implies \lambda = \pm 1$. Similar argument goes for β. Thus, these matrices have eigenvalues either +1 or -1.
- $\alpha_i\beta = -\beta\alpha_i \implies \alpha_i\beta^2 = -\beta\alpha_i\beta \implies \alpha_i = \beta\alpha_i\beta \implies \operatorname{Tr}\alpha_i = -\operatorname{Tr}\alpha_i \implies \operatorname{Tr}\alpha_i = 0$, where we have used the cyclic property of trace. Similar argument goes for β. Hence we obtain that these matrices are also traceless.
- Since the trace, that is, the sum of the eigenvalues, should equal zero, we should definitely have half of the eigenvalues as +1 and the other half to be -1. Thus, these matrices are guaranteed to be *even* dimensional.
- The relations between α_i looks suspiciously similar to the Pauli matrices and we can choose α_i to be σ_i. Alas, we run into a dangerous problem. We note that β cannot be the a two-dimensional matrix, since then it would not be consistent with the anticommutation relation with $\alpha_i \equiv \sigma_i$. Thus, the next option is to try for *four dimensional* matrices.

[1] Well, we do assume that α_i and $\frac{\partial}{\partial x^i}$ commute in some sense

Let us define the following,

$$\alpha_i := \left[\begin{array}{c:c} 0 & \sigma_i \\ \hdashline \sigma_i & 0 \end{array}\right] \qquad \beta := \left[\begin{array}{c:c} \mathbb{1} & 0 \\ \hdashline 0 & -\mathbb{1} \end{array}\right]$$

With this definition, we can indeed verify that all the conditions hold [1] and we obtain a way to proceed further. We have an expression for the equation as,

$$i\frac{\partial \psi}{\partial t} = -i\alpha \cdot \nabla + \beta m \implies i\beta\frac{\partial \psi}{\partial t} = -i\beta\alpha \cdot \nabla + \underbrace{\beta^2}_{\mathbb{1}} m$$

We would like to write the Dirac equation in covariant notation. To do so, let us introduce the following matrices:

$$\gamma^0 := \beta$$
$$\gamma^i := \gamma^0 \alpha_i$$

These matrices are called *Dirac matrices*. Then, using this definition in the above expression, we have:

$$i\gamma^0\partial_0\psi + i\gamma^i\partial_i\psi - m\psi = 0 \implies (i\gamma^\mu\partial_\mu - m)\psi = 0$$

Note that, we had introduced the 4-vector concept in an ad-hoc manner. However, doing this, we finally derive the well-known covariant form of the *Dirac equation*!

16.1. Form of Dirac Matrices

The Dirac matrices that we had defined have the following form:

$$\gamma^i = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}\begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \tag{4}$$

Lecture 17: Dirac in Fourier Mode

We saw that in the previous section that the Dirac equation is essentially:

$$i\partial_t\psi = (-i\alpha \cdot \nabla + \beta m)\psi \qquad \psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix}$$

The solution is a four-component vector since the matrices are 4×4 dimensional. [2] Let the proposed solution be a plane-wave solution, since there is no interaction, and we decompose it into its Fourier modes, that is,

$$\psi(x) = e^{-iEt}\psi(\mathbf{x}) = e^{-iEt}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\psi_p e^{-i\mathbf{p}\cdot\mathbf{x}}$$

Substituting this in the equation above, we have:

- LHS:
$$i\partial_t\psi = i(-iE)\psi(x) = E\psi(x)$$

- RHS:
$$e^{-iEt}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}(-i\alpha \cdot (i\mathbf{p}) + \beta m)\psi_p e^{i\mathbf{p}\cdot\mathbf{x}} = e^{-iEt}\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}(\alpha \cdot \mathbf{p} + \beta m)\psi_p e^{i\mathbf{p}\cdot\mathbf{x}}$$

[1] Note that these matrices are 4×4 matrices. Each element in the matrix are indeed 2×2 matrices.

[2] In non-relativistic QM, during angular momentum, we saw that multi-component wavefunctions imply a non-trivial spin angular momentum, so we expect Dirac equation to describe spinful particles.

Equating both LHS and RHS we get an equation in ψ_p,

$$\begin{aligned}
0 &= (\alpha \cdot \mathbf{p} + \beta m - E)\psi_p \\
&= \left(\begin{pmatrix} 0 & \sigma_i p_i \\ \sigma_i p_i & 0 \end{pmatrix} + \begin{pmatrix} (m-E)\mathbb{1} & 0 \\ 0 & -(E+m)\mathbb{1} \end{pmatrix}\right)\psi_p \\
&= \begin{pmatrix} (m-E)\mathbb{1} & \sigma \cdot \mathbf{p} \\ \sigma \cdot \mathbf{p} & -(m+E)\mathbb{1} \end{pmatrix}\psi_p
\end{aligned}$$

Now, for a solution to exist, the matrix should be *singular*, that is,

$$\det\left(\begin{pmatrix} (m-E)\mathbb{1} & \sigma \cdot \mathbf{p} \\ \sigma \cdot \mathbf{p} & -(m+E)\mathbb{1} \end{pmatrix}\right) = 0$$

We can calculate the determinant in the "usual" sense for block-matrices also, if the elements in the first column commute. Since first element is the identity matrix, it commutes with $\sigma \cdot \mathbf{p}$ and hence, we can write:

$$\begin{aligned}
0 &= \det\left((E^2 - m^2)\mathbb{1} - (\sigma \cdot \mathbf{p})(\sigma \cdot \mathbf{p})\right) \\
&= \det\left((E^2 - m^2)\mathbb{1} - (\sigma_i p_i)(\sigma_j p_j)\right) \\
&= \det\left((E^2 - m^2)\mathbb{1} - (\sigma_i \sigma_j) p_i p_j\right) \\
&= \det\left((E^2 - m^2)\mathbb{1} - |\mathbf{p}|^2 + \sum_{i<j}(\sigma_i\sigma_j + \sigma_j\sigma_i)p_i p_j\right) \\
&= \det\left((E^2 - m^2 - |\mathbf{p}|^2)\mathbb{1}\right)
\end{aligned} \tag{5}$$

From this, we finally obtain that $E^2 - m^2 - |\mathbf{p}|^2 = 0 \implies E = \pm\sqrt{m^2 + |\mathbf{p}|^2}$. We can see that, the equation says that for each momentum $\mathbf{p}$, we have two positive-energy solutions and two negative-energy solutions (each energy is *doubly degenerate*). Now let us find the eigenvectors. Assume it to be of the form,

$$\psi_p = \begin{pmatrix} \phi \\ \xi \end{pmatrix}$$

Then, substituting this in the eigenvalue equation we obtain two equations:

$$\begin{aligned}
0 &= (m-E)\phi + (\sigma \cdot \mathbf{p})\xi \\
0 &= (\sigma \cdot \mathbf{p})\phi - (m+E)\xi
\end{aligned}$$

Solving for ϕ and ξ, we get $\xi = \frac{(\sigma \cdot \mathbf{p})\phi}{m+E}$ and $\phi = \frac{(\sigma \cdot \mathbf{p})\xi}{E-m}$. Thus, we can write the solutions as:

$$\psi_p = \begin{pmatrix} \phi \\ \frac{(\sigma\cdot\mathbf{p})}{m+E}\,\phi \end{pmatrix} \qquad \text{and} \qquad \psi_p = \begin{pmatrix} \frac{(\sigma\cdot\mathbf{p})}{E-m}\,\xi \\ \xi \end{pmatrix} \tag{6}$$

Now, let us first take the particle at rest, that is, $\mathbf{p} = 0$. Then the eigenvalues are just $\pm m$ and the equation reduces to:

$$\begin{pmatrix} m & 0 \\ 0 & -m \end{pmatrix}\begin{pmatrix} \phi \\ \xi \end{pmatrix} = \pm m \begin{pmatrix} \phi \\ \xi \end{pmatrix}$$

Since the matrix is diagonal, the eigenvectors are trivially,

$$(\psi_p)_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (\psi_p)_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (\psi_p)_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad (\psi_p)_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

We can check that, the first two correspond to the positive energy solutions and the other two correspond to the negative energy solutions. Now, let us come back to $\mathbf{p} \neq 0$.The matrix $\sigma \cdot \mathbf{p}$ comes in the expression quite often and calculating that, we find that:

$$\sigma \cdot \mathbf{p} \equiv \begin{pmatrix} p_3 & p_1 - ip_2 \\ p_1 + ip_2 & -p_3 \end{pmatrix}$$

Now, let us take a vector v_m whose m^{th} position has 1 and all other elements are zero.

$$(Mv_m)_i = M_{ij}(v_m)j = M_{ij}\delta_{mj} = M_{im}$$

- CASE 1: $\phi = \begin{pmatrix}1\\0\end{pmatrix} \implies \xi = \frac{1}{E+m}\begin{pmatrix}p_3\\p_1+ip_2\end{pmatrix}$
- CASE 2: $\phi = \begin{pmatrix}0\\1\end{pmatrix} \implies \xi = \frac{1}{E+m}\begin{pmatrix}p_1-ip_2\\-p_3\end{pmatrix}$
- CASE 3: $\xi = \begin{pmatrix}1\\0\end{pmatrix} \implies \phi = \frac{1}{E-m}\begin{pmatrix}p_3\\p_1+ip_2\end{pmatrix}$
- CASE 4: $\xi = \begin{pmatrix}0\\1\end{pmatrix} \implies \phi = \frac{1}{E-m}\begin{pmatrix}p_1-ip_2\\-p_3\end{pmatrix}$

Thus, the final solutions for non-zero momentum are:

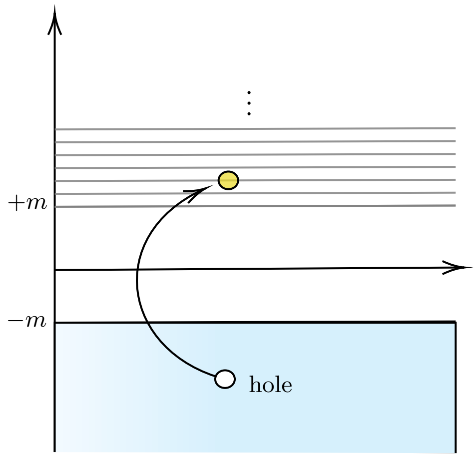
$$(\psi_p)_1 = \mathcal{N}_1\begin{pmatrix}1\\0\\\dfrac{p_3}{E+m}\\\dfrac{p_1+ip_2}{E+m}\end{pmatrix}\quad(\psi_p)_2 = \mathcal{N}_2\begin{pmatrix}0\\1\\\dfrac{p_1-ip_2}{E+m}\\\dfrac{-p_3}{E+m}\end{pmatrix}\quad(\psi_p)_3 = \mathcal{N}_3\begin{pmatrix}\dfrac{p_3}{E-m}\\\dfrac{p_1+ip_2}{E+m}\\1\\0\end{pmatrix}\quad(\psi_p)_4 = \mathcal{N}_4\begin{pmatrix}\dfrac{p_1-ip_2}{E-m}\\\dfrac{-p_3}{E+m}\\0\\1\end{pmatrix}$$

where $\mathcal{N}_i$ are some normalisation constants. The four components wavefunctions are called *Dirac spinors*.

Note that, if $\mathbf{p} = 0$ then the first two solutions correspond to the positive energy while the other two correspond to the negative energy. And thus, we define these solutions corresponding the $E = \sqrt{m^2+|\mathbf{p}|^2}$ and $E = -\sqrt{m^2+|\mathbf{p}|^2}$ respectively.

Note that the positive and negative energy solutions are separated by a gap of $2m$ as $E_+ = \sqrt{|\mathbf{p}|^2+m^2} > m$ and $E_- = -\sqrt{|\mathbf{p}|^2+m^2} < -m$. Also, the energy spectrum is *unbounded* from below.

It is an evident difficulty since any particle satisfying Dirac equation, will inevitably fall down to the lower, negative energy states, causing *destabilisation*.



To sort out this problem, the initial interpretation of the negative energy solutions were that the ground state (vacuum) was such that all the negative energy states were fully filled (if it were not, then transition to lower energy would always have been a possibility!). Thus, the *ground state* contains an infinite number of particles in the negative energy states.

The Dirac equation describes particles with spin-$\frac{1}{2}$ and hence a positive energy particle cannot transit to negative energy state due to the exclusion principle.

Excitations can happen such that a particle with negative energy occupy a positive energy state, leaving a 'hole' in the negative energy domain, which represented an *anti-particle*, having the same mass and *internal quantum numbers*.

Lecture 18: Few Properties of Dirac Equation

We will now look at some properties of the Dirac equation and also of the Dirac matrices. The first to check is the Lorentz covariance, (that is, under a Lorentz transformation, the equation transforms in a similar way), since we started with the relativistic dispersion relation.

18.1. Lorentz Covariance

Let us consider a Lorentz transformation,

$$x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

Let us assume that under the transformation, the wavefunction changes as,

$$\psi'(x') = \psi'(\Lambda x) = S(\Lambda)\psi(x)$$

$S(\Lambda)$ is a *spinor* representation of the Lorentz transformation. What we assume here is that the Lorentz transformation on the wavefunction can be faithfully represented by the matrix S which depends only on the parameter of the transformation Λ and not on the spacetime. Also, since the transformation matrices are *invertible*, we have:

$$\psi(x) = S^{-1}(\Lambda)\psi'(x')$$

$$\begin{aligned}
0 &= (i\gamma^\mu\partial_\mu - m)\psi(x) \\
&= (i\gamma^\mu\Lambda^\nu{}_\mu\partial'_\mu - m)S^{-1}(\Lambda)\psi'(x') \\
&= (iS(\Lambda)\gamma^\mu S^{-1}(\Lambda)\Lambda^\nu{}_\mu\partial'_\mu - mS(\Lambda)S^{-1}(\Lambda))\psi'(x') \\
&= (iS(\Lambda)\gamma^\mu S^{-1}(\Lambda)\Lambda^\nu{}_\mu\partial'_\mu - m)\psi'(x')
\end{aligned}$$

Note that we had used that the matrix S is spacetime independent and thus, commutes with ∂_μ operator. The above equation is Lorentz covariant if we can find an S such that,

$$[S(\Lambda)\gamma^\mu S^{-1}(\Lambda)]\Lambda^\nu{}_\mu = \gamma^\nu \implies S^{-1}(\Lambda)S(\Lambda)\gamma^\mu S^{-1}(\Lambda)\Lambda^\nu{}_\mu S(\Lambda) = S^{-1}(\Lambda)\gamma^\nu S(\Lambda) \implies \Lambda^\nu{}_\mu\gamma^\mu = S^{-1}(\Lambda)\gamma^\nu S(\Lambda) \tag{7}$$

Thus, if we define $\gamma'^\mu \equiv \Lambda^\nu{}_\mu\gamma^\mu = S^{-1}(\Lambda)\gamma^\nu S(\Lambda)$, then we can say that the Dirac matrices transform with one copy of Λ and hence, the vector index on γ, which was introduced in an ad-hoc manner, can actually be taken seriously.

18.2. Helicity

Let us define the following matrices:

$$\Sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu] \qquad \Sigma^i = \frac{1}{2}\epsilon^{ijk}\Sigma^{jk}$$

Then, we can calculate the form of the matrix given by Σ^i since,

$$\begin{aligned}
-2i\Sigma^{jk} &= \gamma^j\gamma^k - \gamma^k\gamma^j \\
&= \begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} - \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix} \\
&= -\begin{pmatrix} \sigma_j\sigma_k & 0 \\ 0 & \sigma_j\sigma_k \end{pmatrix} + \begin{pmatrix} \sigma_k\sigma_j & 0 \\ 0 & \sigma_k\sigma_j \end{pmatrix} \\
&= -\begin{pmatrix} [\sigma_j, \sigma_k] & 0 \\ 0 & [\sigma_j, \sigma_k] \end{pmatrix} \\
&= -2i\epsilon^{jkl}\begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}
\end{aligned}$$

From this, we obtain a neat expression that:

$$\Sigma^{jk} = \epsilon^{jkl}\begin{pmatrix}\sigma_l & 0\\ 0 & \sigma_l\end{pmatrix}$$

Next we use the expression given above:

$$\Sigma^{i} = \frac{1}{2}\epsilon^{ijk}\epsilon^{jkl}\begin{pmatrix}\sigma_l & 0\\ 0 & \sigma_l\end{pmatrix} = \delta_{il}\begin{pmatrix}\sigma_l & 0\\ 0 & \sigma_l\end{pmatrix} = \begin{pmatrix}\sigma_i & 0\\ 0 & \sigma_i\end{pmatrix}$$

Let us calculate the commutator between the Hamiltonian and the quantity $\mathbf{\Sigma}\cdot\mathbf{p}$

$$\begin{aligned}[H,\mathbf{\Sigma}\cdot\mathbf{p}] &\sim [\alpha\cdot\mathbf{p}+\beta m,\mathbf{\Sigma}\cdot\mathbf{p}]\\ &= [\alpha\cdot\mathbf{p},\mathbf{\Sigma}\cdot\mathbf{p}] + m\,[\beta,\mathbf{\Sigma}\cdot\mathbf{p}]\\ &= \begin{pmatrix}0 & \sigma\cdot\mathbf{p}\\ \sigma\cdot\mathbf{p} & 0\end{pmatrix}\begin{pmatrix}\sigma\cdot\mathbf{p} & 0\\ 0 & \sigma\cdot\mathbf{p}\end{pmatrix} - \begin{pmatrix}\sigma\cdot\mathbf{p} & 0\\ 0 & \sigma\cdot\mathbf{p}\end{pmatrix}\begin{pmatrix}0 & \sigma\cdot\mathbf{p}\\ \sigma\cdot\mathbf{p} & 0\end{pmatrix} + 0\\ &= 0\end{aligned}$$

We thus see that this quantity indeed commutes with the Hamiltonian and hence we can measure the value of this operator simultaneously with the Hamiltonian. To provide a more physical measure, we define the *helicity* operator as:

$$\hat{h} := \frac{\hbar}{2}\frac{\mathbf{\Sigma}\cdot\mathbf{p}}{|\mathbf{p}|} \tag{8}$$

Supose our momentum is along the z-direction, then

$$\hat{h} = \frac{\hbar}{2}\Sigma^3 = \frac{\hbar}{2}\begin{pmatrix}1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1\end{pmatrix}$$

which has eigenvalues ± 1. Then, applying this operator on each of the four solutions, we will get eigenvalues ± 1 for positive and ± 1 for the negative solutions and thus the degeneracy is lifted.

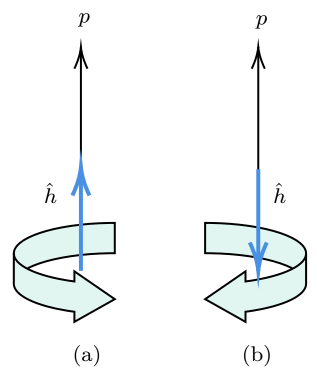


Figure 6: Fig (a) describes when $\hat{h}$ is along the momentum direction, giving positive helicity or right-handedness while Fig (b) describes when $\hat{h}$ is anti-parallel to the momenutm direction, giving negative helicity or left-handedness.

It can be shown that helicity is not Lorentz invariant and this is problematic. Only for massless particles, helicity becomes Lorentz invariant.

18.3. Properties of Dirac Matrices

We first note that,

$$(\gamma^0)^2 = \mathbb{1} \qquad (\gamma^i)^2 = -\mathbb{1}$$

Proof. We have $\gamma^0\gamma^0 = \beta^2 = \mathbb{1}$ and $(\gamma^i)^2 = \beta\alpha_i\beta\alpha_i = -\alpha_i\beta^2\alpha_i = -\alpha_i^2 = -\mathbb{1}$

This is a nice property with which we can differentiate between the spatial and temporal indices. Indeed, the Dirac matrices are somewhat used in relativity too, to describe spacetime.

It turns out that any 4×4 matrix can be written in terms of some combinations of Dirac matrices. In other words, the Dirac matrices form a basis of four dimensional square matrices. Note that, the obvious choice $\mathbb{1}$ and γ^μ cannot generate all the 4×4 matrices since we need 16 elements in the basis. The other elements in the basis are formed by multiplying the Dirac matrices:

- Two at a time, $\gamma^\mu\gamma^\nu$ ...... $\binom{4}{2} = 6$ matrices
- Three at a time, $\gamma^\mu\gamma^\nu\gamma^\alpha$ ...... $\binom{4}{3} = 4$ matrices
- Four at a time, $\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta$ ...... $\binom{4}{4} = 1$ matrix

Thus, we have a total of $1+4+6+4+1 = 16$ elements for the basis and we are happy! If we define,

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

Then the basis set can be written in terms of the 'fifth' Dirac matrix:

$$\Gamma = \{\mathbb{1}, \gamma^\mu, \Sigma^{\mu\nu}, \gamma^5, \gamma^5\gamma^\mu\}$$

Well, these matrices form what is called the *Clifford Algebra*, a fairly complicated mathematical concept. However, the crux of that is the *anti-commutation* relations satisfied by the Dirac matrices. Like, γ^μ's form the Clifford algebra in four dimensions while γ^μ along with γ^5 forms the Clifford algebra in five dimensions. There are higher-dimensional generalisations of the gamma matrices for this purpose.

Having claimed that the Dirac matrices form a basis of 4×4 matrices, have to prove that it is indeed a basis. For this, we need to show the linear independence of each basis element. Also, it would be nice if these basis elements somehow turn out to be 'orthogonal', since it is always convenient to work with orthonormal basis.

For the space of matrices, the most obvious choice to define orthognality is through the trace. That is, matrices A and B are orthogonal if $\mathrm{Tr}(AB) = 0$ [1]. Thus, it becomes necessary to work out the traces of the Dirac matrices.

Lecture 19: Trace and Contraction Identities

In the previous section, we saw that calculating the trace of the Dirac matrices become really important since these define the inner products. Let us look at some of the trace identities followed by the Dirac matrices.

19.1. Trace Identitites

- $\mathrm{Tr}(\gamma^\mu) = 0$

  Proof. We defined $\gamma^0 = \beta$ which was already traceless. From the anti-commutation relation, we know that $\beta\alpha_i = -\alpha_i\beta$ and taking trace on both sides, we get $\mathrm{Tr}\,\gamma^i = 0$

- $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{1}$

  Proof. Let us take the spatial indices first. From the definition of α_i and β, we have the following:

$$\gamma^i\gamma^j + \gamma^j\gamma^i = \beta(\alpha_i\beta)\alpha_j + i \leftrightarrow j = -\beta(\beta\alpha_i)\alpha_j + i \leftrightarrow j = -(\alpha_i\alpha_j + \alpha_j\alpha_i) = -2\delta_{ij}\mathbb{1} = 2\eta^{ij}\mathbb{1}$$

[1] More specifically, we should have used $\mathrm{Tr}(A^\dagger B)$

Now, consider one temporal index, that is,

$$\{\gamma^0, \gamma^i\} = \beta^2\alpha_i + \beta\alpha_i\beta = \alpha_i - \alpha_i\beta^2 = 0$$

Finally, we have $\{\gamma^0, \gamma^0\} = 2\beta^2 = 2\mathbb{1} = 2\eta^{00}\mathbb{1}$. Combining the above, we have the identity.

If we instead take the anti-commutation of the covariant indices, then also the identity remains similar, that is,

$$\{\gamma_\mu, \gamma_\nu\} = 2\eta_{\mu\nu}\mathbb{1}$$

- $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$

  Proof. From the anti-commutation relation of the Dirac matrices, we have:

  $$2\,\mathrm{Tr}(\gamma^\mu\gamma^\nu) = \mathrm{Tr}\{\gamma^\mu, \gamma^\nu\} = \mathrm{Tr}(2\eta^{\mu\nu}\mathbb{1}) = 2\eta^{\mu\nu} \times 4 \implies \mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$$

- $(\gamma^5)^2 = \mathbb{1}$

  Proof. From the definition of γ^5 and the anti-commutation between the gamma matrices, we have,

  $$\gamma^5\gamma^5 = -(\gamma^0\gamma^1\gamma^2\gamma^3)(\gamma^0\gamma^1\gamma^2\gamma^3) = \gamma^1\gamma^2\gamma^3(\gamma^0)^2\gamma^1\gamma^2\gamma^3 = \gamma^1\gamma^2\gamma^3\gamma^1\gamma^2\gamma^3 = \gamma^2\gamma^3(\gamma^1)^2\gamma^2\gamma^3$$
  $$\implies \gamma^5\gamma^5 = -\gamma^2\gamma^3\gamma^2\gamma^3 = \gamma^3(\gamma^2)^2\gamma^3 = -(\gamma^3)^2 = \mathbb{1}$$

- γ^5 is Hermitian.

  Proof. We will use the fact that $\left(\gamma^i\right)^\dagger = -\gamma^i$ for $i = 1, 2, 3$ and $\left(\gamma^0\right)^\dagger = \gamma^0$ along with the anti-commutation relation.

  $$(\gamma^5)^\dagger = -i\left(\gamma^3\right)^\dagger\left(\gamma^2\right)^\dagger\left(\gamma^1\right)^\dagger\left(\gamma^0\right)^\dagger = i\gamma^3\gamma^2\gamma^1\gamma^0$$

  To bring γ^0 to correct position (in front), we need 3 flips, for γ^1 we need 2 flips and for γ^2 we need only one flip, hence total $3 + 2 + 1 = 6$ flips. Each flip introduces a negative sign, so total $(-1)^6 = 1$, hence $(\gamma^5)^\dagger = i\gamma^0\gamma^1\gamma^2\gamma^3 = \gamma^5$.

- $\{\gamma^5, \gamma^\mu\} = 0$

  Proof. Note that, in $\gamma^5\gamma^\mu = i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^\mu$, for exchanging each $\mu \neq \nu$, we introduce a -1. Now, since there are 4 indices, μ must be one of these four indices and hence, we have the same index together, we do not need the -1. Thus, we introduce exactly $(-1)^3 = -1$ while transporting γ^μ to the front. Then we have $\gamma^5\gamma^\mu = -\gamma^\mu\gamma^5$, thus proving the identity.

- $\mathrm{Tr}(\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_{2n+1}}) = 0$

  Proof. As $(\gamma^5)^2 = \mathbb{1}$, let us multiply γ^5 to the above expression,

  $$\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_{2n+1}} = \gamma^5\gamma^5\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_{2n+1}}$$

  Shifting one of the γ^5 to the far right across the $2n + 1$ indices will introduce $(-1)^{2n+1}$ and thus, we get:

  $$\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_{2n+1}} = (-1)^{2n+1}\gamma^5\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_{2n+1}}\gamma^5$$

  Taking trace both sides and taking the cyclic property of the trace, we obtain the following. Thus, the product of odd number of gamma matrices is zero.

- $\mathrm{Tr}\left(\gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu\right) = 4\eta^{\kappa\lambda}\eta^{\mu\nu} - 4\eta^{\kappa\mu}\eta^{\lambda\nu} + 4\eta^{\kappa\nu}\eta^{\lambda\mu}$

Proof. Using the anti-commutation relation, we note the following:

$$\begin{aligned}\{\gamma^\kappa, \gamma^\lambda\gamma^\mu\gamma^\nu\} &= \gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu + \gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\kappa \\ &= \gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu + \gamma^\lambda\gamma^\kappa\gamma^\mu\gamma^\nu - \gamma^\lambda\gamma^\kappa\gamma^\mu\gamma^\nu + \gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\kappa \\ &= \{\gamma^\kappa, \gamma^\lambda\}\gamma^\mu\gamma^\nu - \gamma^\lambda\gamma^\kappa\gamma^\mu\gamma^\nu - \gamma^\lambda\gamma^\mu\gamma^\kappa\gamma^\nu + \gamma^\lambda\gamma^\mu\gamma^\kappa\gamma^\nu + \gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\kappa \\ &= \{\gamma^\kappa, \gamma^\lambda\}\gamma^\mu\gamma^\nu - \gamma^\lambda\{\gamma^\kappa, \gamma^\mu\}\gamma^\nu + \gamma^\lambda\gamma^\mu\{\gamma^\kappa, \gamma^\nu\} \\ &= 2\eta^{\kappa\lambda}\gamma^\mu\gamma^\nu - 2\eta^{\kappa\mu}\gamma^\lambda\gamma^\nu + 2\eta^{\kappa\nu}\gamma^\lambda\gamma^\mu\end{aligned}$$

Now, note that the required quantity is basically half the trace of the above anti-commutator.

$$\begin{aligned}\operatorname{Tr}\left(\gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu\right) = \frac{1}{2}\operatorname{Tr}\{\gamma^\kappa, \gamma^\lambda\gamma^\mu\gamma^\nu\} &= \operatorname{Tr}\left(\eta^{\kappa\lambda}\gamma^\mu\gamma^\nu - \eta^{\kappa\mu}\gamma^\lambda\gamma^\nu + \eta^{\kappa\nu}\gamma^\lambda\gamma^\mu\right) \\ &= \eta^{\kappa\lambda}\operatorname{Tr}\gamma^\mu\gamma^\nu - \eta^{\kappa\mu}\operatorname{Tr}\gamma^\lambda\gamma^\nu + \eta^{\kappa\nu}\operatorname{Tr}\gamma^\lambda\gamma^\mu \\ &= 4\eta^{\kappa\lambda}\eta^{\mu\nu} - 4\eta^{\kappa\mu}\eta^{\lambda\nu} + 4\eta^{\kappa\nu}\eta^{\lambda\mu}\end{aligned}$$

- $\operatorname{Tr}\gamma^5 = 0$

  Proof. Since the Minkowski metric is entire diagonal, $\eta^{ij} = 0$ for $i \neq j$ from which the identity follows using the previous identity.

- $\operatorname{Tr}\left(\gamma^5\gamma^\mu\right) = 0$

  Proof. Note that $\mu \in \{0, 1, 2, 3\}$ and γ^5 contains of all these indices. So, we can bring γ^μ adjacent to the same index in γ^5 by some flips and then it will give $(\gamma^\mu)^2$ which gives ± 1 atmost and the remaining will be a product of odd number of gamma matrices whose trace we had found out to be zero.

- $\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\right) = 0$

  Proof. We will use the fact the γ^5 anti-commutes with every γ^μ. Let us introduce two factors of γ^α in the trace such that $\alpha \neq \mu, \nu$. Note that $(\gamma^\alpha)^2$ will be 1 if $\alpha = 0$, else, it will be -1. In any case, the trace will change by ± 1

$$\begin{aligned}\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\right) = (-1)^{1-\delta_{\alpha,0}}\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\alpha\right) &= (-1)^{1-\delta_{\alpha,0}}(-1)^3\operatorname{Tr}\left(\gamma^\alpha\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\right) \\ &= -(-1)^{1-\delta_{\alpha,0}}\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\alpha\right) \\ &= -\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\right) \\ \implies \operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\right) = 0\end{aligned}$$

  In the last step, we had used the cyclic property of the trace to bring γ^α to the back again.

- $\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta = -4i\epsilon^{\mu\nu\alpha\beta}$

  Proof. If any two of μ, ν, α, β are same, we could bring them together by flipping, introducing atmost ± 1. The remaining will be the product of γ^5 and two other gamma matrices, whose trace we had already shown to be zero.

  Thus, let us assume that all the indices are different. Then in the product $\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta$, each gamma will appear twice. Hence we flip them such that all same index gamma appear together which would give us one of $\pm\mathbb{1}$, hence the trace will be of the form $\pm i\operatorname{Tr}\mathbb{1} = \pm 4i$.

  Suppose μ, ν, α, β appear as an even permutation of $0, 1, 2, 3$. Then we have,

$$\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\right) = -i\operatorname{Tr}\left(\gamma^5 \times i\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\right)$$

  Now the red factor can be brought to the form γ^5 using an even number of exchanges. Since we saw that $(\gamma^5)^2 = \mathbb{1}$, we have $\operatorname{Tr}\left(\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\right) = -4i$.

If μ, ν, α, β appear as an odd permutation of $0, 1, 2, 3$, then the same logic follows, however, the red factor introduces a negative sign since in this case, by an odd number of exchanges, the factor can be brought to the form of γ^5. Hence the trace will give $4i$.

As a summary, we saw that if any two of the four indices are equal, the trace is zero. If indices are even permutation of $0, 1, 2, 3$ then trace $\rightarrow\ -4i$ else if odd permutation, then trace $\rightarrow\ 4i$. This is essentially the same statement as the contravariant Levi-Civita tensor defined as,

$$\epsilon^{\alpha\beta\kappa\lambda} = \begin{cases} +1 & \alpha\beta\kappa\lambda \rightarrow \text{ even-permutation of } 0,1,2,3 \\ -1 & \alpha\beta\kappa\lambda \rightarrow \text{ odd-permutation of } 0,1,2,3 \\ 0 & \text{else} \end{cases}$$

Thus, we prove the identity.

19.2. Contraction Identities

Let us now look at some contraction identities where the Dirac matrices will be contracted. Unlike the trace identitities which gave a number as a final answer, these will give us some matrices.

- $\gamma^\mu \gamma_\mu = 4\mathbb{1}$

  Proof. Since Dirac matrices can be treated as four vectors, we can lower and raise their indices with the metric η. Then,

$$\begin{aligned} \gamma^\mu \gamma_\mu &= \eta_{\mu\nu} \gamma^\nu \gamma^\mu \\ &= \frac{1}{2}(\eta_{\mu\nu} + \eta_{\nu\mu})\gamma^\nu\gamma^\mu \qquad \text{(as metric is symmetric)} \\ &= \frac{1}{2}(\eta_{\mu\nu}\gamma^\nu\gamma^\mu + \eta_{\nu\mu}\gamma^\nu\gamma^\mu) \\ &= \frac{1}{2}(\eta_{\mu\nu}\gamma^\nu\gamma^\mu + \eta_{\mu\nu}\gamma^\mu\gamma^\nu) \qquad \text{(relabelling dummy indices)} \\ &= \frac{1}{2}\eta_{\mu\nu}(\gamma^\nu\gamma^\mu + \gamma^\mu\gamma^\nu) \\ &= \frac{1}{2}\eta_{\mu\nu} \times 2\eta^{\mu\nu}\mathbb{1} \qquad \text{(from anti-commutation of gamma matrices)} \\ &= \delta^\mu{}_\mu \mathbb{1} \\ &= 4\mathbb{1} \end{aligned}$$

- $\gamma^\mu \gamma^\nu \gamma_\mu = -2\gamma^\nu$

  Proof. Using the anti-commutation relation and the previous identity, we will derive this.

$$\gamma^\mu\gamma^\nu\gamma_\mu = \gamma^\mu\gamma^\nu\eta_{\mu\beta}\gamma^\beta = \gamma^\mu\eta_{\mu\beta}\Big(2\eta^{\nu\beta} - \gamma^\beta\gamma^\nu\Big) = 2\gamma^\mu\eta_{\mu\beta}\eta^{\nu\beta} - \gamma^\mu\gamma^\beta\eta_{\mu\beta}\gamma^\nu = 2\gamma^\mu\delta^\nu{}_\mu - \underbrace{\gamma^\mu\gamma_\mu}_{4\mathbb{1}}\gamma^\nu = -2\gamma^\nu$$

- $\gamma^\mu \gamma_\nu \gamma_\lambda \gamma_\mu = 4\eta_{\lambda\nu}\mathbb{1}$

  Proof. We will use the anti-commutation relation and the previous identity together to reduce the expression.

$$\begin{aligned} \gamma^\mu\gamma_\nu\gamma_\lambda\gamma_\mu = \gamma^\mu\gamma_\nu(2\eta_{\lambda\mu} - \gamma_\mu\gamma_\lambda) = 2\eta_{\lambda\mu}\gamma^\mu\gamma_\nu - \gamma^\mu\gamma_\nu\gamma_\mu\gamma_\lambda = 2\gamma_\lambda\gamma_\nu - \eta_{\alpha\nu}\underbrace{\gamma^\mu\gamma^\alpha\gamma_\mu}_{-2\gamma^\alpha}\gamma_\lambda &= 2(\gamma_\lambda\gamma_\nu + \gamma_\nu\gamma_\lambda) \\ &= 4\eta_{\lambda\nu}\mathbb{1} \end{aligned}$$

- $\gamma^\mu \gamma_\nu \gamma_\lambda \gamma_\rho \gamma_\mu = -2\gamma_\rho\gamma_\lambda\gamma_\nu$

Proof. We will again use the anti-commutation relation and the previous identity for this.

$$\gamma^\mu\gamma_\nu\gamma_\lambda\gamma_\rho\gamma_\mu = \gamma^\mu\gamma_\nu\gamma_\lambda(2\eta_{\rho\mu} - \gamma_\mu\gamma_\rho) = 2\gamma_\rho\gamma_\nu\gamma_\lambda - \underbrace{\gamma^\mu\gamma_\nu\gamma_\lambda\gamma_\mu}_{4\eta_{\lambda\nu}\mathbb{1}}\gamma_\rho = 2\gamma_\rho(\gamma_\nu\gamma_\lambda - 2\eta_{\lambda\nu}) = -2\gamma_\rho \times \underbrace{(2\eta_{\lambda\nu} - \gamma_\nu\gamma_\lambda)}_{\gamma_{\lambda\nu}}$$
$$= -2\gamma_\rho\gamma_\lambda\gamma_\nu$$

These are few of the contraction and trace identities that we checked. There exists many such other identities and we can just play with the Dirac matrices to find them 😉!

19.3. Orthonormal Basis

In the Lec 18, we had stated a basis set for the set of 4×4 matrices, however, we have not explicitly shown that it is a basis. Since we are equipped with the traces (which form the inner product for the space) we are now in a position to show that. To show that the set is an orthonormal basis, we need to show two things:

- Linear independence of the basis elements.
- orthognality of the basis elements with respect to the inner product.

Let us start by showing the orthognality. For that, we should painstakingly calculate the traces between the elements in the basis set. However, calculation for only $\binom{5}{2} + 5 = 15$ traces need to be shown as trace is symmetric.

- Term $\longrightarrow \mathrm{Tr}(\mathbb{1}\cdot\Gamma^\alpha)$ where $\Gamma^\alpha \in \Gamma$

  We have $\mathrm{Tr}(\mathbb{1}\cdot\mathbb{1}) = 4$ obviously. From the above derived trace relations, we know $\mathrm{Tr}(\mathbb{1}\cdot\gamma^5) = 0, \mathrm{Tr}(\mathbb{1}\cdot\gamma^\mu\gamma^5) = 0, \mathrm{Tr}(\mathbb{1}\cdot\gamma^\mu) = 0$ and

$$\mathrm{Tr}(\mathbb{1}\cdot\Sigma^\mu\gamma^\nu) = \frac{i}{2}(4\eta^{\mu\nu} - 4\eta^{\nu\mu}) = 0$$

- Term $\longrightarrow \mathrm{Tr}(\gamma^\mu\cdot\Gamma^\alpha)$ where $\Gamma^\alpha \in \Gamma$

  From the trace identity, we have $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$ and $\mathrm{Tr}(\gamma^\mu) = 0$. $\mathrm{Tr}(\gamma^\mu\gamma^5) = 0$ as this is a product of five gamma matrices. $\mathrm{Tr}(\gamma^\mu\Sigma^{\nu\alpha}) = 0$ since it contains the product of three gamma matrices. $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^5) = 0$ from the trace identity.

- Term $\longrightarrow \mathrm{Tr}(\gamma^5\cdot\Gamma^\alpha)$ where $\Gamma^\alpha \in \Gamma$

  We have $\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^5) = -\mathrm{Tr}((\gamma^5)^2\gamma^\mu) = 0, \mathrm{Tr}(\gamma^5\Sigma^{\mu\nu}) = 0$ as $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^5) = 0$ and $\mathrm{Tr}((\gamma^5)^2) = 4$

- Term $\longrightarrow \mathrm{Tr}(\gamma^\mu\gamma^5\cdot\Gamma^\alpha)$ where $\Gamma^\alpha \in \Gamma$

  We have $\mathrm{Tr}(\gamma^\mu\gamma^5\gamma^\nu\gamma^5) = -\mathrm{Tr}(\gamma^\mu\gamma^\nu) = -4\eta^{\mu\nu}$ and $\mathrm{Tr}(\gamma^\mu\gamma^5\Sigma^{\alpha\rho}) = 0$ as product of seven gamma matrices.

- Term $\longrightarrow \mathrm{Tr}(\Sigma^{\alpha\beta}\cdot\Gamma^\alpha)$ where $\Gamma^\alpha \in \Gamma$

  As an exercise, let us calculate this trace explicitly from the definition itself.

$$\mathrm{Tr}\,\Sigma^{\alpha\beta}\Sigma^{\mu\nu} = -\frac{1}{4}\mathrm{Tr}\Big((\gamma^\alpha\gamma^\beta - \gamma^\beta\gamma^\alpha)(\gamma^\mu\gamma^\nu - \gamma^\beta\gamma^\mu)\Big) = -\frac{1}{4}\mathrm{Tr}\Big(\gamma^\alpha\gamma^\beta\gamma^\mu\gamma^\nu\Big) + \frac{1}{4}\mathrm{Tr}\Big(\gamma^\alpha\gamma^\beta\gamma^\nu\gamma^\mu\Big)$$
$$+\frac{1}{4}\mathrm{Tr}\Big(\gamma^\beta\gamma^\alpha\gamma^\mu\gamma^\nu\Big) - \frac{1}{4}\mathrm{Tr}\Big(\gamma^\beta\gamma^\alpha\gamma^\nu\gamma^\mu\Big)$$

  Using the trace identities, this elongates to:

$$-(\eta^{\alpha\beta}\eta^{\mu\nu} - \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu}) + (\eta^{\alpha\beta}\eta^{\nu\mu} - \eta^{\alpha\nu}\eta^{\beta\mu} + \eta^{\alpha\mu}\eta^{\beta\nu})$$
$$+ (\eta^{\beta\alpha}\eta^{\mu\nu} - \eta^{\beta\mu}\eta^{\alpha\nu} + \eta^{\beta\nu}\eta^{\alpha\mu}) - (\eta^{\beta\alpha}\eta^{\nu\mu} - \eta^{\beta\nu}\eta^{\alpha\mu} + \eta^{\beta\mu}\eta^{\alpha\nu})$$

  Note that there are four of each blue and green colored terms while the red terms cancel and hence the trace becomes:

$$\mathrm{Tr}\,\Sigma^{\alpha\beta}\Sigma^{\mu\nu} = 4(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\nu}\eta^{\beta\mu})$$

We thus see that for $\Gamma^a, \Gamma^b \in \Gamma$, $\mathrm{Tr}\left(\Gamma^a\Gamma^b\right) \neq 0$ only when $\Gamma^a = \Gamma^b$ where the equality means that both are same kind of matrices in the Clifford algebra basis. We want to find some kind of definite normalisation which does not depend on the sign of the trace. For that, we will change the basis element $\gamma^\mu\gamma^5 \to i\gamma^\mu\gamma^5$ and we will lower the index, Γ_a, by lowering all the indices of Γ^a using $\eta_{\mu\nu}$. Then, we will find that:

$$\mathrm{Tr}\left(\Gamma_a\Gamma^b\right) = 4\delta^b{}_a$$

Using this definition, the linear independence can be easily shown. Suppose we have:

$$\sum_a C_a\Gamma^a = 0$$

Multiplying the equation both sides with Γ^b and then taking the trace, only the term with C_b will survive on the LHS and will be zero. Thus, $C_a = 0\ \forall\ a$ which implies that the elements are linearly independent.

Since this forms a basis, we can expand any 4×4 matrix in terms of these elements. Thus,

$$M = \sum_a C_a^M\Gamma^a$$

Then multiplying by another element and taking trace, we get:

$$\mathrm{Tr}\,\Gamma_b M = \sum_a C_a^M\,\mathrm{Tr}\,\Gamma_b\Gamma^a = \sum_a 4C_a^M\delta^a{}_b = 4C_b^M \implies \boxed{C_b^M = \frac{1}{4}\,\mathrm{Tr}\,\Gamma_b M}$$

And hence the matrix is expanded as:

$$M = \sum_a \frac{1}{4}\,\mathrm{Tr}(\Gamma_a M)\ \Gamma^a \tag{9}$$

Now let us explicitly introduce the matrix indices which we have suppressed till now. Note that there are two kind of indices here, one is the a index which tells us what type of the basis element it is and now we introduce the matrix indices, which will run from $1, 2, 3, 4$. So, Γ^a_{12} actually means the $(1, 2)$ index of the matrix of type Γ^a. Using matrix indices, the above equation becomes:

$$M_{\alpha\beta} = \frac{1}{4}\sum_a(\Gamma_a)_{\eta\lambda}M_{\lambda\eta}\Gamma^a{}_{\alpha\beta} = \frac{1}{4}\sum_a\left\{(\Gamma_a)_{\eta\lambda}(\Gamma^a)_{\alpha\beta}\right\}M_{\lambda\eta} \implies \boxed{\frac{1}{4}\sum_a(\Gamma_a)_{\eta\lambda}(\Gamma^a)_{\alpha\beta} = \delta_{\alpha\lambda}\delta_{\beta\eta}} \tag{10}$$

This defines a *completeness relation* for these matrices and will be later used to derive some important identitites.

Lecture 20: Transformation of Spinors

To understand how spinors actually transform under Lorentz transformation, we first need to have an idea about what spinors are. Spinors are basically another kind of representation of the Lorentz group loosely speaking. There are different concepts of group and representation theory involved which lead to many different subtleties, however, we shall not digress much into those subtleties and obtain only those ideas which are absolutely needed.

20.1. Spinor Representation of Lorentz group

We know that any representation of the Lorentz group, say $\mathrm{S}[\Lambda]$ can be written in terms of the rotation generators $\mathbf{J}$ and the boost generators $\mathbf{K}$ (from A):

$$\mathrm{S}[\Lambda] = \exp\left(-i\,\boldsymbol{\theta}\cdot\mathbf{J} + i\boldsymbol{\eta}\cdot\mathbf{K}\right)$$

where $\boldsymbol{\theta}$ and $\boldsymbol{\eta}$ are the rotation and boost parameters. The normal boost and rotation generators are not closed under commutation, however if we define[1]:

$$\mathbf{J}^\pm := \frac{\mathbf{J} \pm i\mathbf{K}}{2}$$

[1]This is called *complexification*

Then, one can check that the Lorentz algebra becomes:

$$\begin{aligned}\left[J^{+,i}, J^{+,j}\right] &= i\varepsilon^{ijk} J^{+,k} \\ \left[J^{-,i}, J^{-,j}\right] &= \varepsilon^{ijk} J^{-,k} \\ \left[J^{+,i}, J^{-,j}\right] &= 0\end{aligned}$$

Now, this is familiar to the $SU(2)$ algebra that we had seen from, say the *angular momentum*. In fact, from the complexification of the generators we see two copies of the angular momentum algebra (so we can write $\mathfrak{so}(1,3) = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$) and we can thus label the representations by (j_-, j_+) where $j_\pm \in \{0, \frac{1}{2}, 1, \frac{3}{2} \ldots\}$. We will briefly focus on the most popular representations, namely:

- *Scalar representation* $\longrightarrow (0,0)$
- *Left-handed Weyl spinor representation* $\longrightarrow (1/2, 0)$
- *Right-handed Weyl spinor representation* $\longrightarrow (0, 1/2)$
- *Dirac spinor representation* $\longrightarrow (0, 1/2) \oplus (1/2, 0)$

The scalar representation is the trivial one-dimensional representation where $\mathbf{J}^\pm = 0 \implies \mathbf{J} = 0, \mathbf{K} = 0$ and hence is of not much interest to us. All but the Dirac representation are irreducible representations. Let us now focus on the Weyl representation.

Weyl Representation

Let us consider the left handed spinor representation. The left-handed spinor itself can be denoted by ψ_{L} while any representation of the Lorentz group can be represented by Λ_{L}. In this case, $\mathbf{J}^+ = 0$ while $\mathbf{J}^-$ resembles the representation of spin-half particles and can be taken to be $\mathbf{J}^- \to \frac{\boldsymbol{\sigma}}{2}$ where $\boldsymbol{\sigma}$ is the Pauli vector. Then from the definition,

$$\begin{aligned}\mathbf{J} &= \mathbf{J}^+ + \mathbf{J}^- = 0 + \frac{\boldsymbol{\sigma}}{2} = \frac{\boldsymbol{\sigma}}{2} \\ \mathbf{K} &= -i(\mathbf{J}^+ - \mathbf{J}^-) = i\frac{\boldsymbol{\sigma}}{2}\end{aligned}$$

Note that, the Pauli vector is Hermitian and then so is $\mathbf{J}$ but $\mathbf{K}$ is anti-Hermitian instead, so overall the Weyl representation is not a unitary representation. This is explained by the fact that the Lorentz group is not a *compact* group and by a theorem, any non-trivial finite dimensional irreducible representation cannot be unitary.

Thus, the representation of the Lorentz group and the transformation of a left-handed Weyl spinor becomes:

$$\mathrm{S}[\Lambda_{\mathrm{L}}] = \exp\left((-i\boldsymbol{\theta} - \boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) \qquad \psi_{\mathrm{L}} \to \Lambda_{\mathrm{L}} \psi_{\mathrm{L}}$$

Using a similar argument, we can see that the right-handed Weyl spinor is characterised by:

$$\mathrm{S}[\Lambda_{\mathrm{R}}] = \exp\left((-i\boldsymbol{\theta} + \boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) \qquad \psi_{\mathrm{R}} \to \Lambda_{\mathrm{R}} \psi_{\mathrm{R}}$$

Dirac representation

Now, let us come to the Dirac representation which is a reducible representation and can be written as[1]:

$$\Lambda_{\mathrm{D}} \equiv \begin{pmatrix} \Lambda_{\mathrm{L}} & \\ & \Lambda_{\mathrm{R}} \end{pmatrix} \qquad \Psi \to \Lambda_{\mathrm{D}} \Psi$$

$\triangleright$ Why is the Dirac representation even necessary when we already had the Weyl spinors?

Well, for that consider the parity transformation where $\mathbf{x} \to -\mathbf{x}$. In this case, $\mathbf{K} \to -\mathbf{K}$ since the

[1] We will denote Dirac spinors with capital letters, that is, Ψ while Weyl spinors with small letters, ψ to distinguish between them, since these are inequivalent representations. Of course, it might not always be possible to remember this and the notation might slip sometimes 😢

boost velocity is reversed, however, $\mathbf{J}$ remains same (and thus acts as a *pseudovector*) as intuitively, both position and velocity is reversed.

Thus, under parity $\mathbf{J}^{\pm} \to \mathbf{J}^{\mp}$ and an object in the (j_-, j_+) representation is changed to an object in the (j_+, j_-) representation which is different unless $j_+ = j_-$.

Then, the Weyl spinors (objects of kind $(0, 1/2)$ and $(1/2, 0)$) individually cannot be the basis for a representation of the parity transformation. In this case, it is convenient to work with fields which provide a representation of Lorentz and parity transformations together. Thus, we consider a Dirac spinor written in terms of two Weyl spinors[1],

$$\Psi = \begin{pmatrix} \psi_\mathrm{L} \\ \psi_\mathrm{R} \end{pmatrix}$$

which is a $(0, 1/2) \oplus (1/2, 0)$ object and conforms to parity transformations. Consider an infinitesimal transformation of the Dirac representation, which can be equivalently written as the infinitesimal transformation of the two Weyl representations.

$$\begin{aligned}
\exp\left((-i\boldsymbol{\theta} - \boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) &\approx \mathbb{1} + \left((-i\boldsymbol{\theta} - \boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) = \mathbb{1} - \frac{i}{2}\theta^i\sigma^i - \frac{1}{2}\eta^i\sigma^i \\
\exp\left((-i\boldsymbol{\theta} + \boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) &\approx \mathbb{1} + \left((-i\boldsymbol{\theta} + \boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) = \mathbb{1} - \frac{i}{2}\theta^i\sigma^i + \frac{1}{2}\eta^i\sigma^i
\end{aligned}$$

Substituting this in the Dirac representation, we obtain:

$$\begin{aligned}
\Lambda_\mathrm{D} &= \mathbb{1} - \frac{i}{2}\theta^i \begin{pmatrix} \sigma^i & \\ & \sigma^i \end{pmatrix} - \frac{1}{2}\eta^i \begin{pmatrix} \sigma^i & \\ & -\sigma^i \end{pmatrix} \\
&= \mathbb{1} - \frac{i}{4}\varepsilon^{ijk}\Omega_{jk} \begin{pmatrix} \sigma^i & \\ & \sigma^i \end{pmatrix} - \frac{1}{2}\Omega^{i0} \begin{pmatrix} \sigma^i & \\ & -\sigma^i \end{pmatrix} \\
&= \mathbb{1} - \frac{i}{4}\Omega_{jk}\Sigma^{jk} - \frac{i}{2}\Omega^{i0}\Sigma^{0i} \\
&= \mathbb{1} - \frac{i}{4}\Omega_{jk}\Sigma^{jk} + \frac{i}{2}\Omega_{i0}\Sigma^{0i} \\
&= \mathbb{1} - \frac{i}{4}\Omega_{jk}\Sigma^{jk} - \frac{i}{2}\Omega_{0i}\Sigma^{0i} \\
&= \mathbb{1} - \frac{i}{4}\Omega_{jk}\Sigma^{jk} - \frac{i}{4}\Omega_{0i}\Sigma^{0i} - \frac{i}{4}\Omega_{i0}\Sigma^{i0} \\
&= \mathbb{1} - \frac{i}{4}\Omega_{\rho\sigma}\Sigma^{\rho\sigma}
\end{aligned}$$

where we have used the form of the matrix $\Sigma^{\mu\nu}$ in the chiral basis[2] (since we are working with Weyl representation where the representations Λ act on the Weyl spinors, the chiral basis is an appropriate choice). We had previously found the form of Σ^{jk} in 18.2 which will not change in the chiral basis. We now just find the form of Σ^{0i}:

$$\begin{aligned}
-2i\Sigma^{0i} &= \beta\gamma^i - \gamma^i\beta \\
&= \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} - \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}\begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \\
&= \begin{pmatrix} -\sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix} - \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix} \\
&= -2\begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix}
\end{aligned}$$

[1] A subtlety is that this is only true for the *chiral basis* where γ^0 takes an off-diagonal form

[2] Only γ^0 has a different form in the chiral basis.

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}$$

Thus, we found an expression for the infinitesimal transformation of the spinor generators. Hence, any Dirac spinor representation of the Lorentz group can be found by exponentiating these generators:

$$\mathrm{S}[\Lambda] = \exp\left(-\frac{i}{4}\Omega_{\rho\sigma}\Sigma^{\rho\sigma}\right)$$

As specified earlier, $\mathrm{S}[\Lambda]$ cannot be unitary in general, since Lorentz group is not a compact group and as a result of which, the obvious choice for the inner product of a Dirac spinor is not *Lorentz invariance*. We can explicitly see this as,

$$\Psi'^{\dagger}\Psi' = \Psi^{\dagger}\ \underbrace{\mathrm{S}[\Lambda]^{\dagger}\mathrm{S}[\Lambda]}_{\neq\mathbb{1}}\ \Psi \neq \Psi^{\dagger}\Psi$$

20.2. Constructing Bilinears

We saw that the obvious notion of an *adjoint* is not Lorentz invariant which is problematic. We will now try to construct some tensorial quantitites out of the spinors in a different way, which will be Lorentz invariant/covariant. For that, we need to look at some properties.

Firstly note that γ^{i}'s are anti-Hermitian while γ^0 is Hermitian. Then we have:

$$\gamma^0\gamma^i\gamma^0 = (-\gamma^0)^2\gamma^i = -\gamma^i = \gamma^{i\dagger} \qquad \gamma^0\gamma^0\gamma^0 = \gamma^0 = \gamma^{0\dagger}$$

Hence we have the identity:

$$\boxed{\gamma^0\gamma^\mu\gamma^0 = \gamma^{\mu\dagger}} \tag{11}$$

Now, we will try to see how the generators transform under the adjoint. For that, note:

$$\begin{aligned}
\Sigma^{\mu\nu\dagger} &= \left(\frac{i}{2}\left[\gamma^\mu,\gamma^\nu\right]\right)^{\dagger} \\
&= -\frac{i}{2}\left[\gamma^{\nu\dagger},\gamma^{\mu\dagger}\right] \\
&= -\frac{i}{2}\{(\gamma^0\gamma^\nu\gamma^0)(\gamma^0\gamma^\mu\gamma^0) - (\gamma^0\gamma^\mu\gamma^0)(\gamma^0\gamma^\nu\gamma^0)\} \\
&= -\frac{i}{2}\gamma^0\{\gamma^\nu\gamma^\mu - \gamma^\mu\gamma^\nu\}\gamma^0 \\
&= -\frac{i}{2}\gamma^0\left[\gamma^\nu,\gamma^\mu\right]\gamma^0 \\
&= -\gamma^0\Sigma^{\nu\mu}\gamma^0 \\
&= \gamma^0\Sigma^{\mu\nu}\gamma^0
\end{aligned}$$

Now note the property of the generators:

$$\gamma^0\underbrace{\left(-\frac{i}{4}\Omega_{\rho\sigma}\Sigma^{\rho\sigma}\right)}_{\text{generator of } \mathrm{S}[\Lambda]}\gamma^0 = \frac{i}{4}\Omega_{\rho\sigma}\gamma^0\Sigma^{\rho\sigma\dagger}\gamma^0 = +\ \underbrace{\frac{i}{4}\Omega_{\rho\sigma}\Sigma^{\rho\sigma}}_{\text{generator of } \mathrm{S}^{-1}[\Lambda]} \qquad \text{from the previous property}$$

Translating this to the actual Lorentz representations, we obtain:

$$\gamma^0\ \mathrm{S}^{\dagger}[\Lambda]\ \gamma^0 = \mathrm{S}^{-1}[\Lambda]\ \longleftrightarrow \mathrm{S}^{\dagger}[\Lambda] = \gamma^0\ \mathrm{S}^{-1}[\Lambda]\ \gamma^0$$

From this, we can actually define a new kind of adjoint:

$$\overline{\Psi} := \Psi^{\dagger}\gamma^0$$

Let us now check how this quantity transforms under a Lorentz transformation

$$\overline{\Psi}(x) \longrightarrow \overline{\Psi}'(x') = (\mathrm{S}[\Lambda]\Psi(x))^{\dagger}\gamma^0 = \Psi^{\dagger}(x)\mathrm{S}^{\dagger}[\Lambda]\gamma^0 = \Psi^{\dagger}(x)\gamma^0\mathrm{S}^{-1}[\Lambda] = \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]$$

Hence this *Dirac adjoint* transforms with a copy of $\mathrm{S}^{-1}[\Lambda]$. Using these transformations, we can construction a few tensorial quantitites. For that, remember the basis Γ that we had constructed for 4×4 matrices. Let us construct what we call *bilinears*, as $\overline{\Psi}\ (\ \cdot\ )\ \Psi$. These are called so since these are linear in both Ψ and $\overline{\Psi}$.

- **Basis element**: $\mathbb{1}$

$$\overline{\Psi}\ \mathbb{1}\ \Psi = \overline{\Psi}\Psi \longrightarrow \overline{\Psi}'\Psi' = \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\ \mathrm{S}[\Lambda]\Psi = \overline{\Psi}\Psi$$

  Thus, this quantity is a Lorentz *scalar* since it is invariant under transformation. We denote it by $\mathcal{S}$.

- **Basis element**: γ^μ

$$\begin{aligned}\overline{\Psi}\ \gamma^\mu\ \Psi \longrightarrow \overline{\Psi}'\gamma^\mu\Psi' &= \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\gamma^\mu\mathrm{S}[\Lambda]\ \Psi\\ &= \overline{\Psi}\ \Lambda^\mu{}_\nu\gamma^\nu\ \Psi \quad \text{(using Eq. 7)}\\ &= \Lambda^\mu{}_\nu\big(\overline{\Psi}\ \gamma^\nu\ \Psi\big)\end{aligned}$$

  This quantity transform with one copy of $\Lambda^\mu{}_\nu$ and hence is a *contravariant vector*. We denote it by $\mathcal{V}$

- **Basis element**: $\Sigma^{\mu\nu}$

$$\begin{aligned}\overline{\Psi}\ \Sigma^{\mu\nu}\ \Psi \longrightarrow \overline{\Psi}'\ \Sigma^{\mu\nu}\ \Psi' &= \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\Sigma^{\mu\nu}\mathrm{S}[\Lambda]\ \Psi\\ &= \frac{i}{2}\overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\ [\gamma^\mu,\gamma^\nu]\ \mathrm{S}[\Lambda]\Psi\\ &= \frac{i}{2}\big\{\overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\gamma^\mu\gamma^\nu\mathrm{S}[\Lambda]\Psi - \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\gamma^\nu\gamma^\mu\mathrm{S}[\Lambda]\Psi\big\}\\ &= \frac{i}{2}\big\{\overline{\Psi}\ \big(\mathrm{S}^{-1}[\Lambda]\gamma^\mu\mathrm{S}[\Lambda]\big)\big(\mathrm{S}^{-1}[\Lambda]\gamma^\nu\mathrm{S}[\Lambda]\big)\Psi - \overline{\Psi}\ \big(\mathrm{S}^{-1}[\Lambda]\gamma^\nu\mathrm{S}[\Lambda]\big)\big(\mathrm{S}^{-1}[\Lambda]\gamma^\mu\mathrm{S}[\Lambda]\big)\Psi\big\}\\ &= \frac{i}{2}\Big\{\overline{\Psi}\Big(\Lambda^\mu{}_\alpha\gamma^\alpha\Lambda^\nu{}_\beta\gamma^\beta - \Lambda^\nu{}_\beta\gamma^\beta\Lambda^\mu{}_\alpha\gamma^\alpha\Big)\Psi\Big\}\\ &= \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\frac{i}{2}\overline{\Psi}\ \Big[\gamma^\alpha,\gamma^\beta\Big]\ \Psi\\ &= \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Big(\overline{\Psi}\ \Sigma^{\alpha\beta}\ \Psi\Big)\end{aligned}$$

  This quantity transforms with two copies of Λ and hence, acts as a rank-2 contravariant tensor. We denote it by $\mathcal{T}^{\mu\nu}$

- **Basis element**: γ^5

  γ^5 is defined in the following way using the Levi-Civita tensor:

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = -\frac{i}{4!}\varepsilon_{\alpha\beta\lambda\delta}\gamma^\alpha\gamma^\beta\gamma^\lambda\gamma^\delta$$

$$\overline{\Psi}\ \gamma^5\ \Psi \Longrightarrow \overline{\Psi}'\ \gamma^5\ \Psi' = \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\gamma^5\mathrm{S}[\Lambda]\ \Psi = -\frac{i}{4!}\varepsilon_{\alpha\beta\lambda\delta}\overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\ \gamma^\alpha\gamma^\beta\gamma^\lambda\gamma^\delta\mathrm{S}[\Lambda]\ \Psi$$

  Now, what we did earlier, by introducing $\mathrm{S}[\Lambda]\mathrm{S}^{-1}[\Lambda]$ between each Dirac matrices, we can introduce a copy of Λ. Doing this, we obtain:

$$\overline{\Psi}'\ \gamma^5\ \Psi' = -\frac{i}{4!}\ \overline{\Psi}\varepsilon_{\alpha\beta\lambda\delta}\Lambda^\alpha{}_\mu\Lambda^\beta{}_\nu\Lambda^\lambda{}_\rho\Lambda^\delta{}_\theta\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\theta\ \Psi$$

Fun Fact:
The determinant of a square $n \times n$ matrix can be written in terms of the Levi-Civita tensor as:

$$\det(A) = \frac{1}{n!}\varepsilon_{\mu_1\mu_2\ldots\mu_n}\varepsilon^{\nu_1\nu_2\ldots\nu_n} A^{\mu}{}_{1\nu 1}A^{\mu}{}_{2\nu 2}\ldots A^{\mu}{}_{n\nu n}$$

Contracting the above equation by $\varepsilon_{\nu_1\nu_2\ldots\nu_n}$ we obtain:

$$\varepsilon_{\nu_1\nu_2\ldots\nu_n}\det(A) = \varepsilon_{\mu_1\mu_2\ldots\mu_n}A^{\mu_1}{}_{\nu_1}A^{\mu_2}{}_{\nu_2}\ldots A^{\mu_n}{}_{\nu_n}$$

Using the above fact for Λ, we get:

$$\begin{aligned}\overline{\Psi}'\ \gamma^5\ \Psi' &= -\frac{i}{4!}\overline{\Psi}\ \varepsilon_{\mu\nu\rho\theta}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\theta \det(\Lambda)\ \Psi \\ &= \det(\Lambda)(\overline{\Psi}\ \gamma^5\ \Psi)\end{aligned}$$

This quantity changes almost like a scalar but with the determinant of the transformation. For proper Lorentz element, $\det(\Lambda) = 1$ and hence this is invariant, however, a minus sign is introduced for improper Lorentz transformation. Hence this is instead called a *pseudo-scalar* and is denoted by $\mathcal{P}$

- **Basis element**: $\gamma^\mu\gamma^5$

$$\begin{aligned}\overline{\Psi}\ \gamma^\mu\gamma^5\ \Psi \Longrightarrow \overline{\Psi}'\ \gamma^\mu\gamma^5\ \Psi' = \overline{\Psi}\ \mathrm{S}^{-1}[\Lambda]\gamma^\mu\gamma^5\mathrm{S}[\Lambda]\ \Psi &= \overline{\Psi}\ (\mathrm{S}^{-1}[\Lambda]\gamma^\mu\mathrm{S}[\Lambda])(\mathrm{S}^{-1}[\Lambda]\gamma^5\mathrm{S}[\Lambda])\ \Psi \\ &= \overline{\Psi}\ \Lambda^\mu{}_\nu\gamma^\nu \times \det(\Lambda)\gamma^5\ \Psi \\ &= \det(\Lambda)\Lambda^\mu{}_\nu(\overline{\Psi}\ \gamma^\nu\gamma^5\ \Psi)\end{aligned}$$

This quantity transforms very close to a rank-2 tensor but with the determinant of the transformation and hence changes sign under improper Lorentz transformation. Hence it is called a *pseudo-vector* or *axial vector* and is denoted by $\mathcal{A}$

These five bilinears, corresponding to each element of the basis of 4×4 matrices, is thus useful in constructing general quantities transforming properly under the Lorentz transformation.

Lecture 21: Fierz Identities

In the previous section we saw how different *bilinears* can be created through the use of the basis elements of the Clifford algebra. We also saw the transformation of these quantities and accordingly specified names to them viz. scalars, pseudoscalars, vector, axial vector and tensor. In this section, we will find an identity that will allow us to rearrange the bilinears of the product of two spinors in a different ordering. These can be useful to write a given interaction in some physically meaningful form.

21.1. Outer product of Spinors

Let us consider two Dirac spinors Ψ_1 and Ψ_2. We previously considered terms of type $\overline{\Psi}_1\Psi_2 = \overline{\Psi}_1\ \gamma^0\ \Psi_2$ which is essentially the product of a row vector and a column vector, producing a number .

Let us now consider the *outer product* between them which will result in some matrix. Since the Clifford algebra produces a basis of 4×4 matrices, we can write the outer product as a linear combination of the elements of the basis set Γ. We thus have:

$$\Psi_2\overline{\Psi}_1 = a\mathbb{1} + b_\mu\gamma^\mu + c_{\mu\nu}\Sigma^{\mu\nu} + d_\mu\gamma^\mu\gamma^5 + f\gamma^5$$

We just need to find these coefficients. The general technique will be to multiple the above equation with some element $\Gamma^a \in \Gamma$ and then take the trace. Since the basis elements are orthogonal, the trace will give zero for dissimilar elements and only contribution is from the same basis element. Let us start!

- Multiply by $\mathbb{1}$ and take the trace:

$$\mathrm{Tr}\big(\mathbb{1}\Psi_2\overline{\Psi}_1\big) = a\,\mathrm{Tr}(\mathbb{1}) = 4a \implies a = \frac{1}{4}\mathrm{Tr}\big(\Psi_2\overline{\Psi}_1\big) = \frac{1}{4}\overline{\Psi}_1\Psi_2 \longrightarrow \frac{1}{4}\mathcal{S}_{12}$$

where $\mathcal{S}_{12}$ is the notation to denote the scalar bilinear with spinors Ψ_1 and Ψ_2.

- Multiply by γ^α and take the trace:

$$\mathrm{Tr}\big(\gamma^\alpha\Psi_2\overline{\Psi}_1\big) = b_\mu\,\mathrm{Tr}(\gamma^\alpha\gamma^\mu) = 4\eta^{\alpha\mu}b_\mu \implies b^\alpha = \frac{1}{4}\mathrm{Tr}\big(\gamma^\alpha\Psi_2\overline{\Psi}_1\big) = \frac{1}{4}\overline{\Psi}_1\ \gamma^\alpha\ \Psi_2 \longrightarrow \frac{1}{4}\mathcal{V}_{12}^\alpha$$

- Multiply by γ^5 and take the trace:

$$\mathrm{Tr}\big(\gamma^5\Psi_2\overline{\Psi}_1\big) = f\,\mathrm{Tr}\big((\gamma^5)^2\big) = 4f \implies f = \frac{1}{4}\mathrm{Tr}\big(\gamma^5\Psi_2\overline{\Psi}_1\big) = \frac{1}{4}\overline{\Psi}_1\ \gamma^5\ \Psi_2 \longrightarrow \frac{1}{4}\mathcal{P}_{12}$$

- Multiply by $\gamma^\alpha\gamma^5$ and take the trace:

$$\begin{aligned}\mathrm{Tr}\big(\gamma^\alpha\gamma^5\Psi_2\overline{\Psi}_1\big) &= d_\mu\,\mathrm{Tr}\big(\gamma^\alpha\gamma^5\gamma^\mu\gamma^5\big) = -d_\mu\,\mathrm{Tr}\big(\gamma^\alpha\gamma^\mu(\gamma^5)^2\big) = -d_\mu\,\mathrm{Tr}(\gamma^\alpha\gamma^\mu) = -4\eta^{\alpha\mu}d_\mu\\ \implies d^\alpha &= -\frac{1}{4}\mathrm{Tr}\big(\gamma^\alpha\gamma^5\Psi_2\overline{\Psi}_1\big) = -\frac{1}{4}\overline{\Psi}_1\ \gamma^\alpha\gamma^5\ \Psi_2 \longrightarrow -\frac{1}{4}\mathcal{A}_{12}^\alpha\end{aligned}$$

- Multiply by $\Sigma^{\alpha\beta}$ and take the trace:

$$\mathrm{Tr}\Big(\Sigma^{\alpha\beta}\Psi_2\overline{\Psi}_1\Big) = c_{\mu\nu}\,\mathrm{Tr}\Big(\Sigma^{\alpha\beta}\Sigma^{\mu\nu}\Big)$$

From Lec 19, we have

$$\mathrm{Tr}\,\Sigma^{\alpha\beta}\Sigma^{\mu\nu} = 4(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\nu}\eta^{\beta\mu})$$

Substituting this in the actual term, we obtain:

$$\mathrm{Tr}\Big(\Sigma^{\alpha\beta}\Psi_2\overline{\Psi}_1\Big) = 4c_{\mu\nu}\Big(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\nu}\eta^{\beta\mu}\Big) = 4(c^{\alpha\beta} - c^{\beta\alpha}) = 8c^{\alpha\beta} \implies c^{\alpha\beta} = \frac{1}{8}\overline{\Psi}_1\ \Sigma^{\alpha\beta}\ \Psi_2 \longrightarrow \frac{1}{8}\mathcal{T}_{12}^{\alpha\beta}$$

where we used the fact that $c_{\mu\nu}$ is anti-symmetric since $\Sigma^{\mu\nu}$ are also anti-symmetric.

We found all the components and the outer product of two Dirac spinors is represented by:

$$\boxed{\Psi_2\overline{\Psi}_1 = \frac{1}{4}\Big\{\mathcal{S}_{12}\mathbb{1} + (\mathcal{V}_{12})_\mu\gamma^\mu + \frac{1}{2}(\mathcal{T}_{12})_{\mu\nu}\Sigma^{\mu\nu} - (\mathcal{A}_{12})_\mu\gamma^\mu\gamma^5 + \mathcal{P}_{12}\gamma^5\Big\}}$$

21.2. Rearranging Spinors

Let $\Psi_1, \Psi_2, \Psi_3, \Psi_4$ be four Dirac spinors given to us and consider $(\overline{\Psi}_1\ \Gamma^a\ \Psi_2)(\overline{\Psi}_3\ \Gamma^b\ \Psi_4)$ where a and b can taken the values: $\{\mathcal{S}, \mathcal{V}^\mu, \mathcal{T}^{\mu\nu}, \mathcal{P}, \mathcal{A}^\mu\}$ according as the bilinear that we are considering. We are required to find how this quantity can be expressed with the rearrangement, that is, say $(\overline{\Psi}_1\ \Gamma^a\ \Psi_4)(\overline{\Psi}_3\ \Gamma^b\ \Psi_2)$. For that, let us consider it as a linear combination, that is:

$$(\overline{\Psi}_1\ \Gamma_a\ \Psi_2)(\overline{\Psi}_3\ \Gamma^b\ \Psi_4) = \sum_{c,d}\mathcal{C}_{ad}^{bc}\ (\overline{\Psi}_1\ \Gamma_c\ \Psi_4)(\overline{\Psi}_3\ \Gamma^d\ \Psi_2)$$

Expanding the LHS of the above expression in terms of matrix indices, we get:

$$\sum_{\substack{i,j\\p,q}}\big\{(\overline{\Psi}_1)_i(\Gamma_a)_{ij}(\Psi_2)_j\big\}\Big\{(\overline{\Psi}_3)_p(\Gamma^b)_{pq}(\Psi_4)_q\Big\} = \sum_{\substack{i,j\\p,q}}(\overline{\Psi}_1)_i(\Gamma_a)_{ij}\big\{(\Psi_2)_j(\overline{\Psi}_3)_p\big\}\left(\Gamma^b\right)_{pq}(\Psi_4)_q$$

Now note that we can write:

$$(\Psi_2)_j(\overline{\Psi}_3)_p = \sum_{m,n}\delta_{jm}\delta_{pn}(\Psi_2)_m(\overline{\Psi}_3)_n$$

In Lec. 19.3, we had found a *completeness relation* in Eq. 10, for the elements of the basis and we can use this here.

$$(\Psi_2)_j(\overline{\Psi}_3)_p = \frac{1}{4}\sum_{m,n,c}(\Gamma_c)_{nm}(\Gamma^c)_{jp}(\Psi_2)_m(\overline{\Psi}_3)_n$$

Substituting this in the original expression, we have:

$$\begin{aligned}\text{LHS}: &\frac{1}{4}\sum_{\substack{m,n,c\\i,j,p,q}}(\overline{\Psi}_1)_i(\Gamma_a)_{ij}(\Gamma_c)_{nm}(\Gamma^c)_{jp}(\Psi_2)_m(\overline{\Psi}_3)_n(\Gamma^b)_{pq}(\Psi_4)_q\\ &=\frac{1}{4}\sum_{\substack{m,n,c\\i,j,p,q}}\left\{(\Gamma_c)_{nm}(\Psi_2)_m(\overline{\Psi}_3)_n\right\}(\overline{\Psi}_1)_i(\Gamma_a)_{ij}(\Gamma^c)_{jp}(\Gamma^b)_{pq}(\Psi_4)_q\end{aligned}$$

Note that the bracket term can be written as: $\text{Tr}\big(\Gamma_c\Psi_2\overline{\Psi}_3\big) = \overline{\Psi}_3\Gamma_c\Psi_2$ and the remaining term is written as:

$$\overline{\Psi}_1(\Gamma_a\Gamma^c\Gamma^b)\Psi_4$$

Since $\Gamma_a\Gamma^c\Gamma^b$ is a matrix, from Eq. 9 it can be expanded in terms of the basis elements accordingly as:

$$\Gamma_a\Gamma^c\Gamma^b = \sum_d \frac{1}{4}\text{Tr}\Big(\Gamma_d\Gamma_a\Gamma^c\Gamma^b\Big)\Gamma^d$$

Substituting this in the expression for LHS, we have:

$$(\overline{\Psi}_1\ \Gamma_a\ \Psi_2)(\overline{\Psi}_3\ \Gamma^b\ \Psi_4) = \frac{1}{16}\sum_{c,d}(\overline{\Psi}_3\Gamma_c\Psi_2)\overline{\Psi}_1\,\text{Tr}\Big(\Gamma_d\Gamma_a\Gamma^c\Gamma^b\Big)\Gamma^d\Psi_4 = \frac{1}{16}\sum_{c,d}\text{Tr}\Big(\Gamma_d\Gamma_a\Gamma^c\Gamma^b\Big)(\overline{\Psi}_3\Gamma_c\Psi_2)\overline{\Psi}_1\Gamma^d\Psi_4$$

Renaming the dummy indices $c \leftrightarrow d$ and arranging things accordingly, we obtain:

$$(\overline{\Psi}_1\ \Gamma_a\ \Psi_2)(\overline{\Psi}_3\ \Gamma^b\ \Psi_4) = \frac{1}{16}\sum_{c,d}\text{Tr}\Big(\Gamma^c\Gamma_a\Gamma_d\Gamma^b\Big)(\overline{\Psi}_1\Gamma_c\Psi_4)(\overline{\Psi}_3\Gamma^d\Psi_2)$$

Now, comparing the original linear combination, we obtain the form of the coefficients:

$$\boxed{\mathcal{C}^{bc}_{ad} = \frac{1}{16}\text{Tr}\Big(\Gamma_a\Gamma_d\Gamma^b\Gamma^c\Big)}$$

Let us now see some examples of this technique.

- $a = b = \mathcal{S}$ and so both are scalar bilinears, $\Gamma^a \equiv \mathbb{1}$ and $\Gamma_a \equiv \mathbb{1}$ Using the formula we have:

$$\mathcal{C}^{\mathcal{S}\mathcal{S}}_{cd} = \frac{1}{16}\text{Tr}\Big(\Gamma_c\Gamma^d\Big) = \frac{1}{4}\delta^d{}_c$$

Hence in this case, we can write the terms are:

$$\begin{aligned}\overline{\Psi}_1\Psi_2\overline{\Psi}_3\Psi_4 = \frac{1}{4}\{&(\overline{\Psi}_1\Psi_4)(\overline{\Psi}_3\Psi_2) + (\overline{\Psi}_1\ \gamma_\mu\ \Psi_4)(\overline{\Psi}_3\ \gamma^\mu\ \Psi_2) + (\overline{\Psi}_1\ \gamma_5\ \Psi_4)(\overline{\Psi}_3\ \gamma^5\ \Psi_2)\\ &(\overline{\Psi}_1\ i\gamma_\mu\gamma_5\ \Psi_4)(\overline{\Psi}_3\ i\gamma^\mu\gamma_5\ \Psi_2) + \frac{1}{2}(\overline{\Psi}_1\ \Sigma_{\mu\nu}\ \Psi_4)(\overline{\Psi}_3\ \Sigma^{\mu\nu}\ \Psi_2)\}\end{aligned}$$

The factor of 1/2 has been introduced to avoid double counting since $\Sigma^{\mu\nu}$ has two indices. Thus, we find the rearranged spinors as:

$$\boxed{\mathcal{S}_{12}\mathcal{S}_{34} = \frac{1}{4}\mathcal{S}_{14}\mathcal{S}_{32} + \frac{1}{4}(\mathcal{V}_\mu)_{14}(\mathcal{V}^\mu)_{32} + \frac{1}{4}\mathcal{P}_{14}\mathcal{P}_{32} - \frac{1}{4}(\mathcal{A}_\mu)_{14}(\mathcal{A}^\mu)_{32} + \frac{1}{8}(\mathcal{T}_{\mu\nu})_{14}(\mathcal{T}^{\mu\nu})_{32}}$$

- $a = b = \mathcal{V}$ and so both are vector bilinears. $\Gamma^a \equiv \gamma^\mu$ and $\Gamma_a \equiv \gamma_\mu = \eta_{\mu\nu}\gamma^\mu$. So the coefficient that we need to find out becomes: $\mathrm{Tr}(\gamma_\mu \Gamma_d \gamma^\mu \Gamma^c)$ Let us go case by case:

  $\underline{\Gamma_c = 1}$

$$
\begin{aligned}
\frac{1}{16}\sum_c \mathrm{Tr}(\gamma_\mu \Gamma_d \gamma^\mu)(\bar{\Psi}_1\Psi_4)(\bar{\Psi}_3\Gamma^d\Psi_2) &= \frac{1}{16}\left[\bar{\Psi}_3 \sum_c \mathrm{Tr}(\gamma^\mu\gamma_\mu\Gamma_d)\Gamma^d \ \Psi_2\right](\bar{\Psi}_1\Psi_4) \\
&= \frac{1}{16} \times \bar{\Psi}_3(4\underbrace{\gamma^\mu\gamma_\mu}_{4\mathbb{1}})\Psi_2(\bar{\Psi}_1\Psi_4) \\
&= (\bar{\Psi}_1\Psi_4)(\bar{\Psi}_3\Psi_2) \to \mathcal{S}_{14}\mathcal{S}_{32}
\end{aligned}
$$

  $\underline{\Gamma_c = \gamma_\nu}$

$$
\begin{aligned}
\frac{1}{16}\sum_c \mathrm{Tr}(\gamma_\mu \Gamma_d \gamma^\mu \gamma^\nu)(\bar{\Psi}_1\gamma_\nu\Psi_4)(\bar{\Psi}_3\Gamma^d\Psi_2) &= \frac{1}{16}\left[\bar{\Psi}_3 \sum_c \mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma_\mu\Gamma_d)\Gamma^d \ \Psi_2\right](\bar{\Psi}_1\gamma_\nu\Psi_4) \\
&= \frac{1}{16} \times \bar{\Psi}_3(4\underbrace{\gamma^\mu\gamma^\nu\gamma_\mu}_{-2\gamma^\nu})\Psi_2(\bar{\Psi}_1\gamma_\nu\Psi_4) \\
&= -\frac{1}{2}(\bar{\Psi}_1\gamma_\nu\Psi_4)(\bar{\Psi}_3\gamma^\nu\Psi_2) \to -\frac{1}{2}(\mathcal{V}_\nu)_{14}(\mathcal{V}^\nu)_{32}
\end{aligned}
$$

  $\underline{\Gamma_c = \gamma_5}$

$$
\begin{aligned}
\frac{1}{16}\sum_c \mathrm{Tr}\big(\gamma_\mu \Gamma_d \gamma^\mu \gamma^5\big)(\bar{\Psi}_1\gamma_5\Psi_4)(\bar{\Psi}_3\Gamma^d\Psi_2) &= \frac{1}{16}\left[\bar{\Psi}_3 \sum_c \mathrm{Tr}\big(\gamma^\mu\gamma^5\gamma_\mu\Gamma_d\big)\Gamma^d \ \Psi_2\right](\bar{\Psi}_1\gamma_5\Psi_4) \\
&= \frac{1}{16} \times \bar{\Psi}_3(4\underbrace{\gamma^\mu\gamma^5\gamma_\mu}_{-4\gamma^5})\Psi_2(\bar{\Psi}_1\gamma_5\Psi_4) \\
&= -(\bar{\Psi}_1\gamma_5\Psi_4)(\bar{\Psi}_3\gamma^5\Psi_2) \to -\mathcal{P}_{14}\mathcal{P}_{32}
\end{aligned}
$$

  $\underline{\Gamma_c = i\gamma_\nu\gamma_5}$

$$
\begin{aligned}
\frac{1}{16}\sum_c \mathrm{Tr}\big(\gamma_\mu \Gamma_d \gamma^\mu \ i\gamma^\nu\gamma^5\big)(\bar{\Psi}_1 \ i\gamma_\nu\gamma_5\Psi_4)(\bar{\Psi}_3\Gamma^d\Psi_2) &= -\frac{1}{16}\left[\bar{\Psi}_3 \sum_c \mathrm{Tr}\big(\gamma^\mu\gamma^\nu\gamma^5\gamma_\mu\Gamma_d\big)\Gamma^d \ \Psi_2\right](\bar{\Psi}_1\gamma_\nu\gamma_5\Psi_4) \\
&= -\frac{1}{16} \times \bar{\Psi}_3(4\underbrace{\gamma^\mu\gamma^\nu\gamma^5\gamma_\mu}_{+2\gamma^\nu\gamma^5})\Psi_2(\bar{\Psi}_1\gamma_\nu\gamma_5\Psi_4) \\
&= -\frac{1}{2}(\bar{\Psi}_1\gamma_\nu\gamma_5\Psi_4)(\bar{\Psi}_3\gamma^\nu\gamma^5\Psi_2) \to -\frac{1}{2}(\mathcal{A}_\nu)_{14}(\mathcal{A}^\nu)_{32}
\end{aligned}
$$

  $\underline{\Gamma_c = \Sigma_{\rho\nu}}$

$$
\begin{aligned}
\frac{1}{16}\sum_c \mathrm{Tr}(\gamma_\mu \Gamma_d \gamma^\mu \Sigma^{\rho\nu})(\bar{\Psi}_1\Sigma_{\rho\nu}\Psi_4)(\bar{\Psi}_3\Gamma^d\Psi_2) &= \frac{1}{16}\left[\bar{\Psi}_3 \sum_c \mathrm{Tr}(\gamma^\mu\Sigma^{\rho\nu}\gamma_\mu\Gamma_d)\Gamma^d \ \Psi_2\right](\bar{\Psi}_1\Sigma_{\rho\nu}\Psi_4) \\
&= \frac{1}{16} \times \bar{\Psi}_3(4\underbrace{\gamma^\mu\Sigma^{\rho\nu}\gamma_\mu}_{4(\eta^{\rho\nu}-\eta^{\nu\rho})})\Psi_2(\bar{\Psi}_1\Sigma_{\rho\nu}\Psi_4) = 0
\end{aligned}
$$

Thus we have the final form:

$$
\boxed{\mathcal{V}_{12}\mathcal{V}_{34} = 1 \times \mathcal{S}_{14}\mathcal{S}_{32} + -\frac{1}{2}(\mathcal{V}_\mu)_{14}(\mathcal{V}^\mu)_{32} + -1 \times \mathcal{P}_{14}\mathcal{P}_{32} - \frac{1}{2}(\mathcal{A}_\mu)_{14}(\mathcal{A}^\mu)_{32} + 0 \times (\mathcal{T}_{\mu\nu})_{14}(\mathcal{T}^{\mu\nu})_{32}}
$$

We can continue doing these kind of bs calculations and find more and more rearranged bilinears. However it must be noted that this is a very retarded task to do and one should never pursue this if it is not absolutely required!

	$\mathcal{S}_{14}\mathcal{S}_{32}$	$(\mathcal{V}_{14})_\nu(\mathcal{V}_{32})^\nu$	$(\mathcal{T}_{\rho\nu})_{14}(\mathcal{T}^{\rho\nu})_{32}$	$(\mathcal{A}_\nu)_{14}(\mathcal{A}^\nu)_{32}$	$\mathcal{P}_{14}\mathcal{P}_{32}$
$\mathcal{S}_{12}\mathcal{S}_{34}$	1/4	1/4	1/8	-1/4	1/4
$(\mathcal{V}_\mu)_{12}\mathcal{V}^\mu{}_{34}$	1	-1/2	0	-1/2	-1

Table 1: Table showing the coefficients of the rearranged spinors

Lecture 22: Normalising Spinors

Recall our discussion from Lec 17 where we derived the forms of the Dirac spinor for a momentum p (Eq. 6). This led to positive and negative energy states whose interpretation became problematic. The hole theory itself provided many inconsistencies and now another interpretation is followed. Let the proposed plane-wave solution to the Dirac equation be:

$$\Psi(x) = u(p)e^{-ipx} \qquad \Psi(x) = v(p)e^{+ipx}$$

where the $-$ is interpreted for *particles* and $+$ is interpreted for *anti-particles*. Note that in this case, $E > 0$ always. The distinction between particles and anti-particles comes from the sign of the exponential and hence we do not need to worry separately about negative and positive energy solutions.

We will define the following notation: $\not{a} := \gamma^\mu a_\mu$ henceforth to denote the contraction with the Dirac matrices. Using this, the Dirac equation is written as: $(i\not{\partial} - m)\Psi = 0$. Substituting the above solutions in the Dirac equation, we get:

$$0 = (i\not{\partial} - m)u(p)e^{-ip_\mu x^\mu} = i\gamma^\mu(-ip_\mu)e^{-ip_\mu x^\mu}u(p) - me^{-ip_\mu x^\mu}u(p) = (\not{p} - m)u(p)^{-ip_\mu x^\mu}$$
$$0 = (i\not{\partial} - m)v(p)e^{+ip_\mu x^\mu} = i\gamma^\mu(ip_\mu)e^{-ip_\mu x^\mu}v(p) - me^{ip_\mu x^\mu}v(p) = -(\not{p} + m)v(p)^{+ip_\mu x^\mu}$$

From this, we get two separate equations that should be satisfied by the proposed solutions:

$$(\not{p} - m)u(p) = 0 \qquad (\not{p} + m)v(p) = 0 \tag{12}$$

In the Dirac basis the quantity $\not{p}$ becomes:

$$\not{p} = \gamma^\mu p_\mu = \gamma^0 p_0 + \gamma^i p_i = \gamma^0 p^0 - \gamma^i p^i = \begin{pmatrix} p^0\mathbb{1} & 0 \\ 0 & -p^0\mathbb{1} \end{pmatrix} - \begin{pmatrix} 0 & \sigma^i p^i \\ -\sigma^i p^i & 0 \end{pmatrix} = \begin{pmatrix} p^0\mathbb{1} & -\boldsymbol{\sigma}\cdot\mathbf{p} \\ \boldsymbol{\sigma}\cdot\mathbf{p} & -p^0\mathbb{1} \end{pmatrix} \tag{13}$$

Suppose the solutions be decomposed into a pair of two-component spinors such that:

$$u(p) = \begin{pmatrix} \xi(p) \\ \phi(p) \end{pmatrix} \qquad v(p) = \begin{pmatrix} \chi(p) \\ \eta(p) \end{pmatrix}$$

Substituting this in the above Eq. 12, we get the following:

$$\begin{aligned} (\boldsymbol{\sigma}\cdot\mathbf{p})\xi(p) - (p^0 + m)\phi(p) = 0 &\longrightarrow \phi(p) = \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0 + m}\xi(p) \\ (p^0 + m)\chi(p) - (\boldsymbol{\sigma}\cdot\mathbf{p})\eta(p) = 0 &\longrightarrow \chi(p) = \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0 + m}\eta(p) \end{aligned} \tag{14}$$

Thus the general solutions can be expressed in terms of two arbitrary Weyl spinors ξ and η. For simplicity, let us choose them to be the standard basis vectors of the space, that is,

$$\triangleright\ \xi^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \triangleright\ \xi^2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \triangleright\ \eta^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \triangleright\ \eta^2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The solutions can then be written as (for $s = 1, 2$):

$$u^s(p) = \begin{pmatrix} \xi^s \\ \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s \end{pmatrix} \qquad v^s(p) = \begin{pmatrix} \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s \\ \eta^s \end{pmatrix}$$

22.1. The correct factor

Let us try to find the correct normalisation for the spinors that we have obtained above.

$$u^{r\dagger}u^s = \begin{pmatrix}\xi^{r\dagger} & \xi^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\end{pmatrix}\begin{pmatrix}\xi^s \\ \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s\end{pmatrix} = \xi^{r\dagger}\xi^s + \frac{1}{(p^0+m)^2}\xi^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p})^2\xi^s = \delta^{rs} + \frac{|\mathbf{p}|^2}{(p^0+m)^2}\delta^{rs}$$

We have use the following facts:

- ▷ The Pauli matrices are Hermitian, hence $(\boldsymbol{\sigma}\cdot\mathbf{p})^\dagger = (\boldsymbol{\sigma}\cdot\mathbf{p})$.
- ▷ $(AB)^\dagger = B^\dagger A^\dagger$.
- ▷ $(\boldsymbol{\sigma}\cdot\mathbf{p}) = |\mathbf{p}|^2\mathbb{1}$ (calculated in Eq. 5)
- ▷ $\xi^{r\dagger}\xi^s = \delta^{rs}$ since these are treated as the standard orthonormal basis vectors.

We know that $|\mathbf{p}|^2 = (p^0)^2 - m^2$ and substituting this above, we obtain:

$$u^{r\dagger}u^s = \left(1 + \frac{(p^0)^2 - m^2}{(p^0+m)^2}\right)\delta^{rs} = \frac{2p^0}{p^0+m}\delta^{rs} \tag{15}$$

Exactly same calculations yields for the other solution,

$$v^{r\dagger}v^s = \left(1 + \frac{(p^0)^2 - m^2}{(p^0+m)^2}\right)\delta^{rs} = \frac{2p^0}{p^0+m}\delta^{rs} \tag{16}$$

These are unfortunately not Lorentz invariant and we need to fix some factors to get a Lorentz invariant inner product between the spinors. Okay, so now let's check the orthogonality of the positive and negative frequency solutions.

$$u^{r\dagger}v^s = \begin{pmatrix}\xi^{r\dagger} & \xi^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\end{pmatrix}\begin{pmatrix}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s \\ \eta^s\end{pmatrix} = \frac{2}{p^0+m}\xi^{r\dagger}\ (\boldsymbol{\sigma}\cdot\mathbf{p})\ \eta^s \neq 0 \tag{17}$$

The positive and negative frequency solutions are not orthogonal which is problematic since we expect the solutions for particles and anti-particles to not have any overlap.

We need to find some alternate definition of the inner product. Since the answer has already been found by major stalwarts, we just throw it in here. Consider the inner product $\bar{u}^r v^s = u^{r\dagger}\gamma^0 v^s$

$$u^{r\dagger}\gamma^0 v^s = \begin{pmatrix}\xi^{r\dagger} & \xi^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\end{pmatrix}\begin{pmatrix}\mathbb{1} & \\ & -\mathbb{1}\end{pmatrix}\begin{pmatrix}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s \\ \eta^s\end{pmatrix} = \begin{pmatrix}\xi^{r\dagger} & \xi^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\end{pmatrix}\begin{pmatrix}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s \\ -\eta^s\end{pmatrix} = 0$$

This definition of the inner product restores the *orthogonality* that we had desperately wanted! Let us look at the other inner products, that is, between the same sectors:

$$\begin{aligned}\bar{u}^r u^s = u^{r\dagger}\gamma^0 u^s = \begin{pmatrix}\xi^{r\dagger} & \xi^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\end{pmatrix}\begin{pmatrix}\mathbb{1} & \\ & -\mathbb{1}\end{pmatrix}\begin{pmatrix}\xi^s \\ \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s\end{pmatrix} &= \xi^{r\dagger}\xi^s - \frac{1}{(p^0+m)^2}\xi^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p})^2\xi^s \\ &= \delta^{rs} - \frac{|\mathbf{p}|^2}{(p^0+m)^2}\delta^{rs} \\ &= \frac{2m}{p^0+m}\delta^{rs}\end{aligned}$$

Similar calculation yields, for the negative frequency solution,

$$\bar{v}^r v^s = -\frac{2m}{p^0+m}\delta^{rs}$$

All things look good except the factor p^0+m in the denominator which keeps these inner products from being Lorentz invariant. Hence if we impose the normalisation as:

$$u^s(p) = \sqrt{p^0+m}\begin{pmatrix} \xi^s \\ \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s \end{pmatrix} \qquad v^s(p) = \sqrt{p^0+m}\begin{pmatrix} \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s \\ \eta^s \end{pmatrix}$$

then, we will obtain $\bar{v}^r v^s = -2m\ \delta^{rs}$ and $\bar{u}^r u^s = 2m\ \delta^{rs}$ which is perfectly Lorentz invariant since it involved only the mass m which is a scalar. Henceforth, we will use this normalisation everytime we write the Dirac spinors. With this normalisation then, we get:

$$\boxed{u^{r\dagger}u^s = 2p^0\delta^{rs}} \qquad \boxed{v^{r\dagger}v^s = 2p^0\delta^{rs}} \tag{18}$$

Also note that, from Eq. 17, if we calculate the expression for $\mathbf{p}$ and $-\mathbf{p}$, we get:

$$\boxed{u^{r\dagger}(\mathbf{p})v^s(-\mathbf{p}) = u^{r\dagger}(-\mathbf{p})v^s(\mathbf{p}) = 0} \tag{19}$$

22.2. Outer Products

We looked at inner products and arrived at a nice definition of inner product for Dirac spinors. We now look at the outer product of these spinors.

$$\begin{aligned}
u^s\bar{u}^r = u^s u^{r\dagger}\gamma^0 &= \underbrace{(p^0+m)}_{\text{from normalisation}}\begin{pmatrix} \xi^s \\ \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s \end{pmatrix}\begin{pmatrix} \xi^{r\dagger} & \xi^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m} \end{pmatrix}\begin{pmatrix} \mathbb{1} & \\ & -\mathbb{1} \end{pmatrix} \\
&= \begin{pmatrix} (p^0+m)\xi^s\xi^{r\dagger} & \xi^s\xi^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p})\xi^s\xi^{r\dagger} & \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s\xi^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) \end{pmatrix}\begin{pmatrix} \mathbb{1} & \\ & -\mathbb{1} \end{pmatrix} \\
&= \begin{pmatrix} (p^0+m)\xi^s\xi^{r\dagger} & -\xi^s\xi^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p})\xi^s\xi^{r\dagger} & -\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\xi^s\xi^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) \end{pmatrix}
\end{aligned}$$

Similar for the other solution, we obtain the following form:

$$\begin{aligned}
v^s\bar{v}^r = v^s v^{r\dagger}\gamma^0 &= \underbrace{(p^0+m)}_{\text{from normalisation}}\begin{pmatrix} \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s \\ \eta^s \end{pmatrix}\begin{pmatrix} \eta^{r\dagger}\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m} & \eta^{r\dagger} \end{pmatrix}\begin{pmatrix} \mathbb{1} & \\ & -\mathbb{1} \end{pmatrix} \\
&= \begin{pmatrix} \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s\eta^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) & (\boldsymbol{\sigma}\cdot\mathbf{p})\eta^s\eta^{r\dagger} \\ \eta^s\eta^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) & (p^0+m)\eta^s\eta^{r\dagger} \end{pmatrix}\begin{pmatrix} \mathbb{1} & \\ & -\mathbb{1} \end{pmatrix} \\
&= \begin{pmatrix} \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}\eta^s\eta^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) & -(\boldsymbol{\sigma}\cdot\mathbf{p})\eta^s\eta^{r\dagger} \\ \eta^s\eta^{r\dagger}(\boldsymbol{\sigma}\cdot\mathbf{p}) & -(p^0+m)\eta^s\eta^{r\dagger} \end{pmatrix}
\end{aligned}$$

As ξ and η are basis vectors, these satisfy the completeness relation:

$$\sum_r \eta^r\eta^{r\dagger} = \mathbb{1} \qquad \sum_r \xi^r\xi^{r\dagger} = \mathbb{1}$$

Using this, we get:

$$\sum_r u^r \bar{u}^r = \begin{pmatrix} (p^0+m)\mathbb{1} & -(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p}) & -\frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}(\boldsymbol{\sigma}\cdot\mathbf{p}) \end{pmatrix} = \begin{pmatrix} (p^0+m)\mathbb{1} & -(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p}) & -\frac{|\mathbf{p}|^2}{p^0+m}\mathbb{1} \end{pmatrix} = \begin{pmatrix} (p^0+m)\mathbb{1} & -(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p}) & -(p^0-m)\mathbb{1} \end{pmatrix} = \not{p} + m\mathbb{1}$$

$$\sum_r v^r \bar{v}^r = \begin{pmatrix} \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{p^0+m}(\boldsymbol{\sigma}\cdot\mathbf{p}) & -(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p}) & -(p^0+m)\mathbb{1} \end{pmatrix} = \begin{pmatrix} (p^0-m)\mathbb{1} & -(\boldsymbol{\sigma}\cdot\mathbf{p}) \\ (\boldsymbol{\sigma}\cdot\mathbf{p}) & -(p^0+m)\mathbb{1} \end{pmatrix} = \not{p} - m\mathbb{1}$$

Thus we have two nice identitites concerning the outer products,

$$\boxed{\sum_r u^r \bar{u}^r = \not{p} + m} \qquad \boxed{\sum_r v^r \bar{v}^r = \not{p} - m}$$

We had used the Dirac representation to find these identitites. We could have also used the representation of Dirac matrices in chiral basis which would have given us the same result, since the final expression is independent of the representation of the Dirac matrix.

22.3. Projecting Energy

Let us define the following operator:

$$\Lambda_\pm := \frac{\pm\not{p} + m}{2m}$$

Let us see some properties of these operators:

▷ Idempotence

$$\Lambda_+^2 = \frac{1}{4m^2}(\not{p}+m)(\not{p}+m) = \frac{1}{4m^2}(\not{p}\not{p} + 2m\not{p} + m^2) = \frac{1}{4m^2}(2m^2 + 2m\not{p}) = \Lambda_+$$
$$\Lambda_-^2 = \frac{1}{4m^2}(-\not{p}+m)(-\not{p}+m) = \frac{1}{4m^2}(\not{p}\not{p} - 2m\not{p} + m^2) = \frac{1}{4m^2}(2m^2 - 2m\not{p}) = \Lambda_-$$

where we have used the following:

$$\not{p}\not{p} = \frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)p_\mu p_\nu = \eta^{\mu\nu}p_\mu p_\nu = p^2 = m^2$$

Thus, we find that $\Lambda_\pm$ are idempotent operators.

▷ Orthogonality

$$\Lambda_+\Lambda_- \sim (\not{p}+m)(-\not{p}+m) = (-m^2+m^2) = 0$$
$$\Lambda_-\Lambda_+ \sim (-\not{p}+m)(\not{p}+m) = (-m^2+m^2) = 0$$

Thus, we see that product of the two operators always give zero.

▷ Completeness

$$\Lambda_+ + \Lambda_- = \frac{1}{2m}(\not{p} + m - \not{p} + m) = \mathbb{1}$$

We see that the sum of the operators equal to the identity operator of the space.

The above properties indicate that $\Lambda_\pm$ are *orthogonal projection operators.* To see explicitly, consider the following:

$$\Lambda_+ u^s = \frac{(\not{p}+m)}{2m}u^s = \frac{mu^s + mu^s}{2m} = u^s$$
$$\Lambda_- u^s = \frac{(-\not{p}+m)}{2m}u^s = \frac{-mu^s + mu^s}{2m} = 0$$
$$\Lambda_+ v^s = \frac{(\not{p}+m)}{2m}v^s = \frac{-mv^s + mv^s}{2m} = 0$$
$$\Lambda_+ v^s = \frac{(-\not{p}+m)}{2m}v^s = \frac{mv^s + mv^s}{2m} = v^s$$

where we have used Eq. 12. We see that, starting from an arbitrary spinor, Λ_+ (Λ_-) projects to the positive (negative) frequency sector. Hence these are also called *energy projection operators*.

Lecture 23: Some Other Projections

23.1. Spin Projection

In this section we will be concerned with spins. For the *non-relativistic* case, the *spin projection operator* is given by:

$$\Sigma(\hat{s}) = \frac{1}{2}(1 + \boldsymbol{\sigma}\cdot\hat{\mathbf{s}})$$

Think about it, suppose $\hat{\mathbf{s}} = (0,0,\pm 1)$ then essentially we have a spin state either in $+z$ or $-z$ direction, denoted by say, $|\alpha\rangle$. The operator then becomes:

$$\Sigma_z = \frac{1}{2}(1 \pm \sigma_z)\,|\alpha\rangle = \frac{1}{2}(1 \pm s)\,|\alpha\rangle$$

where $s = \pm 1$ is the eigenvalue of σ_z. If $s = +1$ and $\hat{\mathbf{s}} = (0,0,1)$ or $s = -1$ and $\hat{\mathbf{s}} = (0,0,-1)$, then $\Sigma_z = |\alpha\rangle$, else $\Sigma_z = 0$. This correctly projects an arbitrary spinor to the appropriate spin $\hat{\mathbf{s}}$.

We need to extrapolate this idea to the relativistic case. First, let us begin considering the REST FRAME, where $p^\mu = (p^0, 0)$. Consider the operator,

$$\Sigma(\hat{s}) := \frac{1}{2}\begin{pmatrix} 1+\boldsymbol{\sigma}\cdot\hat{\mathbf{s}} & 0 \\ 0 & 1-\boldsymbol{\sigma}\cdot\hat{\mathbf{s}} \end{pmatrix} = \frac{1}{2}\left[\mathbb{1} + \begin{pmatrix} \boldsymbol{\sigma}\cdot\hat{\mathbf{s}} & 0 \\ 0 & -\boldsymbol{\sigma}\cdot\hat{\mathbf{s}} \end{pmatrix}\right]$$

We define the quantity $\hat{s}^\mu := (0, \hat{\mathbf{s}})$. Then, using the same calculation as in Eq.13, we have:

$$\not{s} = \begin{pmatrix} 0 & -\boldsymbol{\sigma}\cdot\hat{\mathbf{s}} \\ \boldsymbol{\sigma}\cdot\hat{\mathbf{s}} & 0 \end{pmatrix}$$

since $\hat{s}^0 = 0$. Then $\Sigma(\hat{s})$ can be written as $\Sigma(\hat{s}) = \frac{1}{2}\left[\mathbb{1} + \gamma^5\not{s}\right]$. Also, note a few properties:

$$\hat{s}^\mu \hat{s}_\mu = -|\hat{\mathbf{s}}|^2 = -1 \qquad p^\mu s_\mu = 0 \tag{20}$$

Let us see what happens when we act this to the free-wave solutions of the Dirac equation, $u^r(p)$ and $v^r(p)$. Then the solutions become:

$$u^r = \begin{pmatrix} \xi^r \\ 0 \end{pmatrix} \qquad v^r = \begin{pmatrix} 0 \\ \eta^r \end{pmatrix} \longrightarrow \Sigma(\hat{s})u^r = \frac{1}{2}\begin{pmatrix} (1+\boldsymbol{\sigma}\cdot\hat{\mathbf{s}})\xi^r \\ 0 \end{pmatrix} \qquad \Sigma(\hat{s})v^r = \frac{1}{2}\begin{pmatrix} 0 \\ (1-\boldsymbol{\sigma}\cdot\hat{\mathbf{s}})\eta^r \end{pmatrix}$$

For simplicity, let us take $\hat{\mathbf{s}} = (0,0,\pm 1)$ (that is, spin is along $\hat{z}$ axis). We can have an interpretation of u^r and v^r with r indicating the spin, that is, $r = 1$ corresponds to spin $+\frac{1}{2}$ and $r = 2$ corresponds to spin $-\frac{1}{2}$ for the positive energy solution u while the opposite occurs for v.

For $s = \pm 1$, it is now better to denote these solutions as $u(0,s)$ and $v(0,s)$ which mentions the spin explicitly. Hence $u^1 = u(0,+1), u^2 = u(0,-1), v^1 = v(0,-1), v^2 = v(0,+1)$ With this notation, one can verify that:

- If $\hat{\mathbf{s}} = (0,0,1)$ then $\Sigma(\hat{s})u^1 = u^1$
- If $\hat{\mathbf{s}} = (0,0,-1)$ then $\Sigma(\hat{s})u^1 = 0$
- If $\hat{\mathbf{s}} = (0,0,1)$ then $\Sigma(\hat{s})u^2 = 0$
- If $\hat{\mathbf{s}} = (0,0,-1)$ then $\Sigma(\hat{s})u^2 = u^2$
- If $\hat{\mathbf{s}} = (0,0,1)$ then $\Sigma(\hat{s})v^1 = 0$
- If $\hat{\mathbf{s}} = (0,0,-1)$ then $\Sigma(\hat{s})v^1 = v^1$
- If $\hat{\mathbf{s}} = (0,0,1)$ then $\Sigma(\hat{s})v^2 = v^2$
- If $\hat{\mathbf{s}} = (0,0,-1)$ then $\Sigma(\hat{s})v^2 = 0$

We see that in the rest frame, the free-wave solutions are eigenstates of $\Sigma(\hat{s})$ and they correctly project the right spin states, that is,

$$\Sigma(\hat{s})u(0,\hat{s}) = u(0,\hat{s})$$

The equation above has a invariant form, that is, it has no index dependence. We can use this now to go into a general frame by considering $p \to \Lambda(m,0)$ and $\hat{s} \to \Lambda\hat{s}$. Then note that,

$$\begin{aligned} u(p,s) = S(\Lambda)u(0,\hat{s}) &= S(\Lambda)\ \Sigma(\hat{s})\ S^{-1}(\Lambda)S(\Lambda)\ u(0,\hat{s}) \\ &= \frac{1}{2}\left[\mathbb{1} + \gamma^5 S(\Lambda)\ \gamma^\mu\ S^{-1}(\Lambda)\hat{s}_\mu\right]S(\Lambda)u(0,\hat{s}) \\ &= \frac{1}{2}\left[\mathbb{1} + \gamma^5 \left(\Lambda^{-1}\right)^\mu{}_\nu \gamma^\nu \hat{s}_\mu\right]u(p,s) \\ &= \frac{1}{2}\left[\mathbb{1} + \gamma^5\gamma^\nu s_\nu\right]u(p,s) \end{aligned}$$

Hence in a general frame, the spin projection operator takes the form:

$$\Sigma(\pm s) = \frac{1}{2}\left[1 \pm \gamma^5 \not{s}\right]$$

where s^μ is now any vector satisfying Eq.20. We can check that these are indeed projection operators. For that, first let us calculate some quantitites which appear in the expression:

$$\gamma^\mu\gamma^\nu s_\mu s_\nu = (2\eta^{\mu\nu} - \gamma^\nu\gamma^\mu)s_\mu s_\nu = 2s^\nu s_\nu - \gamma^\nu\gamma^\mu s_\mu s_\nu = -2 - \gamma^\mu\gamma^\nu s_\nu s_\mu = -2 - \gamma^\mu\gamma^\nu s_\mu s_\nu \implies \boxed{\gamma^\mu\gamma^\nu s_\mu s_\nu = -1}$$

And we also have the following thing using the above identity:

$$\gamma^5\not{s}\gamma^5\not{s} = \gamma^5\gamma^\mu\gamma^5\gamma^\nu s_\mu s_\nu = -(\gamma^5)^2\gamma^\mu\gamma^\nu s_\mu s_\nu = 1$$

Now, we are in a good position to calculate the projection properties.

- $\Sigma(s)\Sigma(s) = \frac{1}{4}(1 + 2\gamma^5\not{s} + \gamma^5\not{s}\gamma^5\not{s}) = \frac{1}{4}(1 + 2\gamma^5\not{s} + 1) = \Sigma(s)$
- $\Sigma(-s)\Sigma(-s) = \frac{1}{4}(1 - 2\gamma^5\not{s} + \gamma^5\not{s}\gamma^5\not{s}) = \frac{1}{4}(1 - 2\gamma^5\not{s} + 1) = \Sigma(-s)$
- $\Sigma(s)\Sigma(-s) = \frac{1}{4}(1 - \gamma^5\not{s}\gamma^5\not{s}) = 0$
- $\Sigma(s) + \Sigma(-s) = \frac{1}{2}(\mathbb{1} + \gamma^5\not{s} + \mathbb{1} - \gamma^5\ \not{s}) = 0$

We see that $\Sigma(\pm s)$ satisfies all the properties of orthogonal projection and is thus a valid projection operator.

In the previous section, we had seen that the energy projection operators $\Lambda_\pm$ projected states to a particular energy and here we found two operators projecting out to a particular spin state. Then we can project to some state with definite energy and definite spin by successivly acting the energy and spin projection operators together. However, problem might arise if these do not commute since then the answer would depend on their commutator. Hence, let us first see if these commute or not:

$$[\Lambda_\pm, \Sigma(\pm s)] \propto \left[\not{p}, \gamma^5\not{s}\right] = (\gamma^\mu p_\mu)(\gamma^5\gamma^\nu s_\nu) - (\gamma^5\gamma^\nu s_\nu)(\gamma^\mu p_\mu) = -\gamma^5(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)p_\mu s_\nu = -2\eta^{\mu\nu}p_\mu s_\nu = 0$$

where we have used Eq.20. We see that the energy projection and spin projection operators happily commute and we are out of trouble!

23.2. Chirality and Massless Fermions

We will focus on *chirality* now which is a Lorentz invariant property (however, unlike helicity it does not commute with the Hamiltonian). γ^5, having eigenvalues 1 and -1, is called the chirality operator and we define the right and left chiral states as the eigenstates of γ^5, that is:

$$\gamma^5 u_{\mathrm{R}} = u_{\mathrm{R}} \qquad \gamma^5 u_{\mathrm{L}} = -u_{\mathrm{L}}$$

From this definition, we can guess a form of the chirality projection operator:

$$P_{\mathrm{R}} = \frac{1}{2}(\mathbb{1} + \gamma^5) \qquad P_{\mathrm{L}} = \frac{1}{2}(\mathbb{1} - \gamma^5)$$

We can check that

$$\triangleright\ P_{\mathrm{R}} u_{\mathrm{R}} = u_{\mathrm{R}} \qquad \triangleright\ P_{\mathrm{R}} u_{\mathrm{L}} = 0 \qquad \triangleright\ P_{\mathrm{L}} u_{\mathrm{R}} = 0 \qquad \triangleright\ P_{\mathrm{L}} u_{\mathrm{L}} = u_{\mathrm{L}}$$

Also we can check the projection properties:

- $\frac{1}{4}(\mathbb{1} + \gamma^5)(\mathbb{1} + \gamma^5) = \frac{1}{4}(\mathbb{1} + 2\gamma^5 + (\gamma^5)^2) = \frac{1}{2}(\mathbb{1} + \gamma^5)$
- $\frac{1}{4}(\mathbb{1} - \gamma^5)(\mathbb{1} - \gamma^5) = \frac{1}{4}(\mathbb{1} - 2\gamma^5 + (\gamma^5)^2) = \frac{1}{2}(\mathbb{1} - \gamma^5)$
- $(\mathbb{1} + \gamma^5)(\mathbb{1} - \gamma^5) = \mathbb{1} - (\gamma^5)^2 = 0$
- $\frac{1}{2}\big\{(\mathbb{1} + \gamma^5) + (\mathbb{1} - \gamma^5)\big\} = \mathbb{1}$

and thus, this is a valid projection operator. Any spinor can be written in terms of the chiral states as:

$$\Psi = \mathbb{1}\Psi = (P_{\mathrm{L}} + P_{\mathrm{R}})\Psi = \Psi_{\mathrm{L}} + \Psi_{\mathrm{R}} \qquad \text{where,} \quad \Psi_{\mathrm{L}} := P_{\mathrm{L}}\Psi \quad \Psi_{\mathrm{R}} := P_{\mathrm{R}}\Psi$$

In the chiral basis,

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \implies \gamma^5 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix} \longrightarrow P_{\mathrm{R}} = \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{1} \end{pmatrix} \quad P_{\mathrm{L}} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & 0 \end{pmatrix}$$

and we see that, if any arbitrary Dirac spinor Ψ is written in terms of the Weyl spinors as,

$$\Psi = \begin{pmatrix} \psi_{\mathrm{L}} \\ \psi_{\mathrm{R}} \end{pmatrix}$$

then, the chirality projections are respectively

$$\Psi_{\mathrm{R}} = P_{\mathrm{R}}\Psi = \begin{pmatrix} 0 \\ \psi_{\mathrm{R}} \end{pmatrix} \qquad \Psi_{\mathrm{L}} = P_{\mathrm{L}}\Psi = \begin{pmatrix} \psi_{\mathrm{L}} \\ 0 \end{pmatrix}$$

and we were right to name the two-component spinors as *left* and handed Weyl spinors in the chiral basis. However, in the Dirac basis,

$$\gamma^0 = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} \implies \gamma^5 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \longrightarrow P_{\mathrm{R}} = \frac{1}{2}\begin{pmatrix} \mathbb{1} & \mathbb{1} \\ \mathbb{1} & \mathbb{1} \end{pmatrix} \quad P_{\mathrm{L}} = \frac{1}{2}\begin{pmatrix} \mathbb{1} & -\mathbb{1} \\ -\mathbb{1} & \mathbb{1} \end{pmatrix}$$

Hence for any Dirac spinor written in terms of some arbitrary two component spinors ϕ and χ, we have:

$$\Psi_{\mathrm{R}} = P_{\mathrm{R}}\Psi = \frac{1}{2}\begin{pmatrix} \phi + \chi \\ \phi + \chi \end{pmatrix} \qquad \Psi_{\mathrm{L}} = P_{\mathrm{L}}\Psi = \frac{1}{2}\begin{pmatrix} \phi - \chi \\ \chi - \phi \end{pmatrix}$$

Then, if we define $\omega_{\mathrm{R}} = \frac{1}{2}(\phi + \chi)$ and $\omega_{\mathrm{L}} = \frac{1}{2}(\phi - \chi)$, then we get:

$$\Psi_{\mathrm{R}} = \begin{pmatrix} \omega_{\mathrm{R}} \\ \omega_{\mathrm{R}} \end{pmatrix} \qquad \Psi_{\mathrm{L}} = \begin{pmatrix} \omega_{\mathrm{L}} \\ -\omega_{\mathrm{L}} \end{pmatrix} \tag{21}$$

NOTE: ϕ and χ are not Weyl spinors, these are some arbitrary spinors. Above, Ψ_{L} and Ψ_{R} are eigenstates of γ^5 (as can be checked since in Dirac basis, the eigenstates of γ^5 are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$) and thus, have definite chirality. So, the two component spinors with which these are made up of, that is, ω_{L} and ω_{R} are the correct Weyl spinors.

23.2.1. When mass vanishes!

Suppose we have a Dirac spinor in terms of two-component spinors in the Dirac basis, $\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$. Then using Eq. 13 and taking $m = 0$ we have:

$$\left.\begin{aligned} p^0\psi_1 - (\boldsymbol{\sigma}\cdot\mathbf{p})\psi_2 &= 0 \\ p^0\psi_2 - (\boldsymbol{\sigma}\cdot\mathbf{p})\psi_1 &= 0 \end{aligned}\right\} \qquad \boxed{p^0(\psi_1+\psi_2) = (\boldsymbol{\sigma}\cdot\mathbf{p})(\psi_1+\psi_2)} \quad \boxed{p^0(\psi_1-\psi_2) = -(\boldsymbol{\sigma}\cdot\mathbf{p})(\psi_1-\psi_2)} \tag{22}$$

Defining $\psi_{\rm L} = \frac{1}{2}(\psi_1 - \psi_2)$ and $\psi_{\rm R} = \frac{1}{2}(\psi_1 + \psi_2)$, and multiplying both sides of the equation with $(\boldsymbol{\sigma}\cdot\mathbf{p})$, we get:

$$p^0\underbrace{(\boldsymbol{\sigma}\cdot\mathbf{p})\psi_{\rm L}}_{p^0\psi_{\rm L}} = (\boldsymbol{\sigma}\cdot\mathbf{p})^2\psi_{\rm L} = |\mathbf{p}|^2\psi_{\rm L} \longrightarrow (p^0)^2\psi_{\rm L} = |\mathbf{p}|^2\psi_{\rm L}$$
$$p^0\underbrace{(\boldsymbol{\sigma}\cdot\mathbf{p})\psi_{\rm R}}_{-p^0\psi_{\rm R}} = -(\boldsymbol{\sigma}\cdot\mathbf{p})^2\psi_{\rm L} = -|\mathbf{p}|^2\psi_{\rm L} \longrightarrow (p^0)^2\psi_{\rm R} = |\mathbf{p}|^2\psi_{\rm R}$$

For a non-trivial solution, $p^0 = \pm|\mathbf{p}|$ and putting this in Eq. 22,

$$\left.\begin{aligned} \tfrac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{|\mathbf{p}|}\psi_{\rm L} &= \psi_{\rm L} \\ \tfrac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{|\mathbf{p}|}\psi_{\rm R} &= -\psi_{\rm R} \end{aligned}\right\} p^0 = +|\mathbf{p}| \qquad \left.\begin{aligned} \tfrac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{|\mathbf{p}|}\psi_{\rm L} &= -\psi_{\rm L} \\ \tfrac{(\boldsymbol{\sigma}\cdot\mathbf{p})}{|\mathbf{p}|}\psi_{\rm R} &= \psi_{\rm R} \end{aligned}\right\} p^0 = -|\mathbf{p}| \tag{23}$$

Recall from Eq. 8 that the helicity operator was,

$$\hat{h} = \frac{\hbar}{2|\mathbf{p}|}(\boldsymbol{\Sigma}\cdot\mathbf{p}) = \frac{\hbar}{2|\mathbf{p}|}\begin{pmatrix} \boldsymbol{\sigma}\cdot\mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma}\cdot\mathbf{p} \end{pmatrix}$$

Then, if we act $\Psi_{\rm L} = P_{\rm L}\Psi$ and $\Psi_{\rm R} = P_{\rm R}\Psi$ with the helicity operator (and using their form from Eq. 21), we have:

$$\hat{h}(P_{\rm L}\Psi) = \frac{\hbar}{2|\mathbf{p}|}\begin{pmatrix} \boldsymbol{\sigma}\cdot\mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma}\cdot\mathbf{p} \end{pmatrix}\begin{pmatrix} \psi_{\rm L} \\ -\psi_{\rm L} \end{pmatrix} = \frac{\hbar}{2}(P_{\rm L}\Psi)$$
$$\hat{h}(P_{\rm R}\Psi) = \frac{\hbar}{2|\mathbf{p}|}\begin{pmatrix} \boldsymbol{\sigma}\cdot\mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma}\cdot\mathbf{p} \end{pmatrix}\begin{pmatrix} \psi_{\rm R} \\ \psi_{\rm R} \end{pmatrix} = -\frac{\hbar}{2}(P_{\rm L}\Psi)$$

Hence for *particles* the left and right chiral states are also the helicity eigenstates with eigenvalue $\pm\hbar/2$ respectively (while for *anti-particles* the left and right chiral states are helicity eigenstates with eigenvalue $\mp\hbar/2$ which can be checked using the same calculation, hence chirality is negative of helicity for anti-particles).

For massive particles, it is possible to Lorentz boost to different frames of reference to change helicity (however chirality does not change since it is Lorentz invariant) and hence helicity and chirality are equivalent only for *massless* fermions.

Lecture 24: Quantisation of Dirac Field: What not to do!

Till now we have seen many different aspects of the Dirac equations and now we want to quantise it. For that, we first need to figure out a Lorentz invariant Lagrangian for the Dirac field.

24.1. Dirac Lagrangian

The appropriate Lagrangian turns out to be,

$$\mathcal{L}_{\mathrm{D}} = \overline{\Psi}\ (i\not{\partial} - m)\ \Psi$$

where Ψ and $\overline{\Psi}$ are now treated as two independent fields. Let us see whether this Lagrangian gives us the correct equation of motion or not!

$$0 = \partial_\mu\left(\frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial(\partial_\mu\Psi)}\right) - \frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial\Psi} = i\partial_\mu(\overline{\Psi}\gamma^\mu) - (-m\overline{\Psi}) = i\partial_\mu\overline{\Psi}\gamma^\mu + m\overline{\Psi} = i\partial_\mu\Psi^\dagger\gamma^0\gamma^\mu + m\Psi^\dagger\gamma^0$$

We use the identity from Eq. 11 and the invertibility of γ^0 to further simplify this:

$$0 = i\partial_\mu\Psi^\dagger(\gamma^\mu)^\dagger\gamma^0 + m\Psi^\dagger\gamma^0 = (i\partial_\mu\Psi^\dagger(\gamma^\mu)^\dagger + m\Psi^\dagger)\gamma^0 \implies 0 = (i\partial_\mu(\gamma^\mu\Psi)^\dagger + m\Psi^\dagger) \implies \boxed{0 = (i\gamma^\mu\partial_\mu\Psi + m\Psi)}$$

We thus obtain the Dirac equation. Starting with $\overline{\Psi}$ we obtain in an instant:

$$0 = \partial_\mu\left(\frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial(\partial_\mu\overline{\Psi})}\right) - \frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial\overline{\Psi}} = 0 - (i\not{\partial}\Psi - m\Psi) \implies \boxed{(i\gamma^\mu\partial_\mu\Psi + m\Psi) = 0}$$

24.2. Hamiltonian Density

Let us first find the conjugate momentum for the fields.

$$\pi_\Psi = \frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial(\partial_0\psi)} = i\overline{\Psi}\gamma^0 = i\Psi^\dagger(\gamma^0)^2 = i\Psi^\dagger \qquad \pi_{\overline{\Psi}} = \frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial(\partial_0\overline{\psi})} = 0$$

Then using the Legendre transform, we have the Hamiltonian density as:

$$\mathcal{H} = \pi_\Psi\dot{\Psi} + \pi_{\overline{\Psi}}\dot{\overline{\Psi}} - \mathcal{L} = i\overline{\Psi}\gamma^0\partial_0\Psi - i\overline{\Psi}\gamma^\mu\partial_\mu\Psi + m\overline{\Psi}\Psi = -i\overline{\Psi}\gamma^i\partial_i\Psi + m\overline{\Psi}\Psi$$

We had obtained the Lagrangian and the Hamiltonian density for the Dirac field. What we need to do now is to propose some nice guess for the field expressions. We had already found four solutions for a momentum p namely $u^s(p)$ and $v^s(p)$ for $s = 1, 2$. We can thus make the field a linear combination of these four solutions with some arbitrary coefficients:

$$\Psi(x) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\ \frac{1}{\sqrt{2E_{\mathbf{p}}}} \sum_s \left(\hat{a}^s_{\mathbf{p}} u^s(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}} + \hat{b}^s_{\mathbf{p}} v^s(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}}\right)$$

Taking the Hermitian adjoint and multiplying by γ^0 on the right, we get the expansion for the Dirac adjoint,

$$\overline{\Psi}(x) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\ \frac{1}{\sqrt{2E_{\mathbf{p}}}} \sum_r \left(\hat{a}^{r\dagger}_{\mathbf{p}} \bar{u}^r(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{b}^{r\dagger}_{\mathbf{p}} \bar{v}^r(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}}\right)$$

24.3. Total Hamiltonian

We will now try to find an expression for the total Hamiltonian $\hat{H}$:

$$\hat{H} = \int \mathrm{d}^3\mathbf{x}\ \mathcal{H} = \int \mathrm{d}^3\mathbf{x}\ \left(\overline{\Psi}(-i\gamma^i\partial_i + m)\Psi\right)$$

This is an extremely tedious and terrible calculation, one that I wouldn't wish upon anyone. Substituting the expression for $\Psi(\mathbf{x})$ and $\overline{\Psi}(\mathbf{x})$ we get:

$$\hat{H} = \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\frac{1}{2\sqrt{E_{\mathbf{p}}E_{\mathbf{q}}}} \sum_{r,s} \left(\hat{a}^{s\dagger}_{\mathbf{p}} u^s(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{b}^{s\dagger}_{\mathbf{p}} v^s(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}}\right)(-i\gamma^i\partial_i + m)$$
$$\times \left(\hat{a}^r_{\mathbf{q}} u^r(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{x}} + \hat{b}^r_{\mathbf{q}} v^r(\mathbf{q}) e^{-i\mathbf{q}\cdot\mathbf{x}}\right)$$

Note that $\mathbf{p}\cdot\mathbf{x} = x^i p^i = -x^i p_i$ and then $\partial_i \exp(\pm i\mathbf{p}\cdot\mathbf{x}) = \mp i p_i \exp(\pm i\mathbf{p}\cdot\mathbf{x})$. We use this in the above expression to get

$$\hat{H} = \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\frac{1}{2\sqrt{E_\mathbf{p}E_\mathbf{q}}}\sum_{r,s}\left(\hat{a}^{s\dagger}_\mathbf{p}\bar{u}^s(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{b}^{s\dagger}_\mathbf{p}\bar{v}^s(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x}}\right)$$
$$\times\left(\hat{a}^r_\mathbf{q}(-\gamma^i q_i + m)u^r(\mathbf{q})e^{i\mathbf{q}\cdot\mathbf{x}} + \hat{b}^r_\mathbf{q}(\gamma^i q_i + m)v^r(\mathbf{q})e^{-i\mathbf{q}\cdot\mathbf{x}}\right)$$

From Eq. 12, recall that we had the following:

$$(\not{p} - m)u^s(\mathbf{p}) = 0 \implies (\gamma^0 p_0 + \gamma^i p_i - m)u^s(\mathbf{p}) = 0 \implies (-\gamma^i p_i + m)u^s(\mathbf{p}) = E_\mathbf{p}\gamma^0 u^s(\mathbf{p})$$
$$(\not{p} + m)v^s(\mathbf{p}) = 0 \implies (\gamma^0 p_0 + \gamma^i p_i + m)v^s(\mathbf{p}) = 0 \implies (\gamma^i p_i + m)v^s(\mathbf{p}) = -E_\mathbf{p}\gamma^0 v^s(\mathbf{p})$$

Substituting this in the integral above we get:

$$\begin{aligned}
\hat{H} &= \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\frac{1}{2\sqrt{E_\mathbf{p}E_\mathbf{q}}}E_\mathbf{q}\sum_{r,s}\left(\hat{a}^{s\dagger}_\mathbf{p}\bar{u}^s(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{b}^{s\dagger}_\mathbf{p}\bar{v}^s(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x}}\right) \\
&\qquad\times\gamma^0\left(\hat{a}^r_\mathbf{q}u^r(\mathbf{q})e^{i\mathbf{q}\cdot\mathbf{x}} - \hat{b}^r_\mathbf{q}v^r(\mathbf{q})e^{-i\mathbf{q}\cdot\mathbf{x}}\right) \\
&= \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\sqrt{\frac{E_\mathbf{q}}{4E_\mathbf{p}}}\sum_{r,s}\left(\hat{a}^{s\dagger}_\mathbf{p}u^{s\dagger}(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{b}^{s\dagger}_\mathbf{p}v^{s\dagger}(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x}}\right) \\
&\qquad\times(\gamma^0)^2\left(\hat{a}^r_\mathbf{q}u^r(\mathbf{q})e^{i\mathbf{q}\cdot\mathbf{x}} - \hat{b}^r_\mathbf{q}v^r(\mathbf{q})e^{-i\mathbf{q}\cdot\mathbf{x}}\right) \\
&= \int \mathrm{d}^3\mathbf{x} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\sqrt{\frac{E_\mathbf{q}}{4E_\mathbf{p}}}\sum_{r,s}(\hat{a}^{s\dagger}_\mathbf{p}\hat{a}^r_\mathbf{q}u^{s\dagger}(\mathbf{p})u^r(\mathbf{q})e^{i(\mathbf{q}-\mathbf{p})\cdot\mathbf{x}} - \hat{a}^{s\dagger}_\mathbf{p}\hat{b}^r_\mathbf{q}u^{s\dagger}(\mathbf{p})v^r(\mathbf{q})e^{-i(\mathbf{q}+\mathbf{p})\cdot\mathbf{x}} \\
&\qquad+\hat{b}^{s\dagger}_\mathbf{p}\hat{a}^r_\mathbf{q}v^{s\dagger}(\mathbf{p})u^r(\mathbf{q})e^{i(\mathbf{q}+\mathbf{p})\cdot\mathbf{x}} - \hat{b}^{s\dagger}_\mathbf{p}\hat{b}^r_\mathbf{q}v^{s\dagger}(\mathbf{p})v^r(\mathbf{q})e^{i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}})
\end{aligned}$$

Now if we perform the position integration first, due to the exponentials, we would get $\delta^{(3)}(\mathbf{p}\pm\mathbf{q})$ from each term and then if we perform the integration over $\mathbf{q}$, these Dirac deltas would get removed, giving $q = \pm p$. Doing this, our integral becomes:

$$\hat{H} = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2}\sum_r(\hat{a}^{r\dagger}_\mathbf{p}\hat{a}^r_\mathbf{p}u^{r\dagger}(\mathbf{p})u^r(\mathbf{p}) - \hat{a}^{r\dagger}_\mathbf{p}\hat{b}^r_{-\mathbf{p}}u^{r\dagger}(\mathbf{p})v^r(-\mathbf{p})$$
$$+\hat{b}^{r\dagger}_\mathbf{p}\hat{a}^r_{-\mathbf{p}}v^{r\dagger}(\mathbf{p})u^r(-\mathbf{p}) - \hat{b}^{r\dagger}_\mathbf{p}\hat{b}^r_\mathbf{p}v^{r\dagger}(\mathbf{p})v^r(\mathbf{p}))$$

From Eq. 18 and 19, we obtain a nice, cute equation (well, hidden under the cuteness is a huge mess which we have to deal with),

$$\boxed{\hat{H} = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}E_\mathbf{p}\sum_r\left(\hat{a}^{r\dagger}_\mathbf{p}\hat{a}^r_\mathbf{p} - \hat{b}^{r\dagger}_\mathbf{p}\hat{b}^r_\mathbf{p}\right)} \tag{24}$$

Appendices

A. Few BS about Lorentz Group

We will discuss about Lorentz group here. The Lorentz group, denoted by $O(1,3)$, is defined to be the group of all transformations (Lorentz transformation or LT) that preserves the Minkowski metric, that is,

$$\Lambda^{\mathrm{T}}\eta\Lambda = \eta \iff \eta_{\mu\nu}\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta} = \eta_{\alpha\beta} \tag{25}$$

Note that, as long as we are not using the indices, we are talking about the *abstract* group element. However, when we consider the expression using index notation, we are essentially considering the matrix representation of the group element using the *standard basis* of the Minkowski space[1].

From the above definition, we can instantly see that for any $\Lambda \in O(1,3)$, $\det\Lambda = \pm 1$. For $\det\Lambda = +1$, we refer to them as *proper LT* (denoted by L_+) and for $\det\Lambda = -1$ (denote by L_-), we refer to them as *improper LT*.
In Eqn. 25, if we take $\alpha = \beta = 0$, we get:

$$1 = \eta_{\mu\nu}\Lambda^{\mu}{}_{0}\Lambda^{\nu}{}_{0} = \left(\Lambda^{0}{}_{0}\right)^2 - \sum_{i=1}^{3}\left(\Lambda^{i}{}_{0}\right)^2 \implies \left(\Lambda^{0}{}_{0}\right)^2 \geq 1$$

Thus, we can have $\Lambda^{0}{}_{0} \geq 1$ which we refer to as *orthochronous* (denoted by $L^{\uparrow}$) or $\Lambda^{0}{}_{0} \leq -1$ which we call *antichronous* (denoted by $L^{\downarrow}$).

From the above two classifications using the signs of the determinant and $\Lambda^{0}{}_{0}$, we can thus have four disconnected components of the Lorentz group:

- $L_+^{\uparrow}$: These are the proper orthochronous LT. The identity $\mathbb{1}$ is contained in this subgroup. It is denoted by by $\mathrm{SO}^+(1,3)$ and is continuously connected to the identity, that is, we can build up any finite Lorentz transformation by making many consecutive small Lorentz transformations. Sometimes, as an abuse of terminology, this subgroup is itself called the *Lorentz group*, since this contains the only physical transformations.
- $L_-^{\uparrow}$: These are written as a product of the *parity* and $L_+^{\uparrow}$ transformations.
- $L_+^{\downarrow}$: These contain both time and space inversion and hence are written as a product of PT and $L_+^{\uparrow}$ transformation.
- $L_-^{\downarrow}$: These contain only time inversion and hence are written as a product of *time reversal* and $L_+^{\uparrow}$ transformation.

Thus, the entire Lorentz group can be written symbolically:

$$L = L_+^{\uparrow} \bigcup PL_+^{\uparrow} \bigcup PTL_+^{\uparrow} \bigcup TL_+^{\uparrow}$$

Note that except the proper orthochronous LT, none of the other component can form subgroups since these do not contain the identity element.

A.1. Spacetime Representation

For the time being, we will consider only the proper, orthochronous Lorentz group, that is, $\mathrm{SO}^+(1,3)$ which is a Lie group. Since this is continuously connected to the identity, let us consider an infinitesimal transformation from the identity given by,

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu}$$

[1]The Mindkowski space is denoted by $\mathbb{R}^{1,3}$ which is equal to $\mathbb{R}^4$ along with the associated metric $\eta_{\mu\nu}$. $(1,3)$ refers to the metric signature, that is, 1 positive entry and 3 negative entries viz. $\mathrm{diag}(+1,-1,-1,-1)$

Substituting this in Eq. 25 and retaining terms upto the first oder in ω, we have the following:

$$\begin{aligned}\eta_{\alpha\beta} &= \eta_{\mu\nu}(\delta^{\mu}{}_{\alpha} + \omega^{\mu}{}_{\alpha})(\delta^{\nu}{}_{\beta} + \omega^{\nu}{}_{\beta}) \\ &= \eta_{\alpha\beta} + \delta^{\mu}{}_{\alpha}\eta_{\mu\nu}\omega^{\nu}{}_{\beta} + \eta_{\mu\nu}\delta^{\nu}{}_{\beta}\omega^{\mu}{}_{\alpha} \\ &= \eta_{\alpha\beta} + \delta^{\mu}{}_{\alpha}\omega_{\mu\beta} + \delta^{\nu}{}_{\beta}\omega_{\nu\alpha} \\ &= \eta_{\alpha\beta} + \omega_{\alpha\beta} + \omega_{\beta\alpha}\end{aligned}$$

From this, we get that infinitesimal displacement ω must be anti-symmetric, that is,

$$\boxed{\omega_{\alpha\beta} = -\omega_{\beta\alpha}}$$

We know that antisymmetric matrices have six independent elements (since diagonals are zero and upper-diagonal has $(16-4)/2 = 6$ elements and lower diagonal are negative of the upper diagonal) and hence the basis of antisymmetric matrices consists six elements. Of these six, three corresponds to *rotations* while the other three corresponds to *boosts*

Let us consider this basis of the second rank anti-symmetric tensor $\{\mathcal{J}^{\rho\sigma}\}$ which forms the *generators* of the transformation. These are antisymmetric in the indices ρ, σ and hence there are six of them. These can be written as:

$$(\mathcal{J}^{\rho\sigma})^{\mu}{}_{\nu} := i(\eta^{\mu\rho}\delta^{\sigma}{}_{\nu} - \eta^{\mu\sigma}\delta^{\rho}{}_{\nu})$$

As ω is antisymmetric, we can write it as a linear combination of the basis elements, that is,

$$\omega^{\mu}{}_{\nu} = -i\Omega_{\rho\sigma}(\mathcal{J}^{\rho\sigma})^{\mu}{}_{\nu}$$

where $\Omega_{\rho\sigma} \in \mathbb{R}$ (since $\mathcal{J}^{\mu\nu}$ are purely imaginary, $i\mathcal{J}^{\mu\nu}$ becomes real and hence Λ becomes real as it should be) since Lorentz group is a real Lie group. We can write the infinitesimal transformation in the following way:

$$\Lambda = \mathbb{1} - i\Omega_{\rho\sigma}\mathcal{J}^{\rho\sigma}$$

Any finite Lorentz transformation can be generated by exponentiating the generators, that is,

$$\Lambda_{\text{finite}} = \exp\left(-\frac{i}{2}\Omega_{\rho\sigma}\mathcal{J}^{\rho\sigma}\right)$$

Once we found the generators, any element of the Lie algebra of the Lorentz group will be a linear combination of them. It is a good time for us to calculate the Lie bracket since it is an essential characteristic of the Lie algebra. Since any element is a linear combination of the generators, it is sufficient to calculate the Lie bracket for the generators only, which we can do by the explicit form that we wrote earlier.

$$\begin{aligned}[\mathcal{J}^{\mu\nu}, \mathcal{J}^{\rho\sigma}]^{\alpha}{}_{\beta} &= (\mathcal{J}^{\mu\nu}\mathcal{J}^{\rho\sigma})^{\alpha}{}_{\beta} - (\mathcal{J}^{\rho\sigma}\mathcal{J}^{\mu\nu})^{\alpha}{}_{\beta} \\ &= (\mathcal{J}^{\mu\nu})^{\alpha}{}_{\theta}(\mathcal{J}^{\rho\sigma})^{\theta}{}_{\beta} - (\mathcal{J}^{\rho\sigma})^{\alpha}{}_{\theta}(\mathcal{J}^{\mu\nu})^{\theta}{}_{\beta} \\ &= (\eta^{\mu\alpha}\delta^{\nu}{}_{\theta} - \eta^{\nu\alpha}\delta^{\mu}{}_{\theta})(\eta^{\rho\theta}\delta^{\sigma}{}_{\beta} - \eta^{\sigma\theta}\delta^{\rho}{}_{\beta}) - (\eta^{\rho\alpha}\delta^{\sigma}{}_{\theta} - \eta^{\sigma\alpha}\delta^{\rho}{}_{\theta})(\eta^{\mu\theta}\delta^{\nu}{}_{\beta} - \eta^{\nu\theta}\delta^{\mu}{}_{\beta}) \\ &= \left\{\eta^{\mu\alpha}\delta^{\nu}{}_{\theta}\eta^{\rho\theta}\delta^{\sigma}{}_{\beta} - \eta^{\mu\alpha}\delta^{\nu}{}_{\theta}\eta^{\sigma\theta}\delta^{\rho}{}_{\beta} - \eta^{\nu\alpha}\delta^{\mu}{}_{\theta}\eta^{\rho\theta}\delta^{\sigma}{}_{\beta} + \eta^{\nu\alpha}\delta^{\mu}{}_{\theta}\eta^{\sigma\theta}\delta^{\rho}{}_{\beta}\right\} \\ &\quad - \left\{\eta^{\rho\alpha}\delta^{\sigma}{}_{\theta}\eta^{\mu\theta}\delta^{\nu}{}_{\beta} - \eta^{\rho\alpha}\delta^{\sigma}{}_{\theta}\eta^{\nu\theta}\delta^{\mu}{}_{\beta} - \eta^{\sigma\alpha}\delta^{\rho}{}_{\theta}\eta^{\mu\theta}\delta^{\nu}{}_{\beta} + \eta^{\sigma\alpha}\delta^{\rho}{}_{\theta}\eta^{\nu\theta}\delta^{\mu}{}_{\beta}\right\} \\ &= \left\{\eta^{\mu\alpha}\eta^{\rho\nu}\delta^{\sigma}{}_{\beta} - \eta^{\mu\alpha}\eta^{\sigma\nu}\delta^{\rho}{}_{\beta} - \eta^{\nu\alpha}\eta^{\rho\mu}\delta^{\sigma}{}_{\beta} + \eta^{\nu\alpha}\eta^{\sigma\mu}\delta^{\rho}{}_{\beta}\right\} \\ &\quad - \left\{\eta^{\rho\alpha}\eta^{\mu\sigma}\delta^{\nu}{}_{\beta} - \eta^{\rho\alpha}\eta^{\nu\sigma}\delta^{\mu}{}_{\beta} - \eta^{\sigma\alpha}\eta^{\mu\rho}\delta^{\nu}{}_{\beta} + \eta^{\sigma\alpha}\eta^{\nu\rho}\delta^{\mu}{}_{\beta}\right\} \text{complete proof!}\end{aligned}$$

From this, we find out the Lie bracket between the generators of the Lie Algebra of SO$(1,3)$:

$$[\mathcal{J}^{\mu\nu}, \mathcal{J}^{\rho\sigma}] = i(\eta^{\nu\rho}\mathcal{J}^{\mu\sigma} - \eta^{\mu\rho}\mathcal{J}^{\nu\sigma} - \eta^{\nu\sigma}\mathcal{J}^{\mu\rho} + \eta^{\mu\sigma}\mathcal{J}^{\nu\rho})$$

We had obtained the six generators for the Lie algebra and how let us rearrange these into two separate parts:

$$J^{i} = \frac{i}{2}\varepsilon^{ijk}\mathcal{J}^{jk} \qquad K^{i} = i\mathcal{J}^{i0}$$

In terms of these newly defined quantitites, the Lie bracket becomes check!:

$$\begin{aligned}\left[J^i, J^j\right] &= i\varepsilon^{ijk}J^k \\ \left[K^i, K^j\right] &= -i\varepsilon^{ijk}J^k \\ \left[J^i, K^j\right] &= i\varepsilon^{ijk}K^k\end{aligned}$$

This shows that J^i as defined above kinda denotes angular momentum since it satisfies the Lie Algebra of SU(2) while the other relation shows that K^i is a spatial vector. Now let us define two quantitites:

$$\theta^i = \frac{1}{2}\varepsilon^{ijk}\Omega_{jk} \qquad \eta^i = \Omega^{i0}$$

Using this we can write the generators as:

$$\frac{1}{2}\Omega_{\mu\nu}\mathcal{J}^{\mu\nu} = \Omega_{12}\mathcal{J}^{12} + \Omega_{13}\mathcal{J}^{13} + \Omega_{23}\mathcal{J}^{23} + \Omega_{i0}\mathcal{J}^{i0} = \boldsymbol{\theta}\cdot\mathbf{J} - \boldsymbol{\eta}\cdot\mathbf{K}$$

Thus any Lorentz transformation can then be written as:

$$\boxed{\Lambda = \exp\left(-i\ \boldsymbol{\theta}\cdot\mathbf{J} + i\ \boldsymbol{\eta}\cdot\mathbf{K}\right)}$$